

WP4 Deliverable 4.2

Specification of the technical enablers for status, projections and actions

Sa4CPS

ITEA 4 22007

Status: Draft

.....

Edited by Tomáš Bureš (CUNI), Petr Hnětynka (CUNI)

Including contributions from:

Jiří Havlík (IMA)
Sini Olkanen (F-Secure)
Lander Vanhaverbeke (Vintec)
Pedro Faria (ISEP)
Rafael Silva (ISEP)
Joona Vilkkö (Modirum Gespi)
Hannu Lohi (Natlink)
Grégoire Moreau (ADLC)
Yasemin Eker (Necdet Alpata)
Feyza Nur Genç (Necdet Alpata)
Diogo Sousa (Digitalmente)
Ana Lobo (CEL)
Juha Häikiö (Port of Oulu)
Jyrki Tihinen (VTT)
Dilara Bayar Başlık (ACD)
Edanur Başlık (IM)
İbrahim Arif (ERARGE)

Table of Contents

Table of Contents	3
Table of Figures	6
Abstract	7
List of Abbreviations	8
1 Introduction	10
1.1 Scope of the deliverable	11
1.2 Technical enablers covered	12
2 Concept of digital twins	14
2.1 Digital twins & dashboards	15
2.2 Detection and responses	16
2.3 The selection of the digital twin platform	16
3 Integrated Situation Awareness, Risk Assessment, and Operational Response for Drone Operations	18
3.1 Perception, comprehension & risk assessment engine (EN_02)	19
3.2 Operations monitoring dashboard (EN_03)	20
3.3 Response dashboard (EN_04)	21
3.4 State egress adapters (EN_05)	22
4 Unified Operational Awareness and Control for Industrial Forklift Operations	24
4.1 Low-Latency Hybrid Communication Module (EN_06)	25
4.1.1 Setup of the 5G Mesh Network and Indoor Communication Testing	25
4.1.2 5G Test Environment	26
4.2 Sa4CPS Web Platform (EN_07)	27
4.3 3AM Decentralized Backend (EN_08)	30
4.3.1 Triangular Accountability Model (3AM) Decentralized Backend	31
4.4 Digital Twin with Semantic Backend (EN_09)	34
4.5 PHM Backend & Big Data Ingestion Pipeline (EN_10)	35
4.6 Battery PHM (Prognostics & Health Management) Tool (EN_11)	36
4.7 Digital Twin Environment for forklift operations (EN_12)	38
4.8 Optimum Route Planning and Vehicle-to-Vehicle Cooperation Engine (EN_13)	40
4.9 Digital twin-based situational awareness framework (EN_14)	45
4.10 Demonstration site – Port of Oulu	46
5 A language, formal model, and tools to describe, model, and identify situations	49
5.1 Example	49
5.2 Meta-model	50

5.2.1	<i>Objects and inheritance structure</i>	52
5.2.2	<i>Common object properties</i>	52
5.3	<i>Environment description</i>	53
5.4	<i>Situation and behavior modelling</i>	56
6	Behaviour-Based Digital Risk Detection and Situational Awareness	60
6.1	<i>DNS Fingerprinting Data Processing Pipeline (EN_16)</i>	60
6.2	<i>Behavioural data hub (EN_17)</i>	61
7	Real-Time Operational Data Integration and Visualization	63
7.1	<i>Capture platform (EN_18)</i>	63
7.1.1	<i>Data Collection and Protocol Selection</i>	64
7.1.2	<i>Synchronizing Data from Edge to Cloud</i>	64
7.1.3	<i>Accessing Data in the Cloud</i>	65
8	Geospatial Situational Awareness and Event Detection	68
8.1	<i>Map, Map Engine, and Visualization Technologies (EN_19)</i>	69
8.2	<i>AI-Enabled Integration of Location Data into the NSC3 Environment (EN_20)</i>	69
8.3	<i>AI-Based Event Detection and Real-Time Alerting (EN_21)</i>	70
8.4	<i>Integrating Camera Sources to NSC3 Environment (EN_22)</i>	70
8.5	<i>Hardening NSC3 Environment to Zero Trust Authentication and Authorization (EN_23)</i>	70
8.6	<i>NSC3 system description</i>	71
8.6.1	<i>Services in the NSC3 backend</i>	71
8.6.2	<i>Setting up the system for object detections</i>	72
8.6.3	<i>Data flow from video feed to AI-driven detection drawn on a map</i>	75
9	Integrated Status, Decision Support, and Control for Energy Communities and Distribution Networks	78
9.1	<i>Energy Community Operational Management and Governance Interface (EN_24)</i>	79
9.2	<i>Human-in-the-Loop Community Action Validation Interface (EN_25)</i>	80
9.3	<i>Regulatory, Environmental and Community Performance Reporting (EN_26)</i>	80
9.4	<i>Energy Community Digital Twin Backbone (EN_27)</i>	81
9.5	<i>Energy Community Data Aggregation and Visualization Services (EN_28)</i>	82
9.6	<i>Trust-Aware Digital Twin State Confidence Engine (EN_29)</i>	82
9.7	<i>Energy data sharing: Development for continuous delivery of meter data (EN_30)</i>	83
9.8	<i>Energy data sharing: REST APIs (EN_31)</i>	83
9.9	<i>Flexibility services: Load control capabilities (EN_32)</i>	84
9.10	<i>Digital twin for electricity distribution network with distributed generation and flexible load (EN_33)</i>	84
9.11	<i>Dashboard for energy community management (EN_34)</i>	84
9.12	<i>Activation of flexible resources, like storage and electric vehicles (EN_35)</i>	84

9.13	<i>Autonomous detection and mitigation of voltage issues (EN_36)</i>	84
10	Human-Centric Digital Twins, Tracking, and Spatial Situational Awareness	86
10.1	<i>Multicomunicator to mobile, web, car and watch interface (EN_37)</i>	87
10.2	<i>Real-time monitoring services for Multicomunicator environment in server (EN_38)</i>	87
10.3	<i>Data sharing (EN_39)</i>	87
10.4	<i>Autonomous media optimized data traffic between components (EN_40)</i>	87
10.5	<i>3D Map Engine, Server, and Research (EN_41)</i>	88
10.6	<i>Multiplace Digital Twin (EN_42)</i>	88
10.7	<i>Biometric Data Transfer Device (EN_43)</i>	89
10.7.1	<i>Wearable devices and a data pipeline to create situational awareness (Task 4.2)</i>	89
11	Conclusions	92
12	References	93

Table of Figures

Figure 1 WP4 in the overall project architecture	11
Figure 2 Concept of Digital Twin	14
Figure 3 Usage of Digital Twin in the project	15
Figure 4: Architecture of the platform (based on Microsoft Azure Services)	20
Figure 5 Monitoring dashboard (in development)	21
Figure 6: Response page (in development)	22
Figure 7 Setup of the Development Kits	26
Figure 8 General diagram	27
Figure 9 Sa4CPS web platform – Monitoring page.....	28
Figure 10 Sa4CPS web platform – operation history page.....	28
Figure 11 Sa4CPS web platform - PHM tool page	29
Figure 12 Sa4CPS web platform - IPFS page	30
Figure 13 Blockchain-enabled Digital Twin Architecture with Semantic Backend.....	31
Figure 14 API workflow	36
Figure 15 Sa4CPS web platform – PHM tool page.....	37
Figure 16 Factory Environment.....	39
Figure 17 Top View of the Factory	40
Figure 19 Illustration of integrating IoT-based real-time data	41
Figure 20 Visualization of the routing performance analysis	42
Figure 21 Visualization of the optimal transportation route.....	44
Figure 22 Human Detection system on forklift	45
Figure 23 Human detection and distance-based warning generation using ToF cameras and AI during forklift reverse maneuver.....	46
Figure 24 Port of Oulu summarized project story	47
Figure 25 Robot vs Human	50
Figure 26 Robot vs Human as seen by ToF camera	50
Figure 27 A meta-model describing situations	51
Figure 28 Edge vs. Cloud.....	65
Figure 29 Grafana dashboard	65
Figure 30 Capture Data Explorer	66
Figure 31 Creating a task on the NSC3 system.....	72
Figure 32 Attaching an object detection model.....	73
Figure 33 Adding IP camera to the NSC3 system	74
Figure 34 Assign the connected camera a task.....	75
Figure 35 Data flow from video feed to AI driven detection	76
Figure 36 State processing flow.....	83
Figure 37 Architecture of the Data Transfer Device service	89
Figure 38 DTD Processes	91

Abstract

This deliverable, **Specification of the technical enablers for status, projections and actions**, provides a specification-level description of the technical enablers that are being developed in WP4. Building on the requirements defined in D4.1, the deliverable specifies the functionality, scope, and interactions of WP4 enablers supporting status representation, projection of future states, and execution of actions in cyber-physical systems. The specified enablers cover digital twins and dashboards, positioning and tracking, detection and response mechanisms, autonomous and semi-autonomous operations, real-time data integration and visualization, trust- and accountability-supporting components, and human-centric situational awareness solutions. The specifications are presented in a use-case-driven manner across multiple application domains, establishing a coherent baseline for subsequent design, implementation, integration, and validation activities.

List of Abbreviations

Abbreviation	Explanation
3AM	Triangular Accountability Model
ADS	(Beckhoff) Automation Device Specification
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
BLE	Bluetooth Low Energy
BFF	Backend-for-Frontend
CAN	Controller Area Network
CPS	Cyber-Physical System
DID	Decentralized Identifier
DNS	Domain Name System
DSO	Distribution System Operator
DTD	Data Transfer Device
GNSS	Global Navigation Satellite System
HTTP	Hypertext Transfer Protocol
IIoT	Industrial Internet of Things
IoT	Internet of Things
IPFS	InterPlanetary File System
KPI	Key Performance Indicator
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
NATS	Neural Autonomic Transport System
NSC3	National Secure Command and Control Center
NSS	Network Streaming Service
OGC	Open Geospatial Consortium
OPC-UA	Open Platform Communications – Unified Architecture
PHM	Prognostics and Health Management
PLC	Programmable Logic Controller
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDK	Software Development Kit

SOC	State of Charge
SOH	State of Health
SPARQL	SPARQL Protocol and RDF Query Language
ToF	Time-of-Flight
UC	Use Case
UNS	Unified Namespace
URP	Universal Render Pipeline
UTM	Unmanned Traffic Management
VC	Verifiable Credential
WFS	Web Feature Service
WMS	Web Map Service
WMTS	Web Map Tile Service

1 Introduction

The Sa4CPS project addresses the challenge of achieving secure and trustworthy situational awareness in complex cyber-physical systems operating across multiple domains, stakeholders, and operational environments. The core project objective, as defined in the Full Project Proposal (FPP), is to enable systems that can reliably perceive their environment, understand and interpret evolving situations, project future developments, and support or execute appropriate actions under explicit security and trust assumptions. This objective is pursued through a structured work plan that separates conceptual foundations, technical enablers, and validation activities while maintaining strong integration across work packages.

Within this structure, WP1 and WP2 establish the conceptual and architectural foundations of Sa4CPS. WP2, in particular, defines the overarching Sa4CPS architecture and coordinates the technical contributions of the solution-oriented work packages. WP3, WP4, and WP5 then develop the concrete technical capabilities that populate this architecture, following a situational-awareness pipeline perspective. WP3 focuses on perception and comprehension, including sensing, data acquisition, and initial interpretation of the environment. WP4 addresses the subsequent stages of this pipeline by providing enablers for representing the current situation, projecting future states, and reasoning about or executing actions. WP5 complements these capabilities by addressing trust, security, and privacy, ensuring that situational awareness is not only technically correct but also reliable, verifiable, and usable across organizational boundaries. WP6 provides the experimentation and validation framework in which the integrated results of WP1–WP5 are evaluated in realistic operational settings.

Within this context, WP4 has a central integrative role. It is responsible for transforming heterogeneous, potentially uncertain inputs into operationally meaningful representations of system status, for enabling reasoning about risks and future developments, and for supporting responses that keep the system within safe and intended operating conditions. WP4 explicitly covers digital twins and dashboards, positioning and tracking of objects, autonomous and semi-autonomous operations, detection of unexpected events and emergencies, reasoning about appropriate responses, and the presentation of overlapping operational views through 3D maps and tools. These capabilities are organized in WP4 through five tasks (T4.1–T4.5), which together define the functional scope within which the WP4 enablers are developed.

Deliverable D4.2 is positioned at a critical point in the WP4 timeline. It follows D4.1, which establishes the technological requirements for WP4, and precedes subsequent deliverables that address detailed design, implementation, and evaluation. As such, D4.2 serves as the specification-level baseline for WP4: it refines the requirements identified in D4.1 into concrete technical enablers, clarifies their responsibilities and scope, and establishes how they contribute to the WP4 objectives without yet committing to detailed implementation decisions. In this sense, D4.2 forms the contractual and technical bridge between requirements definition and solution realization within WP4, and it provides the common reference point for later design and validation activities.

This deliverable provides the specification-level description of the technical enablers developed in WP4. The deliverable builds on the technological requirements identified in D4.1 and translates them into structured specifications that define the functionality, scope, and interactions of the WP4 enablers.

The structure of D4.2 reflects this role. Rather than presenting a single monolithic solution, the deliverable specifies a coherent set of enablers that collectively support status monitoring, projection, and action, while being instantiated in different application domains. The specification is explicitly use-case driven, but it maintains a common technical logic across sections: each use-case slice exercises a subset of the same

WP4 capability space, shaped by domain-specific constraints, operational rhythms, and stakeholder requirements. This approach aligns with the FPP's intent that WP4 results should be reusable and composable across domains rather than tailored to isolated vertical solutions.

1.1 Scope of the deliverable

The scope of this deliverable is to provide a technical specification of the technical enablers related to status monitoring, projection of future states, and execution of actions, as addressed by the tasks of WP4. These enablers are specified in a use-case-driven manner, ensuring applicability across multiple Sa4CPS use cases, while maintaining a common and coherent technical structure.

In particular, D4.2:

- Refines the requirements defined in D4.1 into concrete technical enablers.
- Describes the responsibilities, inputs, outputs, and interactions of these enablers.
- Establishes a common specification baseline for further design and development work.

This deliverable does not cover detailed system design or implementation aspects, which are addressed in subsequent WP4 deliverables.

Figure 1 illustrates the overall concept architecture of the project, with a focus on WP4. The figure also provides an overview of areas where each use case makes contributions (the colored dots correspond to the color of each use case in the top row). The yellow triangles indicate areas affected by the project story.

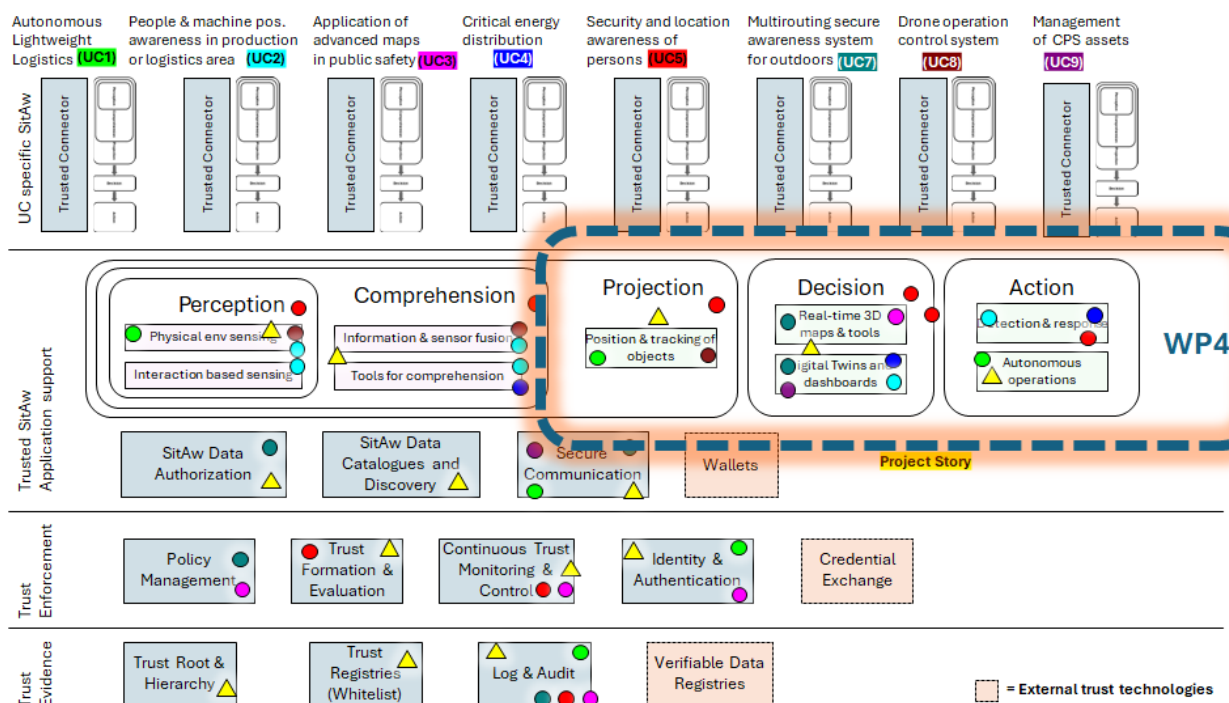


Figure 1 WP4 in the overall project architecture

1.2 Technical enablers covered

This deliverable specifies a comprehensive set of **technical enablers being developed within WP4** that support **status representation, projection of future states, and execution of actions**. The enablers are specified at a functional and technical level and are organised according to the main capability areas of WP4 and their instantiation across different application domains and use cases.

The scope of the technical enablers covered in this deliverable includes:

- **Digital twin concepts and foundations** (Section 2), defining the role of digital twins and dashboards as authoritative representations of physical and digital entities, system state, and operational context, including semantic enrichment and historical state management.
- **Integrated situation awareness, risk assessment, and operational response for drone operations** (Section 3), specifying enablers that combine status monitoring, risk detection, projection, and response mechanisms tailored to safety-critical and highly dynamic drone operation environments.
- **Unified operational awareness and control for industrial forklift operations** (Section 4), covering a comprehensive set of enablers for industrial logistics, including communication infrastructure, digital twins, tracking, autonomous and cooperative operations, detection and response mechanisms, accountability-supporting backends, and operational dashboards, with validation in realistic industrial and port environments.
- **Formal language, model, and tools for situation description and identification** (Section 5), providing cross-cutting enablers for modelling, describing, and identifying situations in cyber-physical systems, supporting systematic reasoning about people–machine interactions and safety-relevant situations at runtime.
- **Behaviour-based digital risk detection and situational awareness** (Section 6), specifying enablers for detecting and analysing risk situations based on behavioural patterns, illustrated through cybersecurity-oriented and anomaly-detection scenarios.
- **Real-time operational data integration and visualization** (Section 7), defining enablers for edge-to-cloud data ingestion, real-time and historical data processing, and standardized interfaces that support dashboards, analytics, and integration with external systems.
- **Geospatial situational awareness and event detection** (Section 8), addressing enablers for positioning, spatial reasoning, geospatial and 3D map-based visualization, and detection of spatially defined events in safety- and mission-critical environments.
- **Integrated status, decision support, and control for energy communities and distribution networks** (Section 9), specifying enablers that combine digital twin–based state representation, decision support, and controlled actions in distributed, multi-stakeholder energy systems.
- **Human-centric digital twins, tracking, and spatial situational awareness** (Section 10), covering enablers that treat humans as first-class entities, including tracking, wearable and biometric data integration, multi-device interaction, and human-in-the-loop situational awareness capabilities.

The enablers are specified across multiple use cases, including industrial logistics, drone operations, port operations, emergency response, energy communities, cybersecurity, and human-centric monitoring

scenarios. Each enabler is described in terms of its functional scope, technical architecture, and contribution to the relevant WP4 tasks and use-case objectives.

Together, the specified enablers form an integrated toolbox that enables WP4 to move from consolidated system status to informed projections and controlled actions, while remaining modular, reusable, and applicable across heterogeneous cyber-physical environments.

2 Concept of digital twins

ID	Enabler Name	Related Tasks	Addressed Requirements	Related Use Case
EN_01	Digital twin	T4.1, T4.4	REQ_COMMON_5	ALL

To achieve the goals of WP4, its tasks (esp. T4.1 and T4.4), and to provide solutions required by the project story, the project has adopted digital twins in the trust backend system. The aim of the proposed concept of digital twins (developed by VTT) is to facilitate communication, automate database operations, monitor personnel attributes, and alert of attribute threshold violations. Digital twins are digital presentations of either physical world entities or digital world entities. Digital twins are typically divided into three subtypes, which are presented in Figure 2.

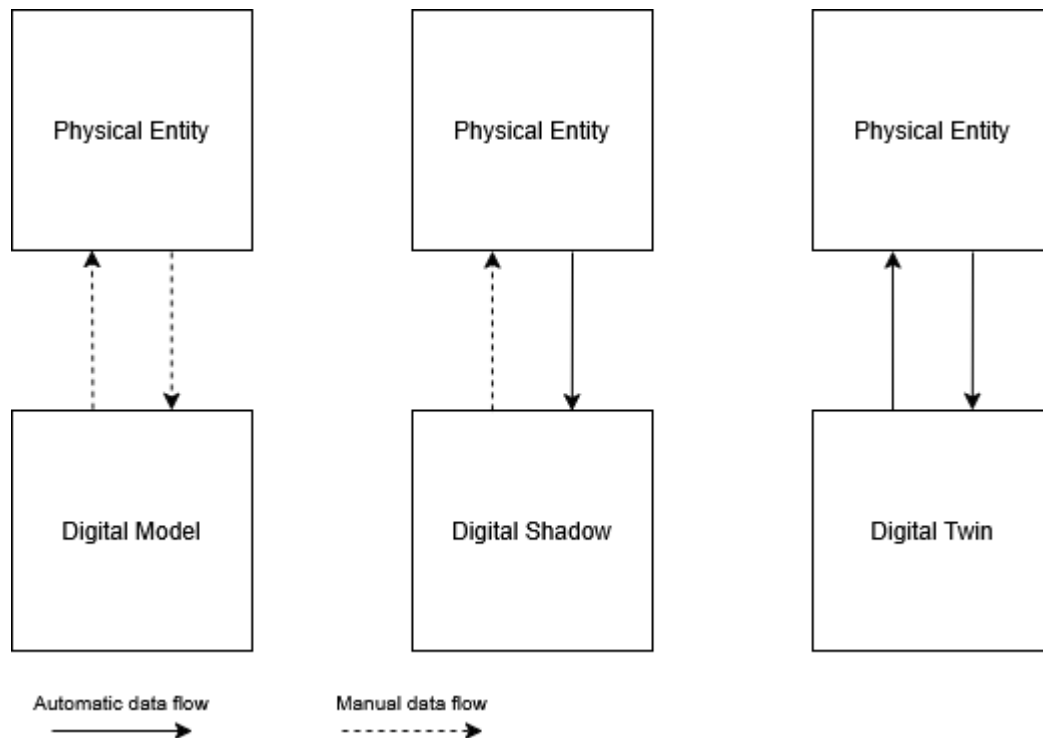


Figure 2 Concept of Digital Twin

A *digital model* has only manual data traffic with its counterpart or no data traffic at all. The digital model can exist even before the counterpart and may be used as a data storage for design and planning data. When data flows unidirectionally from the counterpart to the digital twin, it is referenced as a *digital shadow*. This digital twin type is suitable for planning, monitoring, and prediction purposes. When data flows automatically from the counterpart to the digital twin and vice versa, it is a full-fledged *digital twin*. Changes in digital twins can change the behavior of the counterpart. Digital twins are suitable for monitoring, predicting, and controlling the counterpart.

A digital twin platform concept was introduced into the Sa4CPS project's trust system backend. Digital twin platforms offer various services, such as a database and database operations, connectivity and support for various communication protocols, data visualization, and authorization. Digital twin platforms are available as open-source and commercial solutions. Digital twin platforms provide a safe and systematic way to access and update digital twin data by offering authorization and communication interfaces. Digital twin platforms also preserve the digital twin's data even when the connection to the real-world counterpart is lost. Platforms may also save the historical data of the twins, and may be viewed and visualized subsequently. Artificial intelligence can also be utilized with digital twin platforms for tasks such as data analysis and anomaly detection. Figure 3 shows how digital twins and digital twin platforms are intended to be used in the project. In this case, Real World Entities are people and machines in the port environment.

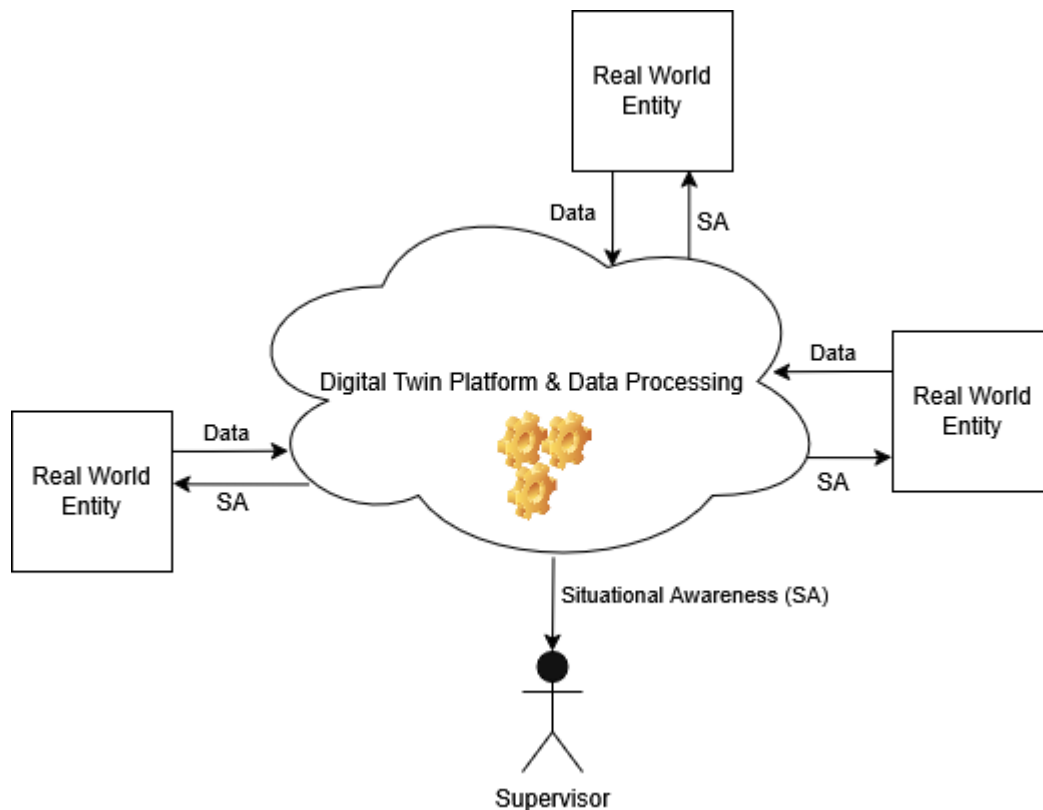


Figure 3 Usage of Digital Twin in the project

2.1 Digital twins & dashboards

Before entering the port area, the visitor requests the port supervisor to grant them the required credentials. Upon granting the credentials, the visitor's digital twin is also created. The visitor updates their data, such as location data. The digital twin can provide situational awareness to visitors by monitoring geofence violations and warning the visitor before the violation occurs. Port personnel may have more data stored on their digital twins, for example, ownerships of devices and access to their digital devices.

Port supervisor can utilize the data of digital twins for situation awareness purposes, e.g., 3D maps with locations of entities in the area. Usage of digital twins also enables machine-learning and AI based monitoring and emergency response.

2.2 Detection and responses

Digital twin platforms can be utilized to monitor the digital twins and their attributes. For example, it is possible to set a threshold value for an attribute. If the value of the attribute violates the threshold value, an automatic alert is created and sent to the authority, who will then take the necessary measures.

It is also possible to integrate machine learning algorithms and artificial intelligence into digital twin platforms for anomaly detection and prediction. It might be possible to use the historical data of digital twins as training data for the said technologies.

2.3 The selection of the digital twin platform

The first platform introduced (Eclipse Ditto) had very limited features. For example, there is no built-in rule engine to define the platform behavior. The missing functionalities were replaced with workaround scripts. However, as the number of scripts grows, the complexity of the backend system increases, and maintainability decreases. The need for a replacement platform was realized, and along with it, a need for a comparison study of open-source digital twin platforms. A master's thesis was written on the subject during the summer and autumn of the year 2025.

The study was initiated by searching for and collecting names of open-source digital twin platforms. Only non-archived, general-purpose platforms in English were selected for the study, and a total of five platforms were selected. In the first phase of the study, the maturity of the chosen platforms was evaluated with the Qualification and Selection of Open Source Software (QSOS) framework. The framework helps the user to evaluate various points defining maturity and rank them from 0 to 2. The two most mature platforms (ThingsBoard and OpenRemote) were then chosen for the next phase, where their functionalities were tested with a test scenario based on prior experiences with the Ditto platform. The test scenario included tasks, such as connecting to an external MQTT broker, digital twin creation, reading, updating, and deleting operations via the MQTT connection, dispatching digital twin attribute change notifications, and data visualization.

The results of the test scenario revealed that neither platform fully passed the test scenario. ThingsBoard has an in-built MQTT broker. The free community edition cannot be connected to an external MQTT broker. Only the paid professional version can do it. The creation and deletion of digital twins is not possible via MQTT. ThingsBoard divides digital twin data into telemetry and three attribute types. Attributes can be updated via the MQTT connection, but change notification subscription is not possible. Client-side attributes can be updated and requested via the MQTT connection. Shared attributes can only be requested and subscribed to via the MQTT connection. ThingsBoard offers great visualization features. The rule engine is versatile and requires user familiarization. OpenRemote can connect to an external MQTT broker, but it is not possible to create or delete digital twins via the MQTT connection. All digital twin data is attributes, and they can be updated and subscribed to via MQTT. OpenRemote's rule engine is not quite as versatile as ThingsBoard's rule engine, but it is easier to use. Visualization features are more basic than the other platform's features.

The study concludes that we need to discuss how restricted MQTT capabilities can be worked with. If it is not feasible to create and delete digital twins via the HTTP API or manually, migration to another platform

is not possible. If the restrictions are acceptable, OpenRemote is a better choice due to the connectivity to external MQTT brokers and simpler digital twin data structure. Migrating to OpenRemote would fix the shortcomings of Ditto and remove the need for workaround scripts.

3 Integrated Situation Awareness, Risk Assessment, and Operational Response for Drone Operations

Drone operations constitute a use case characterized by mobile autonomous assets, shared airspace constraints, and the need for continuous supervision and timely intervention to maintain operational safety. In line with T4.1 (Digital twins & dashboards), the enablers described in this section (being developed by ADLC) establish a unified operational view of drone fleets by consolidating system status, spatial context, and operational constraints into real-time visualizations. This supports WP4's goal of showing the current situation in a form that is understandable and actionable by operators, enabling informed supervision during both normal operation and abnormal or degraded conditions.

Beyond visualization, drone operations impose additional requirements that shape how situational awareness, risk assessment, and response must be realized. The highly dynamic nature of airspace operations, combined with fleet-level coordination and variable connectivity conditions, requires continuous interpretation of operational state rather than static monitoring. Situational awareness must therefore integrate live telemetry, mission context, external constraints, and system health into a coherent, evolving operational picture that can serve both automated reasoning components and human operators in the Operations Control Centre.

The enablers presented in Sections 3.1–3.4 address these needs in a complementary manner by supporting perception and comprehension of the operational situation, enabling assessment of emerging risks and abnormal conditions, and facilitating timely and controlled responses. Together, they form a coherent set of UC8-specific enablers that realize the Sa4CPS platform capabilities for drone operations, while remaining aligned with the common architectural principles and interfaces defined at the platform level.

ID	Enabler Name	Related Tasks	Addressed Requirements	Related Use Case
EN_02	Perception, comprehension & risk assessment engine	T4.2, T4.4	REQ_UC8.3, REQ_UC8.4, REQ_UC8.6 REQ_UC8.7 REQ_UC8.8 REQ_UC8.9 REQ_UC8.10 REQ_UC8.11	UC8
EN_03	Operations monitoring dashboard	T4.1	REQ_UC8.1, REQ_UC8.2 , REQ_UC8.12 REQ_UC8.13 REQ_UC8.9 REQ_UC8.11	UC8
EN_04	Response dashboard	T4.1, T4.4	REQ_UC8.7	UC8
EN_05	State egress adapters	T4.2	REQ_UC8.5	UC8

3.1 Perception, comprehension & risk assessment engine (EN_02)

Within drone operations, this enabler provides the core capability for interpreting heterogeneous operational data at the level of individual flights and fleets. By continuously fusing inputs from multiple sources into a unified internal state, it enables consistent reasoning about the current operational condition despite variability in data quality, timing, and availability.

The embedded risk assessment mechanisms allow abnormal patterns and emerging issues to be identified early, based on both instantaneous conditions and their evolution over time. This makes EN_02 the primary source of enriched state and risk information for downstream components, supplying the inputs required for monitoring, visualization, and response without embedding decision logic directly into user-facing or integration-oriented enablers.

This enabler is implemented as a distributed, event-driven processing layer built using cloud applications and message queues, and a stateful worker architecture.

A set of ingress services connect to heterogeneous external data providers using their native protocols (MQTT, HTTP, WebSocket,...). These ingress services normalize all inputs into a unified internal data model and publish them onto several message queue topics. This decouples external volatility from internal processing and guarantees scalability, fault isolation, and replayability.

A flight-oriented worker system subscribes to these topics and instantiates an independent processing context per flight. Each worker continuously:

- fuses multi-source data into a coherent operational state,
- evaluates rule-based and threshold-based detection conditions,
- detects abnormal patterns (failures, deviations, data loss, spoofing indicators),
- persists state snapshots for resilience and traceability.

The rule engine executes both instantaneous checks (e.g., GNSS loss, contingency volume exit) and temporal checks (e.g., depletion trends, data silence, progressive divergence), enabling both detection and prediction. All outputs (state, risk levels, and events) are republished as streams, enabling downstream visualization, response orchestration, and external sharing.

The design choice of per-flight isolation plus event streaming ensures deterministic behavior, auditability, and safe scalability to large multi-drone fleets.

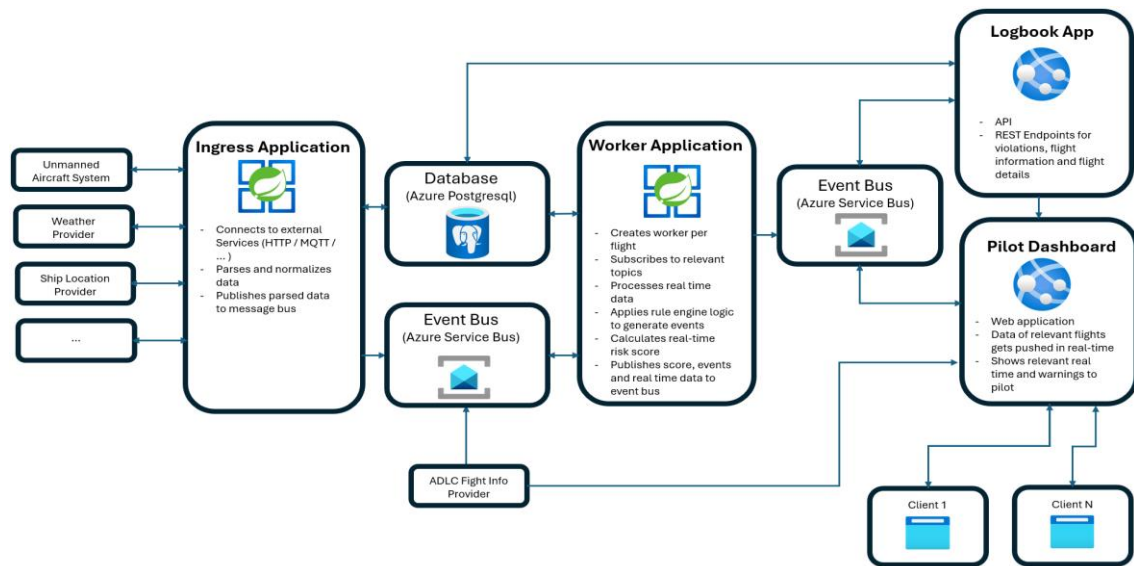


Figure 4: Architecture of the platform (based on Microsoft Azure Services)

3.2 Operations monitoring dashboard (EN_03)

This enabler translates live streams of telemetry, risk indicators, and events into a consolidated, operator-oriented view that emphasizes clarity and prioritization. By combining map-based visualization with severity-based alerting, visual encoding of risk and health, and progressive disclosure of details, it reduces cognitive load while preserving access to relevant context. The strict separation between core processing and visualization ensures that operational safety logic remains unaffected by user interface complexity or load, reinforcing the role of EN_03 as a presentation-layer component rather than a decision-making one.

The enabler is implemented as a real-time, web-based digital twin built with a Backend-for-Frontend (BFF) architecture. The backend aggregates live streams (risk, telemetry, events) from relevant processing streams and structured historical/contextual data from the other services. It exposes only pilot-relevant abstractions to the frontend and pushes updates via WebSockets, guaranteeing sub-second refresh without polling.

The frontend renders a map-based digital twin showing spatial, temporal, and risk dimensions simultaneously: drone state, trajectories, airspace volumes, surrounding traffic, system health, and alerts. Cognitive load is reduced through:

- severity-based alert prioritization,
- visual encoding of risk and health,
- consistent spatial metaphors (volumes, paths, envelopes),
- and progressive disclosure (alert details on demand).

Authentication and access control are handled through Azure Active Directory, ensuring that only authorized operators can view or interact with operational data.

Figure 5 Monitoring dashboard (in development)

In drone operations, this enabler structures the handling of abnormal situations by linking detected events and assessed risk levels to predefined operational procedures. By presenting situation-specific checklists, recommended pilot actions, and ranked response options, it supports operators in applying appropriate responses that are consistent with the current flight context, operational phase, and severity of the situation. The focus on procedural guidance rather than automated execution ensures that responses remain aligned with established operational practices while retaining human control in safety-critical scenarios.

The response backend subscribes to abnormal event streams and enriched state topics and maps them to predefined procedural models representing standard and emergency operating procedures. These procedures are parameterized by flight context, risk severity, and operational phase.

- situation-specific checklists,
- recommended pilot actions,
- system-suggested responses ranked by urgency and impact.

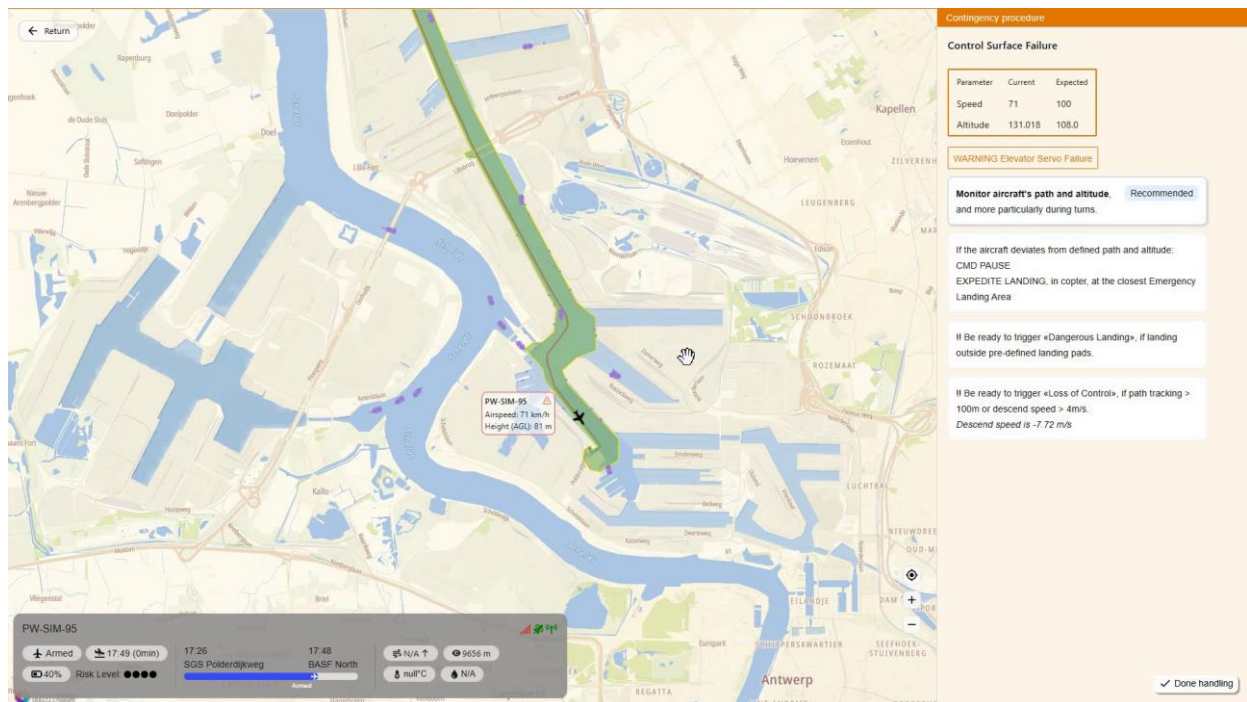


Figure 6: Response page (in development)

3.4 State egress adapters (EN_05)

This enabler addresses the need to expose selected elements of the operational state to external systems that depend on timely and reliable awareness of drone status and position. By republishing filtered and standardized views of internal state to UTM and authority systems, it enables coordination and conflict avoidance while enforcing strict constraints on what information is shared and how it is transmitted. The architectural separation of these outbound adapters from core processing ensures that interoperability requirements are met without introducing external dependencies into the internal safety and reasoning mechanisms established by the preceding enablers.

Additional adapters continuously monitor the operational state and push updates to external parties such as UTM systems, enabling airspace managers and other operators to take into account the status and location of drones using ADLC's platform to avoid conflicts with their operations.

This enabler is implemented as a set of outbound adapters that subscribe to the internal operational state and republish filtered, standardized views to external stakeholders.

Adapters transform internal representations into formats and protocols expected by UTM and authority systems, enforcing:

- data minimization,
- schema compliance,
- security policies,
- and transmission guarantees.

They operate independently from core processing, ensuring that external system failures cannot affect internal safety logic.

4 Unified Operational Awareness and Control for Industrial Forklift Operations

Industrial forklift operations in factory and warehouse environments constitute a representative UC1 scenario, characterized by mobile cyber-physical assets operating in shared and constrained spaces, strict safety requirements, and the need for continuous situational awareness and timely intervention. Multiple WP4 enablers contribute to this use case by addressing complementary aspects of status representation, tracking, autonomous operation, and response, together forming an integrated operational picture of in-factory logistics.

In line with T4.1 (Digital twins & dashboards), several enablers establish digital twins and monitoring views that represent the current state of forklifts, surrounding infrastructure, and operational context. These enablers provide operators with unified, real-time visibility into asset status, task execution, and environmental conditions, supporting WP4's objective of clearly showing the current situation. By consolidating heterogeneous system state into coherent dashboards and digital representations, they enable informed supervision across normal operations and exceptional situations.

At the same time, UC1 relies on T4.2 (Positioning & tracking of objects) to maintain continuous awareness of forklift locations, movements, and spatial relationships. Tracking-related enablers ensure that the position and trajectories of mobile assets remain available to higher-level WP4 functions, forming the basis for collision avoidance, task coordination, and spatial reasoning. This positioning information is not treated in isolation, but as an integral part of the overall system status.

UC1 is further supported by T4.3 (Autonomous operations) and T4.4 (Detection and response) through enablers that enable local or near-real-time reactions to evolving operational conditions. These include autonomous or semi-autonomous task execution, detection of unsafe situations or abnormal behavior, and support for timely responses that keep operations within safe and efficient bounds. Rather than focusing on low-level control, these enablers operate on consolidated system state to support decision-making and action at the operational level.

Taken together, the UC1-related enablers illustrate how WP4 integrates digital twins, tracking, autonomous behavior, and response mechanisms to keep complex industrial logistics operations under control. By combining continuous status representation with interpretation and action support, they directly contribute to WP4's overarching goal of managing dynamic cyber-physical systems in real-world industrial environments.

Beyond these task-level contributions, the enablers in this section also address the enabling infrastructure required to sustain such integrated awareness and control at scale. Low-latency communication mechanisms ensure that operational state changes propagate reliably and within the temporal bounds required for safety-critical coordination. Decentralized backend components introduce integrity, accountability, and traceability for task execution and state transitions, complementing operational awareness with verifiable evidence. Data ingestion pipelines and prognostics and health management capabilities extend the operational picture with condition- and energy-related perspectives that are essential for sustained and efficient vehicle operation. Fleet-level routing and vehicle-to-vehicle cooperation mechanisms further elevate situational awareness from individual forklifts to coordinated behavior across shared operational spaces. The integration of these complementary enablers is demonstrated and validated in a realistic industrial setting at the Port of Oulu, where communication, digital twins, analytics, coordination, and response mechanisms are exercised together under real-world operational constraints.

ID	Enabler Name	Related Tasks	Addressed Requirements	Related Use Case
EN_06	Low-Latency Hybrid Communication Module	T4.2, T4.3	REQ_UC1.14	UC1
EN_07	Sa4CPS Web-Based Monitoring Platform	T4.1	REQ_UC1.13	UC1
EN_08	3AM Decentralized Backend	T4.3, T4.4	REQ_UC1.26	UC1
EN_09	Digital Twin with Semantic Backend	T4.1, T4.2, T4.3, T4.4	REQ_UC1.29, REQ_UC1.30	UC1
EN_10	PHM Backend & Big Data Ingestion Pipeline	T4.1, T4.2, T4.3, T4.4	REQ_UC1.18, REQ_UC1.20	UC1
EN_11	Battery PHM (Prognostics & Health Management) Tool	T4.4	REQ_UC1.23	UC1
EN_12	Digital Twin Environment for forklift operations	T4.1, T4.2, T4.4, T4.5		UC1
EN_13	Optimum Route Planning and Vehicle-to-Vehicle Cooperation Engine	T4.2, T4.3, T4.4	REQ_UC1.31, REQ_UC1.32, REQ_UC1.33, REQ_UC1.41, REQ_UC1.42	UC1
EN_14	Digital twin-based situational awareness framework	T4.1	REQ_UC1.3, REQ_UC1.10	UC1

4.1 Low-Latency Hybrid Communication Module (EN_06)

To assess the feasibility of a low-latency hybrid communication architecture for autonomous mobile robots operating in industrial environments, a series of experimental studies and field tests were conducted. These experiments focused on evaluating the performance, reliability, and robustness of a 5G-enabled mesh-based communication system under indoor and operational conditions where conventional wireless connectivity, such as Wi-Fi, may be intermittent or unavailable.

The experimental setup included both controlled indoor tests and real-world field evaluations carried out on an autonomous mobile robot. Key performance indicators such as communication continuity, latency, packet delivery reliability, and data integrity were analyzed to determine the suitability of the proposed hybrid communication module for mission-critical industrial applications.

By demonstrating how hybrid, 5G-enabled connectivity can maintain reliable and bounded-latency communication under realistic indoor conditions, this enabler establishes a foundational infrastructure for the WP4 capabilities described in this section. It ensures that operational state and event information required by digital twin, tracking, and coordination enablers can be exchanged consistently, even in environments where network conditions are variable.

4.1.1 Setup of the 5G Mesh Network and Indoor Communication Testing

In this study, the setup of a 5G Mesh Network and the evaluation of its indoor communication performance were carried out. Development kits were programmed with sink, router, and non-router roles based on the

Wirepas guidelines, and a Raspberry Pi was used as the gateway. The application was developed in C using the Wirepas SDK, compiled via Docker, and deployed to the devices. After deployment, the network was monitored using Wirepas tools.

During the testing phase, message transmission was evaluated in an indoor environment with the sink and router fixed and the non-router mobile. It was observed that messages reached the non-router by hopping through the router, and that distance and latency performance varied depending on environmental conditions such as walls and noise.

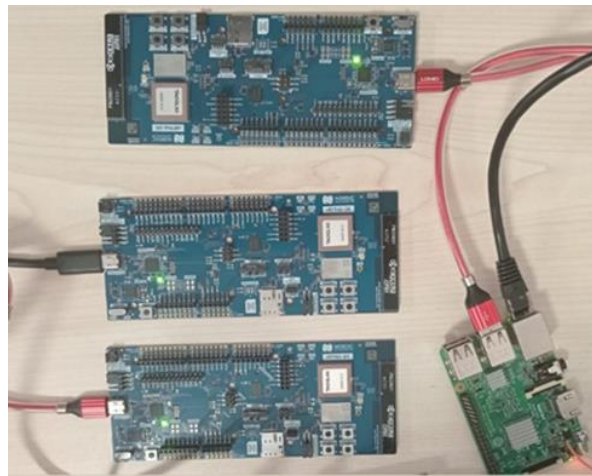


Figure 7 Setup of the Development Kits

4.1.2 5G Test Environment

The purpose of the conducted field tests is to evaluate to what extent a mesh-based cellular network can ensure communication continuity on an autonomous mobile robot operating in an industrial environment, particularly in scenarios where the Wi-Fi connection is interrupted.

Within this scope, the robot was operated along a route approximately 120 meters in length consisting of four task points, and the tests were conducted under two separate scenarios: with the mesh network active and inactive. When the mesh network was active, it was observed that the robot continued to receive tasks and operate even when the Wi-Fi connection was interrupted, with no communication disruptions. In contrast, in the scenario without the mesh network, the robot was unable to receive tasks following the Wi-Fi interruption and had to be manually moved to a location with stronger signal strength.

During the tests, wireless network outages lasting approximately 2 to 2.5 minutes were experienced; however, in the scenario where the mesh network was active, data transmission was observed to continue uninterrupted. An analysis of the messages transmitted via MQTT showed that out of approximately 16,000 messages, only 0.24% experienced delays greater than 500 milliseconds, corresponding to an average of 1–2 delayed messages per minute. For the data received via UART, it was determined that approximately 89.7% of about 71,000 messages were transmitted correctly, with an average interval of approximately 48 milliseconds between correctly received messages.

In tests conducted with messages of larger data sizes, data corruption was observed during the transmission of 185-byte packets, with only 4 out of 10 trials completing without errors. This indicates that

the current message structure and transmission mechanism need to be improved for large data packets. In latency and signal strength measurements, end-to-end transmission delays were mostly found to range between 0 and 250 milliseconds, while the received signal levels were measured between -78 dBm and -90 dBm. Overall, the results demonstrate that the mesh-based cellular network provides high communication continuity and reliability against Wi-Fi interruptions in industrial environments, offers sufficient performance for small data packets, but requires optimization for larger data packets.

4.2 Sa4CPS Web Platform (EN_07)

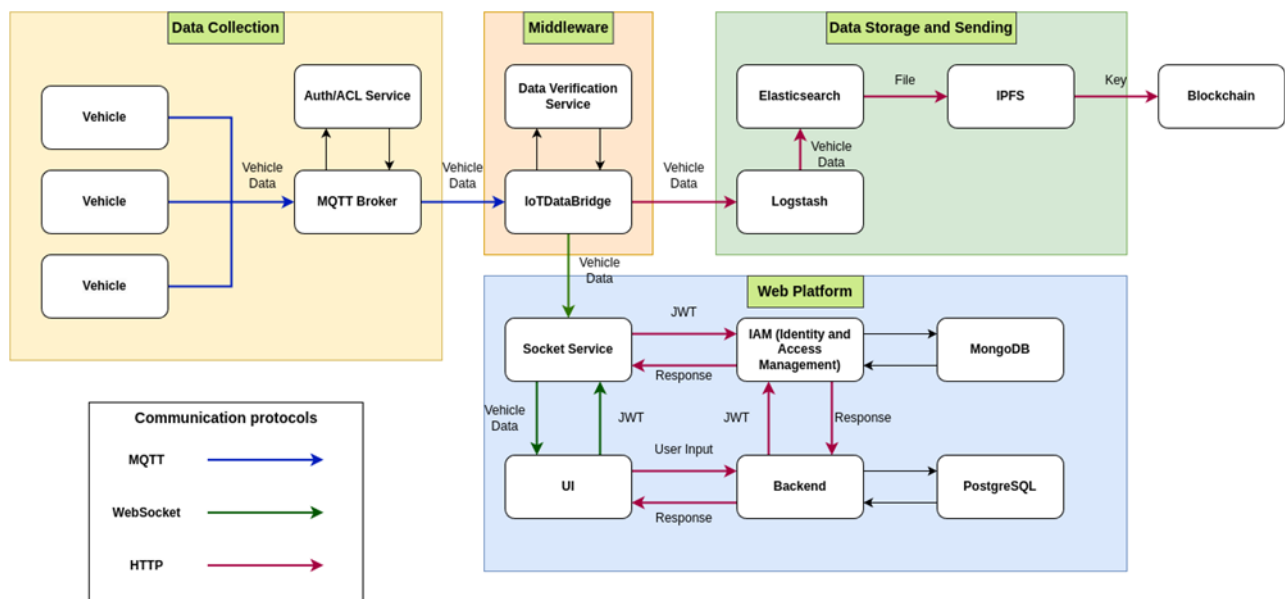


Figure 8 General diagram

The Sa4CPS web platform was developed to enable end-to-end monitoring, analysis, and historical evaluation of the operational states of forklifts. Built on a React-based architecture, the platform includes registration, login, and token-based authentication mechanisms to ensure secure user access, along with authorization controls that restrict platform content to authenticated users only.

The platform is designed to process data received from forklifts in real time. Real-time location, task, and connectivity status data are collected from forklifts via an MQTT infrastructure and transmitted to the web interface through WebSocket channels. This allows users to monitor forklifts live on a map and distinguish whether the data is transmitted over Wi-Fi or the 5G infrastructure. Historical forklift data is stored in a database, enabling subsequent analysis.

PHM (Prognostics and Health Management) capabilities have been developed on the platform to monitor battery and system health. Battery-related parameters such as state of charge, state of health, temperature, cell-level voltages, current and voltage values, and energy consumption are visualized in real time. This enables users to observe battery behavior and anticipate potential risks.

To enhance operational visibility, a dashboard structure presenting summary metrics has been integrated into the platform. This dashboard displays the overall status of forklifts, connectivity information, battery metrics, and operational indicators on a single screen, allowing users to quickly assess the system at a high level.

In addition, a feature has been developed to visualize the historical routes followed by forklifts on the map. Based on the selected forklift, time range, and data density, users can review past movements at different scales, ranging from daily to yearly views.

The platform has been extended beyond monitoring to also include secure data management capabilities. Data stored on IPFS related to forklifts can be listed, downloaded, or shared via copyable links directly through the platform.

To improve user experience, loading screens have been added during waiting periods, along with success and error notifications for operation results. Furthermore, a sustainable and enterprise-compliant theming infrastructure has been developed, and platform usability has been enhanced through support for both dark and light themes.

Summary, the SA4CPS platform has been developed as a scalable and secure industrial monitoring solution that integrates real-time monitoring, historical data analysis, battery and system health tracking, operational reporting, and distributed data access within a single web-based interface.

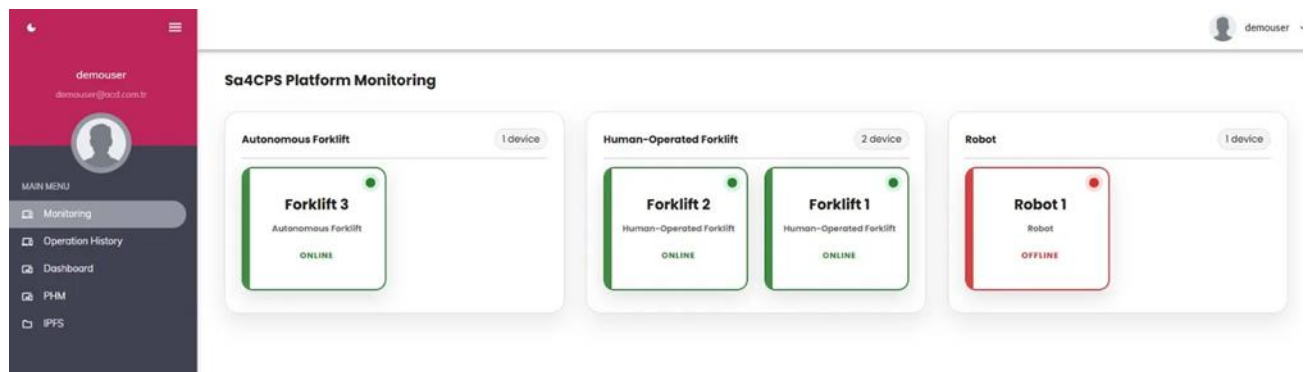


Figure 9 Sa4CPS web platform – Monitoring page

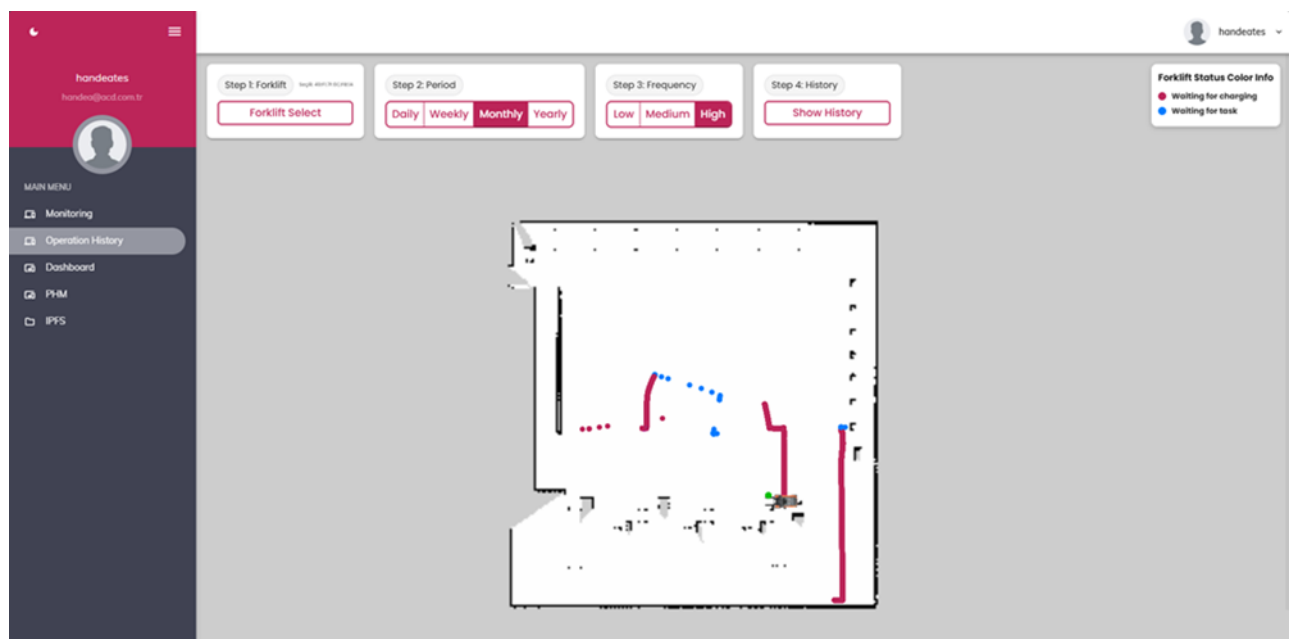


Figure 10 Sa4CPS web platform – operation history page

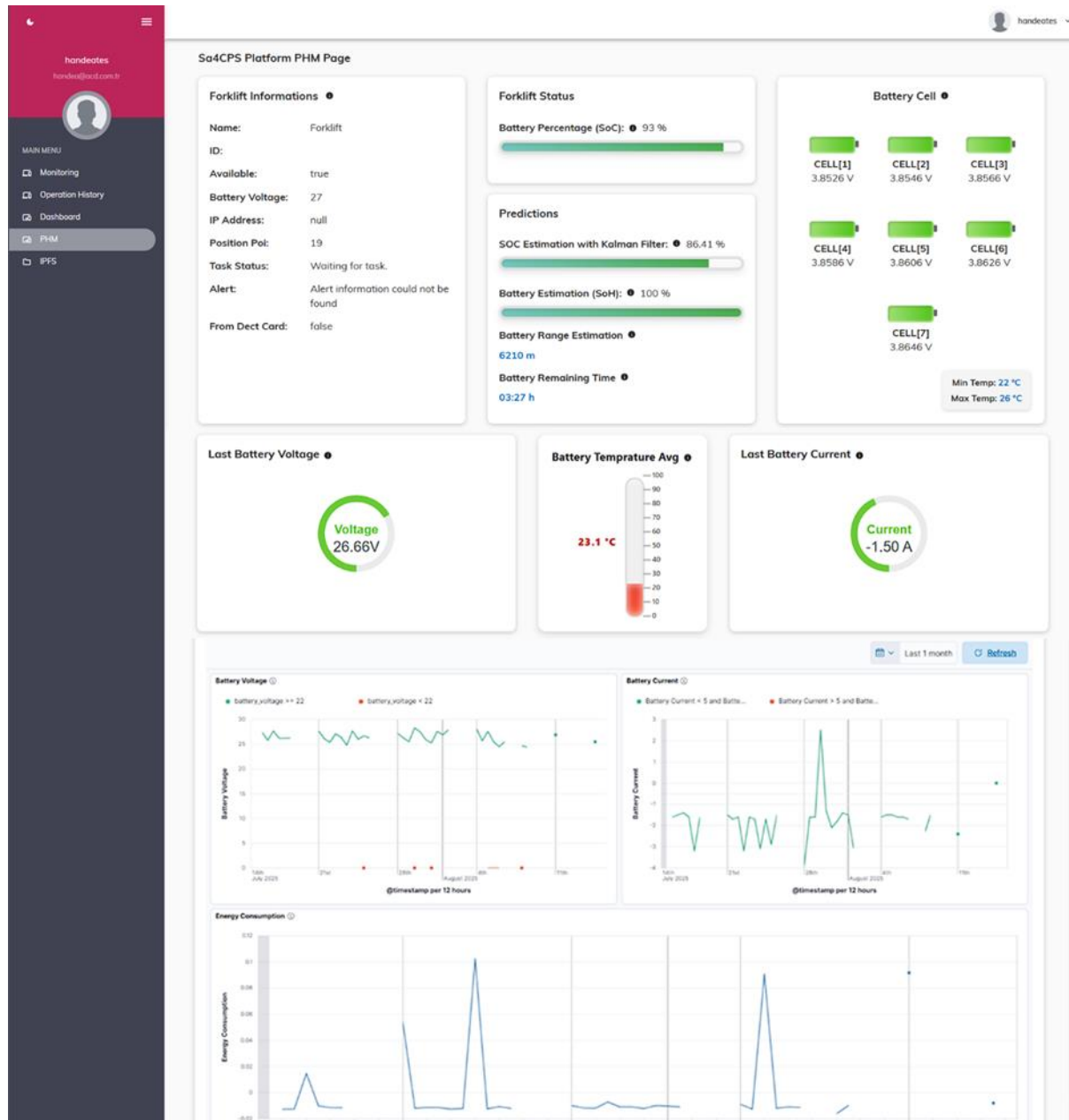


Figure 11 Sa4CPS web platform - PHM tool page

Forklift ID	Uploaded By	Mime	Size (MB)	Start	End	Block Number	Hash	URL
1	IoTDataBridge	text/csv	12.24	01.10.2025 00:00:00	02.10.2025 00:00:00	301924	0xe286e29eb7a2eb9e252ef...	
1	IoTDataBridge	text/csv	12.24	01.10.2025 00:00:00	02.10.2025 00:00:00	388074	0x29a4e39288ab9db0c2eaf...	
1	IoTDataBridge	text/csv	40.85	02.10.2025 00:00:00	03.10.2025 00:00:00	301975	0x6c933cb3350e188f73d23...	
1	IoTDataBridge	text/csv	40.85	02.10.2025 00:00:00	03.10.2025 00:00:00	388123	0x3249e00af072ebd9f9bd...	
1	IoTDataBridge	text/csv	42.60	03.10.2025 00:00:00	04.10.2025 00:00:00	302026	0x96a858af3cedcf09c2d...	
1	IoTDataBridge	text/csv	42.60	03.10.2025 00:00:00	04.10.2025 00:00:00	387702	0xc5790b0e220ad338d2a...	
1	IoTDataBridge	text/csv	42.60	03.10.2025 00:00:00	04.10.2025 00:00:00	388174	0xb2070c25dc86a644880...	
1	IoTDataBridge	text/csv	42.99	04.10.2025 00:00:00	05.10.2025 00:00:00	302077	0x5518905f6d9d5561c4bd...	
1	IoTDataBridge	text/csv	42.99	04.10.2025 00:00:00	05.10.2025 00:00:00	388225	0xb96a0b0c2930438d22f...	
1	IoTDataBridge	text/csv	43.55	05.10.2025 00:00:00	06.10.2025 00:00:00	302126	0x2e42870f2e086a06a37be...	

Figure 12 Sa4CPS web platform - IPFS page

4.3 3AM Decentralized Backend (EN_08)

This enabler provides the backend foundation for maintaining a consistent and accountable view of operational state across distributed logistics environments, focusing on status monitoring, operational projections, and accountability in distributed logistics and cyber-physical operations. It addresses the need for reliable system state awareness and traceable operational execution in environments involving heterogeneous data sources, mobile assets, and multiple stakeholders, where operational decisions and actions must be linked to verifiable execution histories. By combining a decentralized accountability layer based on blockchain technologies with a semantic digital twin backend, operational data originating from vehicles and sensors can be securely recorded, structured, and reasoned over. Positioned between edge-level data acquisition and higher-level situational awareness and coordination functions, the 3AM decentralized backend (being developed by ERARGE) complements the other WP4 enablers by providing integrity, traceability, and a shared reference point for understanding system behaviour over time, without embedding application-specific control logic into the infrastructure layer.

The overall architecture, illustrated in Figure 13 combines edge-level data acquisition, a decentralized blockchain-based backend, and a semantic digital twin infrastructure. Operational data originating from vehicles (forklifts and distribution trucks), sensors, and external information sources are securely transmitted via a secure gateway and ingested through a message-based interface. These data streams are then processed by two complementary backend layers: a decentralized accountability layer based on blockchain technologies, and a semantic digital twin layer enabling structured representation, reasoning, and projection of system behaviour.

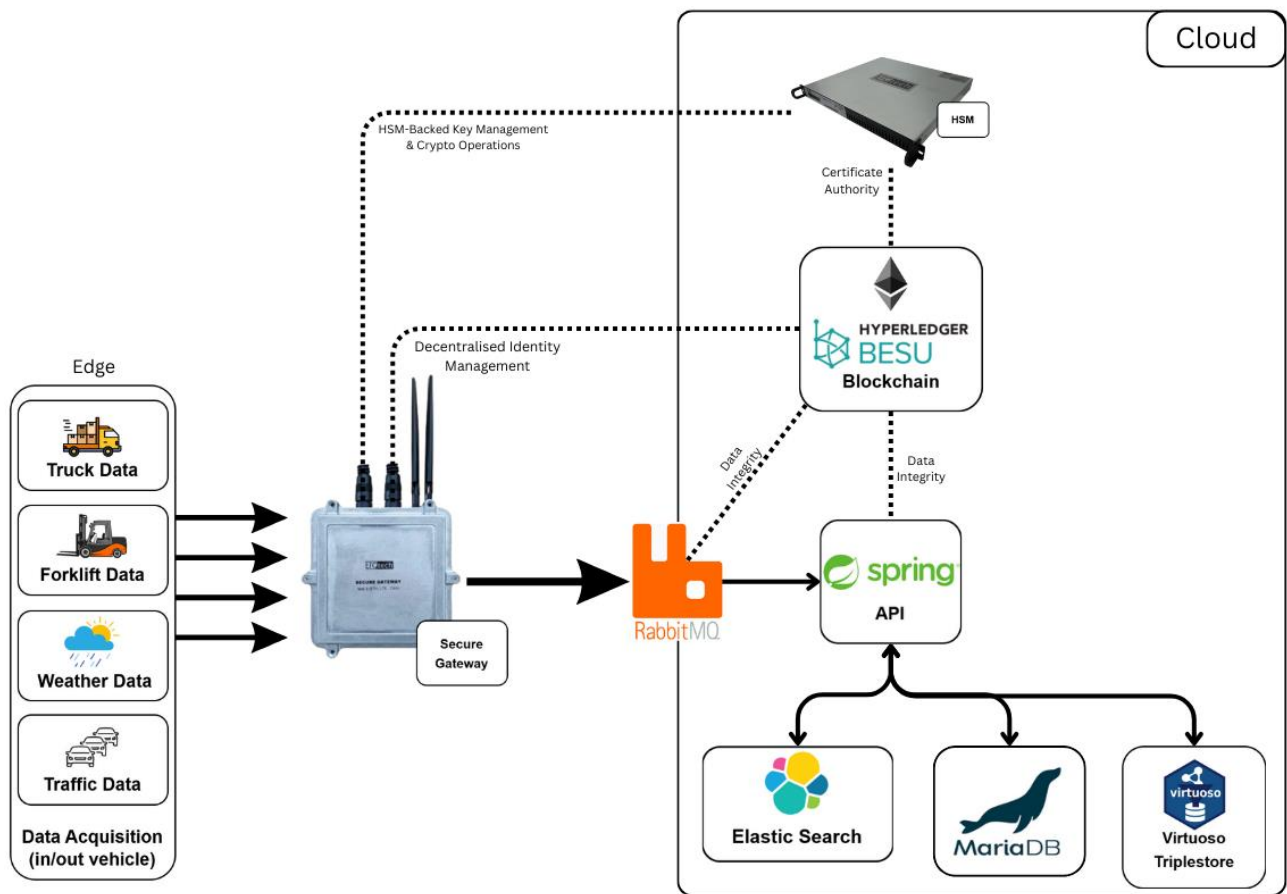


Figure 13 Blockchain-enabled Digital Twin Architecture with Semantic Backend

The 3AM Decentralized Backend provides an immutable and distributed operational ledger that records task lifecycle events, asset status changes, and operational evidence. By leveraging decentralized identity concepts and smart contracts, it enables traceable attribution of actions and supports auditable monitoring of logistics operations. This layer establishes a trusted operational history that can be consumed by higher-level monitoring and analytics services.

Complementing this, the Digital Twin with Semantic Backend transforms heterogeneous operational data into a unified, semantically enriched system representation. By integrating relational, time-series, and blockchain-anchored data into a knowledge graph, the digital twin enables real-time system state awareness, semantic correlation of events, and knowledge-driven operational projections.

Together, these enablers form a coherent backend infrastructure for UC1, in which verifiable operational data captured by the decentralized backend is interpreted and exploited by the semantic digital twin. While the architectural integration of trust, security, and privacy mechanisms is reflected in the overall design, their detailed specification and validation are addressed under WP5.

4.3.1 Triangular Accountability Model (3AM) Decentralized Backend

The 3AM Decentralized Backend provides a decentralized operational backbone enabling immutable status tracking, traceable task execution, and accountability for logistics operations in UC1. It establishes

a shared, tamper-proof operational history across vehicles, operators, and backend services, supporting both real-time monitoring and retrospective analysis.

Functional Scope

The enabler supports the following UC1 operational capabilities:

- Task lifecycle monitoring
 - Creation, assignment, execution, and completion of logistics tasks
 - State transitions recorded as immutable events
- Operational accountability
 - Attribution of actions to vehicles, operators, and system components
 - Non-repudiable linkage between actors and executed operations
- Evidence anchoring
 - Integrity verification of task-related documents and artefacts
 - Proof of existence and timestamping for operational evidence
- Input for operational projections
 - Reliable historical event data for trend analysis and forecasting

Technical Architecture and Technologies

The enabler, as depicted as Blockchain block in Figure 13 is implemented using the following technologies and architectural elements:

- Hyperledger Besu (permissioned blockchain)
 - Acts as a decentralized ledger for operational events
 - Stores task state changes, identity references, and document hashes
 - Ensures immutability, traceability, and consistency across participants
- Web API Framework
 - Entry point for edge devices, operators, and backend services
 - Translates operational actions into blockchain transactions
 - Exposes query endpoints for monitoring and analytics components
- Decentralized Identifiers (DIDs)
 - Represent vehicles, operators, and backend services
 - Enable persistent identity resolution across system lifecycles
 - Used strictly for operational attribution (no personal data storage)
- Soulbound Tokens (SBTs)

- Non-transferable identity tokens bound to DIDs
 - Represent operational roles and long-term system identities
 - Ensure that recorded actions remain permanently linked to the responsible entity
- Verifiable Credentials (VCs)
 - Cryptographically signed operational assertions
 - Used for task assignments, execution acknowledgements, and status confirmations
 - Enable validation of operational claims before consumption by monitoring services
- Task Registry Smart Contract
 - Maintains task identifiers and task membership
 - Associates operational events and documents with specific tasks
 - Enforces task-level segregation at the blockchain layer
- Document Hash Anchoring
 - Operational documents are hashed off-chain
 - Hashes and metadata are stored on-chain
 - Any document modification invalidates the recorded fingerprint

Edge-to-Backend Data Flow

- Operational events originate from vehicles and edge systems
- Events are transmitted via SGW-3 Secure Gateway
- Message-based ingestion enables low-latency event handling
- Validated events and document hashes are committed to the blockchain
- Blockchain data is consumed by monitoring dashboards and digital twin services

PRIGM HSM and CANAT devices are architecturally integrated to support trusted event origination and authenticated access at the cyber-physical boundary; their security and cryptographic functions are specified under WP5.

Contribution to UC1 Objectives

- **Status monitoring:** immutable, near real-time visibility of task and asset states
- **Operational projections:** reliable historical datasets enabling forecasting and optimisation
- **Accountability:** auditable, non-repudiable linkage between actions, actors, and evidence

4.4 Digital Twin with Semantic Backend (EN_09)

This enabler provides a knowledge-driven representation of logistics operations, enabling system state awareness, semantic reasoning, and operational projections based on verified operational data. It's an interpretative layer that transforms operational data into a coherent and queryable representation of system behaviour. By organizing information about assets, tasks, locations, and events into a semantically enriched digital twin, it enables consolidated system state awareness, contextual interpretation of multi-source data, and knowledge-based reasoning over operational evolution. The semantic backend supports correlation of events across time and assets, detection of deviations, and projection of future states based on historical patterns, while remaining tightly coupled to verifiable operational inputs. Through its explicit integration with the decentralized backend, semantic states are directly linked to immutable operational records, ensuring that monitoring and projections are derived from traceable and accountable execution histories. In this role, the digital twin complements the accountability infrastructure by providing a structured, reasoning-oriented view of logistics operations that can be consumed by monitoring, analysis, and decision-support components across the UC1 stack.

Functional Scope

The enabler supports the following UC1 capabilities:

- **System state representation**
 - Unified view of assets, tasks, locations, and events
 - Continuous synchronisation with operational data streams
- **Semantic comprehension**
 - Contextual interpretation of multi-source data
 - Correlation of tasks, assets, and events across time
- **Operational projection**
 - Detection of abnormal situations and deviations
 - Projection of future states based on historical patterns
- **Provenance-aware monitoring**
 - Semantic states linked to blockchain-anchored operational records

Technical Architecture and Technologies

The enabler, as depicted in Figure 13 is realised through a hybrid semantic data management stack:

- **Virtuoso Triplestore**
 - Hosts the digital twin knowledge graph
 - Stores semantically enriched representations of UC1 entities
 - Enables efficient SPARQL querying and reasoning
- **UC1 Ontology**
 - Custom semantic model for:
 - Assets and vehicles
 - Tasks and workflows
 - Events and status transitions
 - Spatial and temporal relations

- Ensures consistent interpretation across system components
- **SPARQL Reasoning and Query Layer**
 - Real-time access to digital twin state
 - Semantic inference over task progress and system behaviour
 - Support for anomaly detection and future state estimation
- **Hybrid Storage Backend**
 - **MariaDB:** static and reference data (asset models, configurations)
 - **Elasticsearch:** dynamic and time-series operational data
 - Semantic identifiers unify relational and time-series data at query time

Integration with Decentralized Backend

- Blockchain-anchored task events and document references are linked into the semantic graph
- Each semantic assertion can be traced back to an immutable blockchain record
- Digital twin states are therefore derived exclusively from verifiable operational inputs

Contribution to UC1 Objectives

- **Status monitoring:** semantically enriched, unified system view
- **Operational projections:** knowledge-based forecasting and anomaly detection
- **Accountability:** traceability between semantic states and immutable operational evidence

4.5 PHM Backend & Big Data Ingestion Pipeline (EN_10)

To provide data for the PHM (Prognostics & Health Management) page, a Node.js–based backend service was developed. The system is designed with a REST API + Socket.IO architecture, enabling both request–response communication and real-time data streaming to the frontend (see Figure 14).

Data Collection and Integration

- Robot-generated data are collected by the backend through MQTT brokers.
- Parameters such as battery voltage, current, individual cell voltages, temperature values, and task states are subscribed to from multiple MQTT topics.
- All incoming data are routed into a centralized data processing pipeline for further analysis.

Backend Processing & Data Infrastructure

- The backend aggregates raw and computed data into a single structured API response.
- Elasticsearch–based historical data processing is used to store and analyze time-series battery data.
- Processed and computed data are streamed to the frontend in real time via Socket.IO, using the `mqtt_data` event.
- This architecture ensures reliable, scalable, and low-latency delivery of both real-time and historical PHM data.

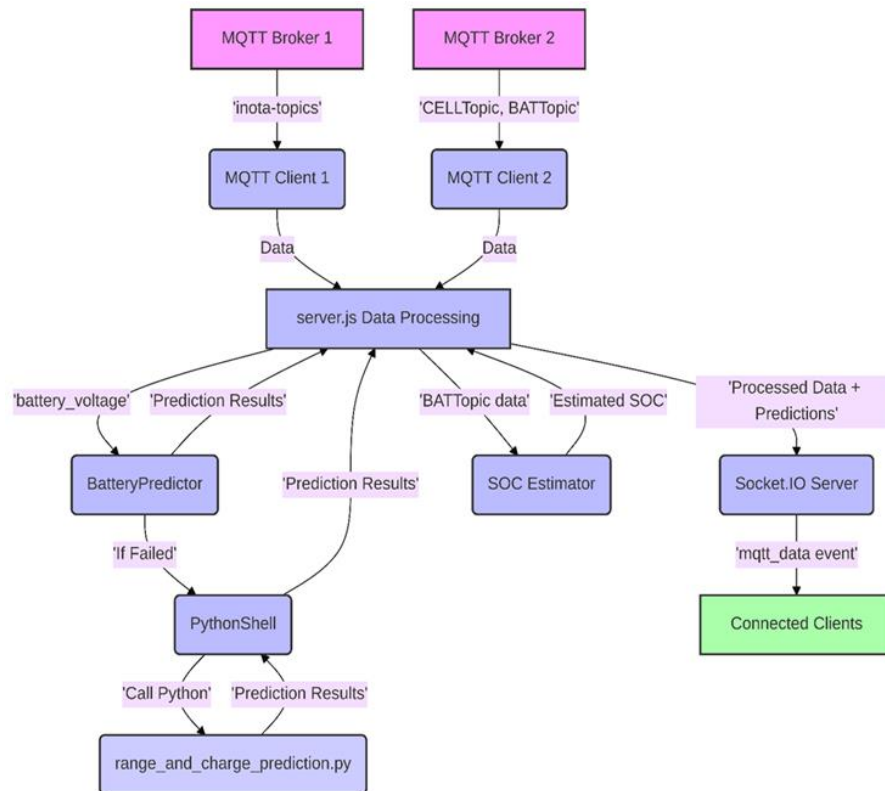


Figure 14 API workflow

4.6 Battery PHM (Prognostics & Health Management) Tool (EN_11)

The Battery PHM Tool focuses on battery condition monitoring, health assessment, and predictive analytics, combining real-time measurements with estimation and modeling techniques.

Battery Monitoring and Calculations;

Using incoming battery voltage and current data, the backend computes:

- Energy consumption
- State of Charge (SOC)
- State of Health (SOH)

Additional monitoring features include:

- Cell-level voltage analysis, enabling individual battery cell monitoring.
- Temperature data processing, where minimum, maximum, and average temperature values are calculated.

Estimation and Modeling Studies;

- Battery SOC estimation is performed using a Kalman Filter–based approach.
- Remaining operating time and estimated battery range are calculated using regression-based prediction models.
- The estimation process is executed by the backend prediction service.
- If the primary prediction mechanism fails, the system automatically switches to an alternative Python-based prediction service, executed via PythonShell, ensuring fault tolerance and uninterrupted prediction capability.

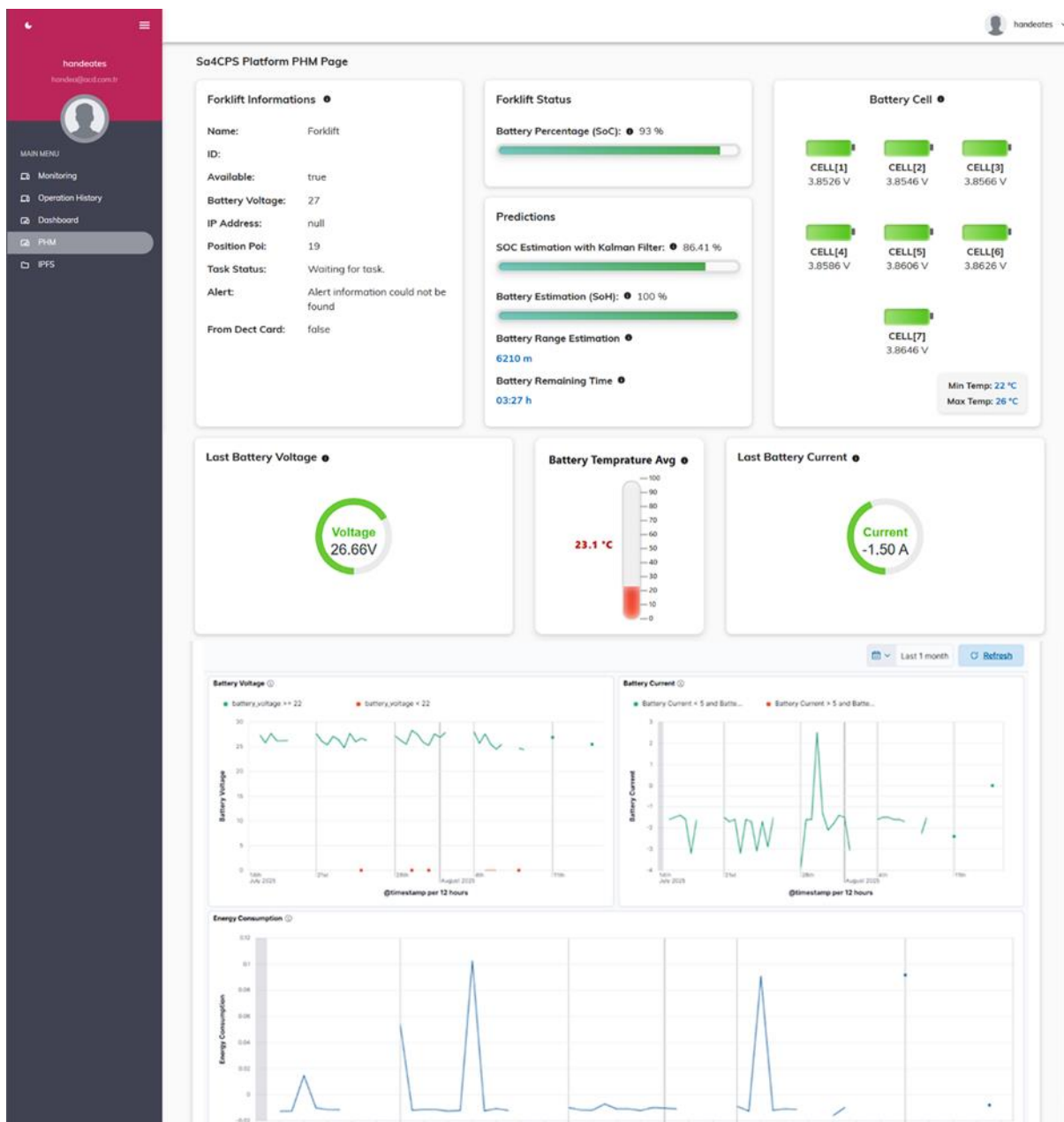


Figure 15 Sa4CPS web platform – PHM tool page

Feature	Description
Forklift Informations	Displays general information about the forklifts and connection details.
Forklift Status	Displays the current battery charge level as a percentage (%) relative to its capacity.
SOC Estimation with Kalman Filter	Estimates the battery State of Charge (SOC) using a Kalman Filter. This method reduces measurement noise, enabling more accurate results.
Battery Estimation (SOH)	Displays the battery State of Health (SoH). This value represents the battery's overall condition and capacity compared to a new battery.
Battery Range Estimation	Estimates how far the robot can travel with the current battery charge.
Battery Remaining Time	Displays the estimated remaining operating time before the battery is fully depleted.
Battery Cell	Displays the individual voltage values of each battery cell.
Last Battery Voltage	Displays the most recently measured battery voltage in volts (V).
Battery Temperature Avg	Displays the average temperature of the battery pack in degrees Celsius (°C).
Last Battery Current	Displays the change in battery current over time.
Battery Voltage Graph	Displays the change in battery voltage over time. The green color indicates safe voltage levels, while the red color indicates critically low voltage.
Battery Current Graph	Displays battery current trends over time. Negative values represent discharge (consumption), while positive values represent charging. Color coding helps indicate abnormal current levels.
Energy Consumption	Displays the energy consumption pattern based on battery voltage and current. It is typically measured per second and converted into Wh (Watt-hours).

4.7 Digital Twin Environment for forklift operations (EN_12)

To eliminate simulation issues encountered in the Gazebo environment and remove ROS dependency, the forklift simulation was migrated to the Unity game engine, and a realistic digital factory environment is under development.

- A physics-based simulation infrastructure was established using Unity 6 and the Universal Render Pipeline (URP).
- The forklift's driving and load-lifting mechanisms are controlled via custom ForkliftDrive and ForkliftLift scripts.
- Physical interactions are realistically simulated using Unity's built-in PhysX engine along with Rigidbody and Wheel Collider components.
- A detailed digital factory building, approximately 60–70 meters wide and 100 meters long, with interactive racking systems, was created.
- The forklift's capabilities for load handling, dropping, and maneuvering were modeled, enabling console-based monitoring of operational states such as Idle, MovingEmpty, and CarryingLoad.

- MQTT publishing, multi-forklift support, and advanced mapping systems were integrated into the Unity-based simulation to enhance digital twin capabilities.
- Anti-roll (tip-over prevention) forces and center-of-mass optimizations were applied to improve forklift stability and driving realism.
- MQTT communication was implemented in C# using the `MqttClient` class.
- A centralized controller, `ForkliftControllerManager`, was developed to manage multiple forklifts, camera transitions, and map visualizations.



Figure 16 Factory Environment

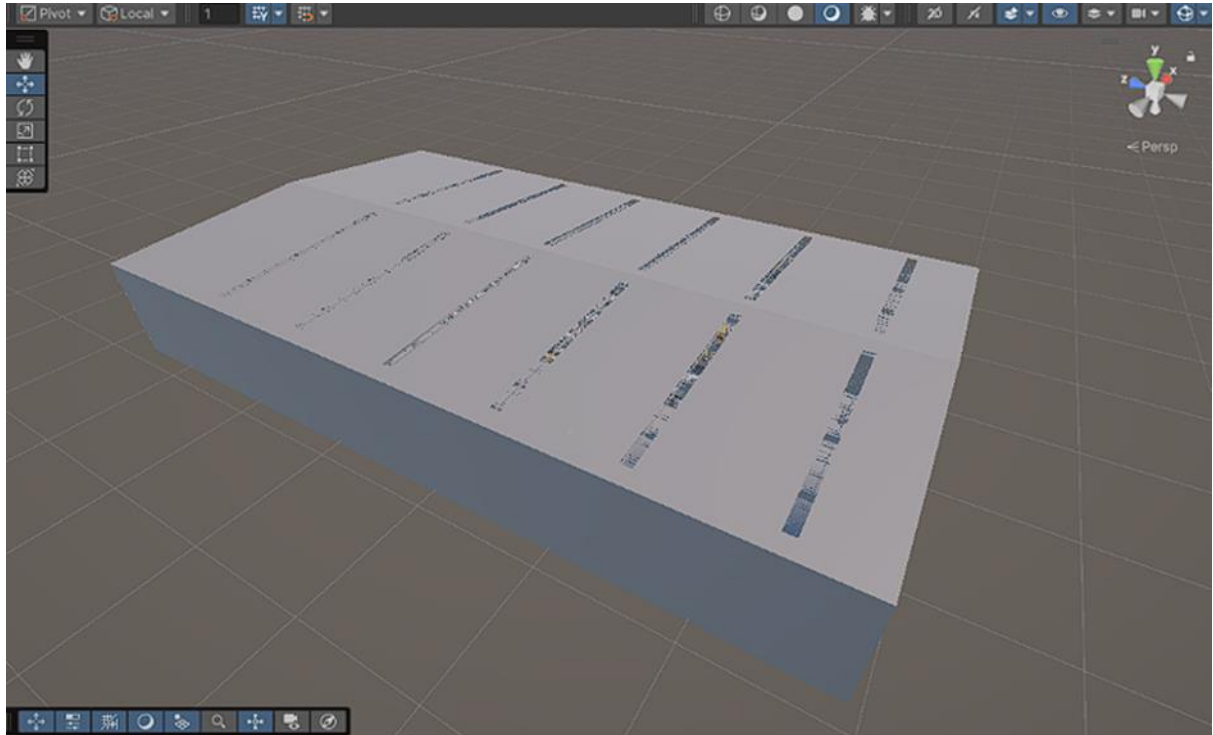


Figure 17 Top View of the Factory

4.8 Optimum Route Planning and Vehicle-to-Vehicle Cooperation Engine (EN_13)

Optimum Route Planning and ML-based Decision Support Engine is a decision-support enabler designed to compute, update, and validate optimal transportation routes for autonomous and semi-autonomous vehicles operating in industrial environments. This enabler, provided by ALPATA, focuses on dynamic route optimization and machine-learning-based decision support for connected and autonomous industrial logistics systems. The contribution is validated in UC1 – Connected and Autonomous Logistics, with a specific focus on real-time routing, congestion prediction, and adaptive decision-making in shared factory environments.

The enabler integrates real-time IoT data processing, semantic data interpretation, and optimization and machine learning algorithms to support status assessment, projection, and action execution, in line with the technical scope of this deliverable.

The enabler continuously processes real-time IoT data streams, including:

- vehicle position data (e.g., GPS),
- beacon-based proximity and zone information,
- vehicle operational data such as speed, payload, and status.

Based on this data, the enabler dynamically adapts routing decisions to improve operational efficiency, safety, and robustness under changing environmental and operational conditions. The enabler features the following technical characteristics:

Real-time Data Acquisition and Processing. The enabler implements an event-driven data processing pipeline that ingests real-time data from IoT devices and vehicle sensors. Incoming data is buffered, filtered, and synchronized to enable low-latency reactions to changes such as congestion, obstacle appearance, or safety zone violations.

Dynamic Route Optimization. Routing decisions are computed dynamically by considering vehicle positions, transport tasks, and environmental constraints. The system supports continuous re-routing, enabling vehicles to adapt their paths in response to real-time operational changes.

Machine Learning-based Delay and Congestion Prediction. To improve proactive decision-making, the enabler integrates machine learning models, including supervised learning and reinforcement learning approaches, to predict: potential delays, congestion formation, operational bottlenecks. Predicted conditions are fed back into the routing logic to enable proactive route adaptation instead of purely reactive behavior.

Thus, in the frame of UC1 the enabler (a) generates dynamic routes for autonomous forklifts and transport vehicles based on real-time operational data, (b) reduces delays and empty runs through adaptive routing, (c) improves safety by considering proximity data and safety zones, and (d) supports human-machine collaboration in shared factory environments.

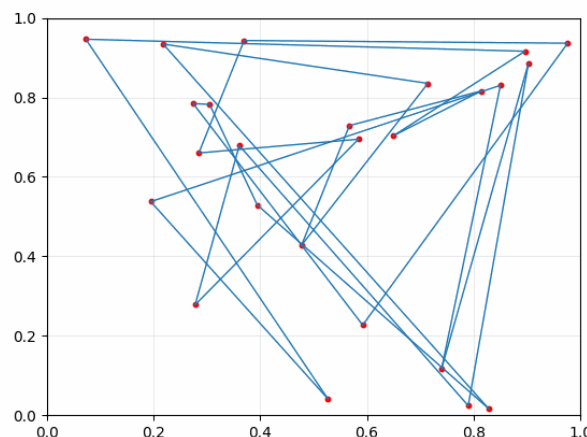


Figure 19 Illustration of integrating IoT-based real-time data

Figure 19 illustrates how the developed software operates by integrating IoT-based real-time data (e.g., GPS positions and beacon signals) to generate dynamic and optimized routes for vehicles operating in an industrial environment. The system continuously processes location and operational data collected from vehicles and on-site infrastructure to compute optimal routing alternatives.

The routing model is designed to increase operational efficiency, reduce delays, and optimize in-factory transportation flows by applying optimal route planning algorithms. Within the scope of the project, an extensive literature review has been conducted, relevant algorithms and their working principles have been identified, and a model structure suitable for the Otokar industrial logistics scenario has been defined.

At the current stage, real operational data provided by Otokar have been collected and analyzed to serve as system inputs. The software is designed to listen to and interpret the Otokar operational scenario, including constraints and requirements, and to execute routing and decision-support functions based on a model adapted to these real-world conditions.

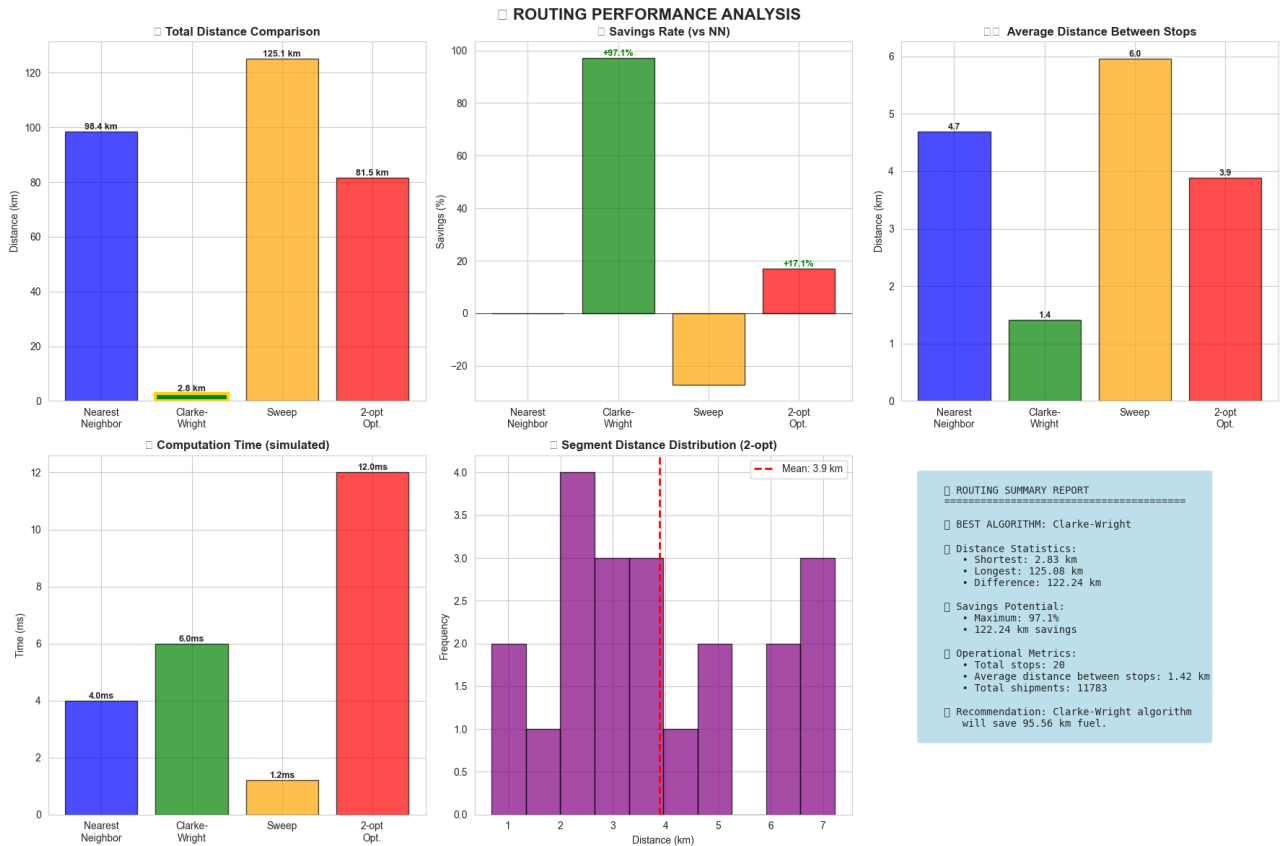


Figure 20 Visualization of the routing performance analysis

Routing Performance Analysis – Multi-Criteria Algorithm Evaluation

This set of visualizations (Figure 18) presents the comparative performance analysis of the routing software developed within the ALPATA Optimal Route Planning Enabler. During the development phase, more than ten different routing algorithms and optimization strategies were evaluated, including heuristic, savings-based, and local optimization approaches.

For clarity and comparability, the four best-performing algorithms were selected and visualized in detail:

- Nearest Neighbor
- Clarke-Wright Savings
- Sweep
- 2-opt Optimization

The evaluation is conducted using operational and computational performance metrics, such as total route distance, average distance between stops, computational complexity, and potential savings.

Total Distance Comparison

The first panel compares the total route distances obtained by each selected algorithm. This metric is directly related to fuel consumption, travel time, and operational cost, and therefore represents a key performance indicator for logistics operations. The algorithm producing the shortest total distance is visually highlighted, indicating the most efficient routing solution for the evaluated scenario.

Savings Rate Analysis (Baseline: Nearest Neighbor)

In the second panel, the Nearest Neighbor algorithm is used as a baseline reference, and the distance savings achieved by the other algorithms are calculated as percentages. This comparison quantitatively demonstrates the operational improvements provided by more advanced routing methods. Positive savings are explicitly highlighted to emphasize algorithms that offer meaningful efficiency gains.

Average Distance Between Stops

The third panel illustrates the average segment length for each algorithm, calculated as the total route distance divided by the number of stops. This metric provides insight into route smoothness and operational stability. Shorter average distances between stops indicate more balanced routes, which are particularly important in human-machine cooperative logistics environments.

Computation Time and Algorithmic Complexity (Simulated)

The fourth panel presents simulated computation times based on the theoretical algorithmic complexity of each method. Rather than focusing on absolute execution times, this analysis highlights the scalability characteristics of the algorithms, comparing $O(n^2)$ and $O(n \log n)$ behaviors in the context of real-time and near-real-time applications.

Segment Distance Distribution (2-opt)

The fifth panel shows the distribution of segment distances for the optimized route obtained using the 2-opt algorithm. The histogram analysis helps identify route uniformity and detect potential distance spikes between consecutive stops. The mean segment distance is explicitly indicated to support a clearer interpretation of route homogeneity.

Decision-Support Oriented Performance Summary

The final panel provides a text-based performance summary designed for decision-support purposes. It consolidates key results, including:

- Best-performing algorithm
- Shortest and longest route distances
- Maximum potential savings
- Operational metrics such as number of stops and total shipments
- Estimated fuel and distance savings



OTOKAR OPTIMAL ROUTE MAP (Clarke-Wright)

Total Distance: 2.83 km | Number of Stops: 20

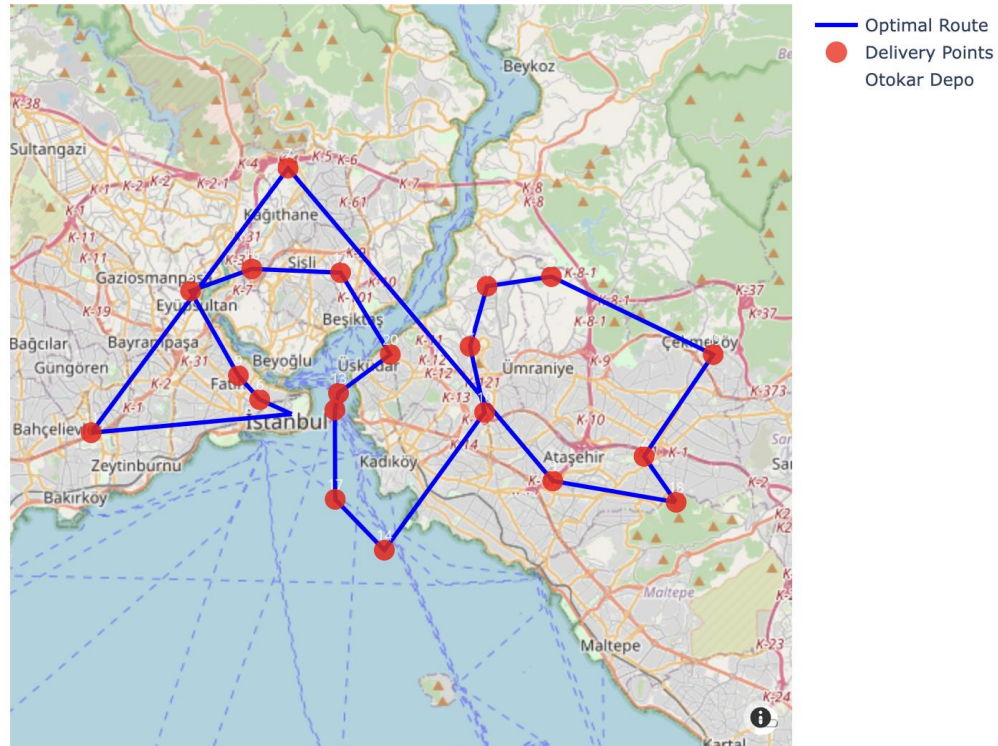


Figure 21 Visualization of the optimal transportation route

Otokar Optimal Route Map – Best Algorithm-Based Route Visualization

The second visualization presents the optimal transportation route generated for the Otokar use case using the best-performing algorithm identified in the comparative analysis. The interactive map displays the depot location, delivery points, and the resulting optimal route connecting these locations.

The visualization consists of three layers (traces): route line, delivery points, and depot location. It supports interactive exploration through zooming, panning, and point-level information display. For each delivery point, shipment volume and priority information are shown interactively.

This map demonstrates that the ALPATA enabler is capable of:

- Processing real-time location data,
- Selecting the most suitable routing algorithm among numerous alternatives,
- Transforming analytical results into operationally meaningful and visually interpretable outputs.

4.9 Digital twin–based situational awareness framework (EN_14)

This enabler addresses the need to bring together operational data, digital twin representations, and analytical services into a unified situational awareness framework that supports monitoring and optimisation of industrial vehicle operations. Building on the underlying digital twin and data integration capabilities defined earlier in this section, it focuses on presenting an integrated view of system status, identifying anomalies and inefficiencies, and supporting informed operational decisions. The framework leverages digital twin data to analyse vehicle behaviour, task execution, and movement patterns, enabling improved planning and optimisation of logistics operations while maintaining awareness of safety- and performance-relevant conditions. The digital twin–based situational awareness framework (being developed by Otokar) thus acts as a convergence point where status information, analytical results, and operational insights are combined to support supervision and coordination at the operational level. It tracks the real-time status of forklifts, transported parts, and operational areas.

Using location data, AI-based task assignment and route optimization mechanisms are studied to improve transportation time and operational efficiency. In addition, camera-based perception and anomaly detection solutions are investigated to prevent forklift–human collisions.

These enablers support a safe, predictable, and well-controlled logistics environment within the factory.



Figure 22 Human Detection system on forklift

Within the scope of Otokar's contribution, a human detection and warning system for in-factory forklift operations (Figure 22) has been developed to improve operational safety. Time-of-Flight (ToF) cameras are mounted on both the front and rear sides of the forklift to detect humans and measure distance in real time.

All sensor data is processed inside an onboard hardware enclosure located within the forklift cabin. This enclosure contains a Raspberry Pi running the AI software, an Ethernet switch, and a 24V power converter. Cameras and sensors are connected to this unit via Ethernet.

An AI-based human detection algorithm identifies people approaching the forklift, especially during reverse maneuvers (Figure 23). When the distance between the forklift and a detected person falls below a predefined threshold, the system generates a “Warning” state, followed by a “Danger” state at a closer distance.

In the “Danger” state, a red warning light inside the operator cabin is activated to alert the driver. This system directly supports the Detection & Response and Autonomous Operations enablers defined in WP4.

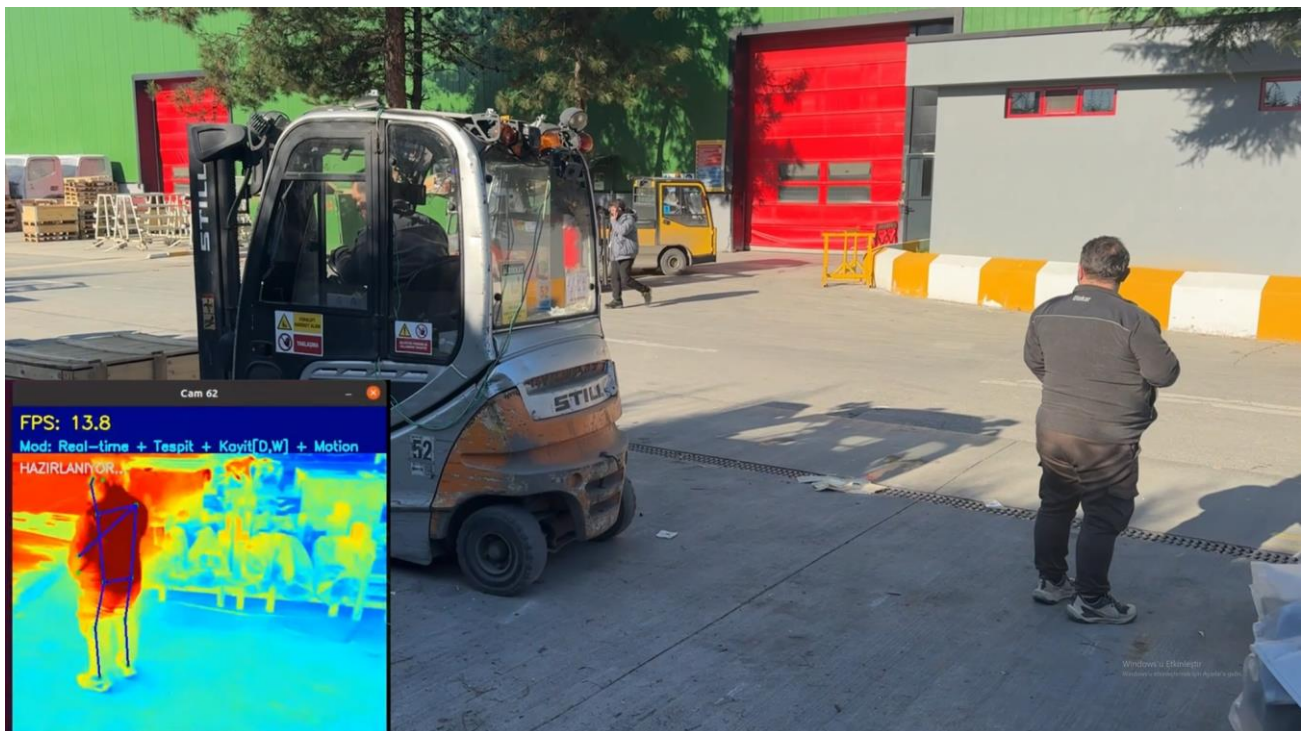


Figure 23 Human detection and distance-based warning generation using ToF cameras and AI during forklift reverse maneuver

4.10 Demonstration site – Port of Oulu

The Port of Oulu, as the port authority, has the primary responsibility for developing port infrastructure, which includes both physical and digital infrastructure. Within the Sa4CPS project, work related to technical enablers for status, projections, and actions is carried out in close collaboration with the project's technology partners. The Port of Oulu does not independently implement Sa4CPS technical enablers; rather, its role is to act as a co-development partner engaging with technology companies throughout the development process and providing input on requirements and specifications for technical enablers designed for critical environments.

The project storyline developed by the Port of Oulu has been instrumental in identifying the port area's critical needs concerning the technical enablers defined within SA4CPS. The summarized project story is illustrated in Figure 24. Each phase-level figure is accompanied by a description of the corresponding activities, which also serve as the high level requirements necessary for that phase to be realized.

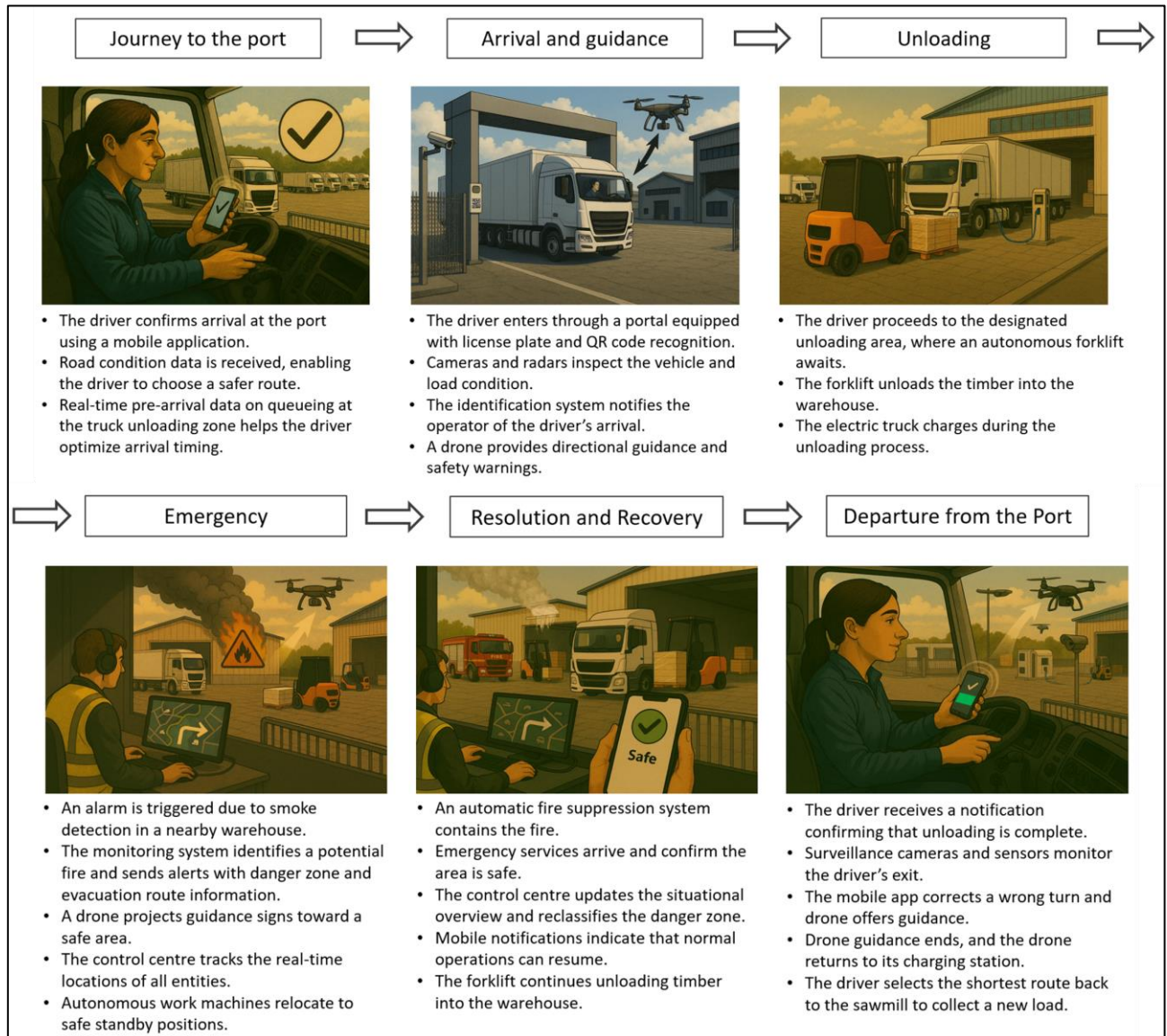


Figure 24 Port of Oulu summarized project story

The table below outlines the first set of project-story-related high-level requirements that serve as a basis for the specification of the technical enablers for status, projections, and actions.

Req ID	Requirement Name	Description
REQ_STORY.1	Road and Weather Conditions Information	The system must provide information about road and weather conditions, such as slippery road surfaces, on the way to the port area and within the port area.
REQ_STORY.2	Unloading Time Notification	The application must offer information about the next possible unloading time to the driver.

REQ_STORY.3	License Plate Recognition	The system must recognize the vehicle's license plate at the port gate upon arrival.
REQ_STORY.4	Cabin Occupancy Check	Cameras must check that there are no other persons in the truck cabin at the port gate.
REQ_STORY.5	Vehicle Scanning Condition	Cameras and sensors must scan the vehicle at the port gate to detect any deviations in load securing, load condition, or vehicle condition before arrival.
REQ_STORY.6	Driver Arrival Notification	The identification system must send information about the driver's arrival at the port area to the operator, who has prepared for the arrival based on the advance notice and ensure that the autonomous work machine is waiting in the warehouse.
REQ_STORY.7	Drone Assistance for Route Instructions	The system must enable the driver to order a drone for assistance, which projects route instructions and warnings onto the road.
REQ_STORY.8	Stop Sign Projection	The drone or fixed projector must project a stop sign onto the ground to make the driver stop the truck and allow the container handler to pass. When it is safe to continue, the drone must project the information onto the ground.
REQ_STORY.9	Correct Route Notification	The mobile application must notify the driver to return to the correct route if they turn in the wrong direction at an intersection.
REQ_STORY.10	Area Monitoring System	The area monitoring system must track the movements of vehicles, work machines, and people in the area and intervene if necessary.
REQ_STORY.11	Hazard Alert and Communication	The system sends a hazard alert and action instructions to all persons in the area if necessary. The system allows communication and sharing of information about the progress of events.

These needs are linked to the wide range of Sa4CPS use cases presented in Chapter 2 and to the components of the Sa4CPS concept. In regular technical meetings of the Sa4CPS project, the requirements for technical enablers have been reviewed to ensure they address the needs of the critical port environment as part of the overall Sa4CPS framework.

In alignment with WP4 objectives - which focus on defining and specifying technical enablers for situational awareness, projections, and actions - the Port of Oulu has concentrated on providing port-specific input across the key technical areas defined in WP4 tasks. The Port's role is to ensure that the requirements and operational perspectives of the port environment are adequately represented and integrated into the development.

5 A language, formal model, and tools to describe, model, and identify situations

ID	Enabler Name	Related Tasks	Addressed Requirements	Related Use Case
EN_15	A language, formal model, and tools to describe, model, and identify situations	T4.1, T4.2, T4.3, T4.4, T4.5	REQ_UC2.3 REQ_UC2.17 REQ_UC2.19 REQ_UC2.20 REQ_UC2.21 REQ_COMMON_1 REQ_COMMON_2 REQ_COMMON_4	UC2

In this section, we describe the means for modeling dynamic (and potentially hazardous) situations that can arise in human-machine interactions. The section introduces a language, a formal model, and supporting tools for describing, modelling, and identifying situations in cyber-physical systems. The proposed approach focuses on capturing the relevant entities, their properties, and relationships in a structured form that enables automated reasoning about situations of interest at runtime.

In production and logistics environments, situational awareness depends not only on knowing the positions of people and machines, but on understanding how their spatial relationships, movement patterns, and operational contexts evolve over time. Human workers, mobile machines, and fixed infrastructure interact in shared spaces where potentially hazardous situations emerge from combinations of location, behavior, and intent rather than from individual signals in isolation. Addressing such situations requires a representation that can integrate heterogeneous sensor inputs, abstract them into semantically meaningful entities and relations, and support reasoning about their joint state.

The language and formal model presented in this section provide this abstraction layer. By defining a common structure for representing physical entities, sensed properties, and dynamic relationships, they enable systematic identification of situations such as unsafe proximity, anomalous behavior, or deviations from expected operational patterns. The accompanying tools support the instantiation and evaluation of these models at runtime, allowing higher-level detection and response mechanisms to operate on well-defined situation constructs rather than raw data streams. In this way, Section 5 complements the platform-level enablers described earlier by supplying the formal foundation required for consistent, explainable, and extensible situation awareness across people-machine interaction scenarios in production and logistics domains.

5.1 Example

As a particular example, the section utilizes the use case UC2 (People and machine position awareness in production or logistics area), which focuses on improving safety and operational efficiency in dynamic industrial environments where humans and machines work in close proximity. Namely, UC2 employs a robotic arm as the machine, which operates in a workshop where humans also work. Figure 25 shows the

robot and its “hazardous area” (in red), where the robot can harm a human. If a human appears in this area, the robot has to stop its operations or operate only in a part of the area.



Figure 25 Robot vs Human

To detect positions of humans and machines, inexpensive sensors like ToF cameras and UWB sensors are employed. The ToF cameras have very small resolution (e.g., 8x8) but also provide a distance to observed objects. Figure 26 illustrates the situation as it can be seen by the ToF camera placed on the ceiling above the robot. The left part of the figure only shows the grid of the camera, while the right part illustrates the “image” seen by the camera (the darker shades of gray represent a smaller distance to an object).

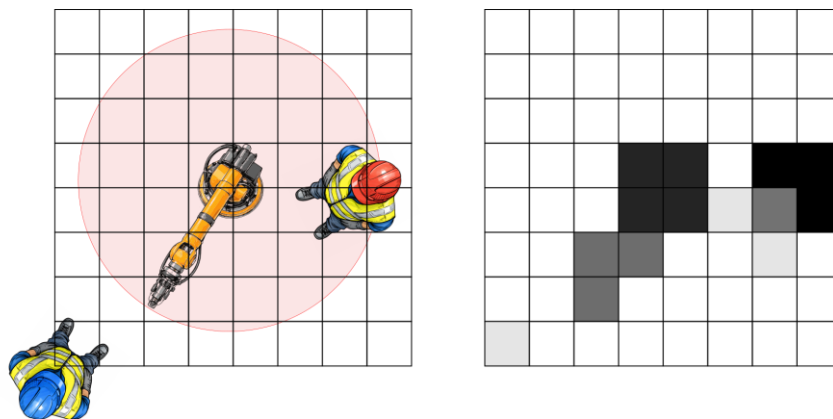


Figure 26 Robot vs Human as seen by ToF camera

5.2 Meta-model

To describe the sensors, robots, and other equipment in the workshop, as well as the dynamic situation, we define a specialized model. The model is determined via a meta-model, as shown in Figure 27 (a simplified version of the meta-model is presented for easy readability).

We use the terms “model” and “meta-model” as defined by OMG¹. The model “is information selectively representing some aspect of a system based on a specific set of concerns”, and the meta-model describes what can be used in a model.

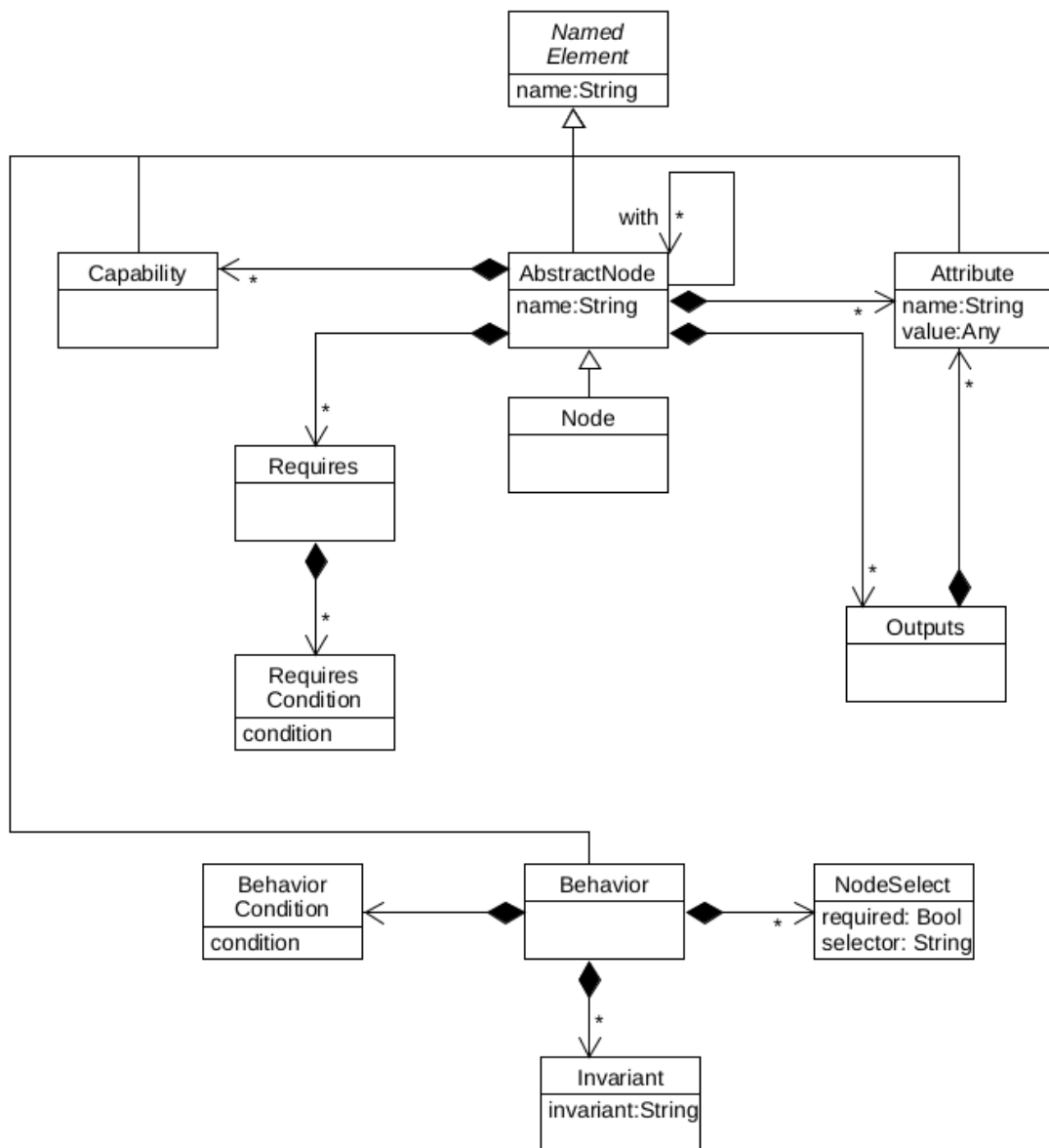


Figure 27 A meta-model describing situations

Rather than introducing the meta-model purely in abstract terms, we explain it through a concrete example that captures a representative fragment of Use Case UC2 (people and machine position awareness in production or logistics environments). This example-driven presentation is intentional: it reflects the fact

¹ OMG: MDA Guide revision 2.0, <http://www.omg.org/cgi-bin/doc?ormsc/14-06-01>, (2014).

that the meta-model is designed to support reasoning about concrete operational situations arising from interactions between people, machines, sensors, and infrastructure in shared physical spaces.

At its core, the meta-model defines the abstract syntax of situation descriptions, that is, the set of modelling concepts and the relationships between them. The abstract syntax determines what can be expressed about a system—such as the existence of entities, their capabilities, their configurations, and their interactions—independently of any particular textual notation. To make the model usable in practice, this abstract syntax must be mapped to a concrete syntax. Instead of introducing a custom domain-specific language with a dedicated parser and tooling, the approach adopts YAML as the concrete syntax. YAML provides a human-readable, structured representation that is widely supported and easy to integrate into existing toolchains, while remaining expressive enough to encode the required modelling concepts.

5.2.1 Objects and inheritance structure

The modelling approach is based on a prototype-oriented view of system entities, inspired by prototype-based programming languages. In contrast to class-based approaches—where types are defined first and instances are created afterwards—prototype-based modelling emphasizes reuse and incremental specialization. Objects are defined by inheriting properties from other objects that serve as prototypes, and inheritance relationships can evolve dynamically.

Three kinds of objects are used to capture different modelling concerns:

- **Nodes** represent concrete, physical entities in the system, such as sensors, robots, or computing units deployed in the environment.
- **Abstract-nodes** define reusable prototypes that capture common properties and capabilities shared by multiple nodes. They provide a means to factor out common structure, such as hardware characteristics or connectivity features.
- **Traits** represent orthogonal and potentially dynamic aspects of nodes. Unlike abstract-nodes, traits can be attached to or removed from nodes at runtime, allowing the model to reflect changes in configuration, operational mode, or functionality depending on the current situation.

This separation allows the model to distinguish between structural aspects (what a device fundamentally is), configurational aspects (how it is currently set up), and situational aspects (what role it plays at a given moment).

5.2.2 Common object properties

All three kinds of objects share a common set of properties, which provides a uniform modelling interface:

- The **name** uniquely identifies the object within the model and is mandatory.
- **Capabilities** act as semantic descriptors of what an object can do or what kind of role it can play. Capabilities are not required to be unique and serve as the primary mechanism for selecting and reasoning about objects in situations. They can be seen as a sort of object type.
- **Attributes** capture configuration parameters and static or slowly changing properties, such as hardware characteristics or spatial location. They are represented as name-value pairs.
- **Outputs** represent values produced by the object at runtime, such as sensor readings or computed results. Outputs are conceptually distinct from attributes in that they reflect observed or generated state rather than configuration. They are represented as name-value pairs.

- The optional **requires** clause constrains inheritance by specifying which capabilities must be present for another object to inherit from this one. This ensures that traits or abstract-nodes are only applied to compatible entities.

Together, these properties enable the model to describe not only what entities exist, but also what they can provide, how they are configured, and what information they generate.

5.3 Environment description

Listing 1 illustrates how a concrete environment is described using this modelling approach. It specifies multiple sensors (namely, ToFs and radars) and a robot (a robotic arm). The abstract-nodes are:

- a) rasperrypi5 representing a Raspberry Pi 5 computer,
- b) network-autoconfig representing a computer with autoconfiguration of its network connection,
- c) ToF representing a computer with the ToF sensor connected to it,
- d) radar representing a computer with the radar sensor connected to it,
- e) robot representing a computer with the robotic arm connected to it.

This set of abstract-nodes captures reusable hardware and connectivity building blocks.

Traits are then used to represent alternative configurations of the same physical sensor, such as high-resolution versus low-power operating modes, which can be activated depending on the situation. There are two traits in the use-case:

- a) hires-config, which is a configuration of the ToF sensor to provide readings with a high framerate,
- b) lowpower-config, which is a configuration of the ToF sensor to operate with low power settings.

These two are defined as traits, allowing objects (ToF sensors) to switch between them dynamically.

Finally, the multiple actual nodes are defined; two Raspberry Pi 5 computers with connected ToFs, one with the radar sensor, and one with the robotic arm. All of these nodes are specified via the abstract-node specifications.

Concrete nodes are defined by composing these abstract-nodes through the always-with relationship, thereby instantiating actual deployed devices with specific locations in space. This composition-based definition makes the physical deployment explicit while avoiding duplication of common configuration details.

```
abstract-node:
  name: rasperrypi5
  capabilities:
    - computer
    - armcpu
    - sdcard
  attributes:
    ram: 4GB
    disk: 64GB

abstract-node:
  name: network-autoconfig
```

```
capabilities:
  - network-connected
attributes:
  network-type: WIFI
  ip: dhcp
  gateway: auto
  hostname: $"node.name"-node

abstract-node:
  name: ToF
  capabilities:
    - sensor
    - ToF
    - ToF_8x8
    - VL53L8
  attributes:
    firmware: tof_standard_v2.1.elf
    connection: serial

abstract-node:
  name: radar
  capabilities:
    - sensor
    - radar
    - XE125
  attributes:
    firmware: xe125_standard_v1.elf
    connection: serial

abstract-node:
  name: robot
  capabilities:
    - actuator
    - robotic-hand
    - ABCD1234
  attributes:
    connection: serial
    dimension:
      x: 1000
      y: 1000
      z: 2000

trait:
  name: hires-config
  capabilities:
```

```
- ToF-resolution-config
requires: # capabilites
- ToF
attributes:
  resolution: 16
  framerate: 20
  connection: SPI
  multitarget: True

trait:
  name: lopower-config
  capabilities:
    - ToF-resolution-config
  requires: # capabilites
    - ToF
  attributes:
    resolution: 64
    framerate: 10
    connection: I2C
    multitarget: False

node:
  name: RPi1
  attributes:
    location:
      x: 1000
      y: 2000
      z: 2000
      #coord: "50.088366,14.403524"
  always-with:
    - rasperrypi5
    - network-autoconfig
    - ToF

node:
  name: RPi2
  attributes:
    location:
      x: 3000
      y: 5000
      z: 2000
  always-with:
    - rasperrypi5
    - network-autoconfig
    - ToF
```



```
node:
  name: RPi3
  attributes:
    location:
      x: 1000
      y: 10000
      z: 2000
  always-with:
    - rasperrypi5
    - network-autoconfig
    - radar

node:
  name: ROBOT
  attributes:
    location:
      x: 5000
      y: 10000
      z: 2000
  always-with:
    - rasperrypi5
    - network-autoconfig
    - robot
```

Listing 1 Environment description

5.4 Situation and behavior modelling

While Listing 1 focuses on the static structure of the environment, Listing 2 introduces dynamic situations, expressed through additional traits and behaviors. The traits declare functional roles that nodes can assume, along with the outputs they produce and the states they can be in. These traits bridge the gap between raw devices and higher-level semantic functions, such as detecting human presence or executing robotic actions.

In Listing 2, specifications of dynamic situations are shown. First, there are two additional traits:

- a) human-position-detection, which declares the functionality of detecting a human position in the space. It can be applied only to a node that has the sensor capability and either ToF or radar capability. It declares the outputs that will be produced by the node with this trait applied.
- b) stationary-robotic-hand-work, which declares the functionality of an actuator, a robotic hand in particular.

Both traits are then applied to the particular nodes.

Through behaviors, the model captures what is happening, and under what conditions and with what constraints. For example, situations can express time-dependent operational modes, adaptive sensor configurations, or safety-relevant interactions such as a human entering the operating area of a robot.

There are multiple behaviors defined, which specify dynamic situations. The behavior features are:

- a) nodes that select which nodes take a role in the situation,
- b) condition specifying the situation based on the attributes of the selected nodes,
- c) invariants specifying the attributes' values and nodes' traits in the specified situation.

```
trait:
  name: human-position-detection
  requires:
    all: [ sensor ]
    one-of: [ ToF, radar ]
  capabilities:
    - service
    - human-detection
  attributes:
    state:
      - name: working
      - name: stopped
  outputs: # like attributes but these are produced by the object
    position: 3d-area
    timestamp: timestamp

trait:
  name: stationary-robotic-hand-work
  capabilities:
    - service
    - robotic-hand-working
  requires:
    all: [ actuator, robotic-hand ]
  attributes:
    state:
      - name: stopped
      - name: working
      params: # parameters of the state that must be set
        in-area: 3d-area

node:
  name: RPi1
  with:
    - human-position-detection

node:
  name: ROBOT
  with:
```

- stationary-robotic-hand-work

behavior:

name: robot-regular-work

nodes:

- name: robot
required: True
select:
node: ROBOT

invariants:

- "ROBOT.state = work (in-area = ROBOT.dimension)"

behavior:

name: human-detection-regular-work

nodes:

- name: sensor
required: True
select:
any: [human-detection]

condition: "6 < datetime.now().hour < 18"

invariants:

- "sensor.state = working"
- "sensor HAS-TRAIT hires-config"

behavior:

name: human-detection-low-power

nodes:

- name: sensor
required: True
select:
any: [human-detection]

condition: "! (6 < datetime.now().hour < 18)"

invariants:

- "sensor.state = working"
- "sensor HAS-TRAIT lopower-config"

behavior:

name: human-near-robot

nodes:

- name: detector
required: True
select:
any: [human-detection]

```
- name: robot
  required: True
  select:
    node: ROBOT
  condition: "detector.position within ROBOT.dimension"
  invariants:
    - "ROBOT.state = working (safe-area = (ROBOT.dimension -
detector.position))"
```

Listing 2 Situations description

Similarly, other situations in UC2 (such as unauthorized persons in the area) and other use cases can also be described.

These descriptions will be processed by the SA4CPS deployment engine, which, based on the descriptions, configures individual devices (such as sensors and robots), deploys the relevant software packages to these devices, and manages them during execution.

As a result, the formal model serves as a link between high-level situational concepts and concrete system behaviour, enabling consistent interpretation, automated configuration, and systematic reasoning across heterogeneous cyber-physical systems. The same modelling approach can be applied to additional situations in UC2 (e.g., unauthorized access) as well as to other use cases, providing a reusable and extensible foundation for situation awareness across production and logistics domains.

6 Behaviour-Based Digital Risk Detection and Situational Awareness

This section describes the enablers to support behaviour-based digital risk detection and situational awareness in UC5. These enablers (being developed by F-Secure) focus on transforming low-level behavioural and network signals into higher-level situational understanding that enables detection, interpretation, and response to emerging digital risks over time. By combining large-scale data processing pipelines, machine-learning-based fingerprinting, and a unified behavioural data hub, the proposed enablers provide the technical foundation for monitoring evolving digital situations, supporting projections of risk development, and enabling informed actions at both device and ecosystem levels.

ID	Enabler Name	Related Tasks	Addressed Requirements	Related Use Case
EN_16	DNS Fingerprinting Data Processing Pipeline	T4.1, T4.4	REQ_UC5.7 REQ_UC5.8	UC5
EN_17	Behavioural data hub	T4.1, T4.4	REQ_UC5.7 REQ_UC5.8	UC5

6.1 DNS Fingerprinting Data Processing Pipeline (EN_16)

A scalable, cloud-native data processing pipeline has been developed to support DNS fingerprinting research and model development. The pipeline facilitates 1) continuous intake and processing of DNS traffic samples from various sources, 2) transforming raw data into training datasets, and 3) training and validation of DNS fingerprint detection ML models suitable for deployment across diverse environments.

Pipeline Architecture

The pipeline implements a sequence of stages:

- 3) The pipeline accommodates continuous DNS data ingestion from a variety of sources (e.g., opt-in clients, control devices, pilot routers, etc.), handling variable volume and velocity constraints inherent in real-world network traffic sample collection. Data samples are ingested into cloud storage infrastructure, which provides a centralized repository accessible to data scientists while maintaining data integrity and traceability throughout the entire pipeline workflow.
- 3) Data processing is implemented in distinct incremental steps:
 - a. **Preprocessing** applies to client isolation, normalization, and quality filtering to ensure data consistency and remove any incomplete records.
 - b. **Clustering** combines DNS queries based on set criteria into contextual clusters, representing higher level events that initiated the burst of DNS data. These bursts are the most relevant signals in detecting DNS fingerprints.
 - c. **Manual curation** (optional) allows a human sensor to inspect and/or manipulate automatically created clusters of DNS data. Manual curation of clusters provides a fast track to create fingerprints for e.g., recently emerged malicious applications that need detection as soon as possible.

- 3) Grouping organizes aggregated cluster data by relevant dimensions (e.g., similarity) to facilitate targeted model training. Grouping prepares processed datasets for machine learning experimentation, enabling eventual rapid iteration on detection model development and validation.

Data Accessibility and Experimentation

Processed data is maintained at multiple pipeline stages, allowing data scientists to access intermediate representations suited to different objectives. This staged approach supports both exploratory analysis and controlled validation workflows, enabling model training on appropriately prepared datasets and validation of results with minimal data wrangling.

Model Outputs

The pipeline produces DNS fingerprint detection models tailored for different purposes. On-device detection models are optimized for mobile platforms, respecting computational and memory constraints while maintaining detection accuracy for client-side applications. Router-based detection models are designed for deployment at customer premises equipment (CPE), enabling network-wide visibility and detection capabilities at the network edge for any (W)LAN connected devices, including IoT.

Implementation

The pipeline is implemented on Amazon Web Services using managed service building blocks to minimize operational overhead and support scaling to production volumes. AWS Lambda provides serverless computation for data intake and transformation steps. Amazon S3 serves as the primary data lake for raw datasets. Amazon DynamoDB enables efficient state management and metadata tracking throughout the pipeline. Amazon SageMaker provides experimentation infrastructure for model development, training, and validation.

6.2 Behavioural data hub (EN_17)

The Behavioural Data Hub represents a shift from traditional, perimeter-based cybersecurity toward a human-centric protection model. Rather than treating security as a set of isolated tools (firewalls, scanners, or warnings), the hub places the user at the center and understands risk as something that emerges from behaviour, context, and exposure over time.

Modern users move fluidly across multiple digital moments every day—browsing online, communicating via messages, managing personal data, maintaining devices, and responding to awareness signals. Each of these moments generates behavioural signals that, in isolation, may seem benign, but when connected, reveal meaningful risk trajectories.

The Behavioural Data Hub acts as a unifying layer that:

- Aggregates behavioural signals across browsing, messaging, personal data usage, device hygiene, and awareness interactions
- Normalizes these signals into comparable, privacy-respecting features
- Connects protection touchpoints (e.g., scam detection, password vaults, device scanning) to observable behavioural outcomes

To ensure that user-centered behavioral security remains trustworthy and ethically grounded, the following privacy-by-design and trust enablers are integral components of the Behavioral Data Hub's implementation:

- Consent and purpose-limited collection
- Data minimization + on-device feature extraction
- Pseudonymization and linkability controls
- Privacy-preserving analytics for benchmarking (“compared to peers”)

Extending the Behavioral Data Hub’s privacy-by-design foundations, the following features operationalize risk scoring, modeling, and governance to transform behavioral insights into robust and trustworthy security outcomes:

- Feature store
- Hybrid risk engine (rules + ML)
- Model governance and evaluation
- Explainability for user-facing insights

The following features support the Integration with “Our protection abilities” by translating privacy-by-design and risk governance principles into actionable security functions:

- Messaging protection signals (SMS + instant messaging)
On-device classification, link reputation checks, scam pattern features, and minimal-content collection by default.
- Browsing/online protection signals (browser/desktop extensions)
- Personal data protection (identity monitoring, password vault)
- Device hygiene (scanning, OS status, device lock, privacy advisor)
- Awareness & motivation (scam scanner/deep fake detection, news/education)

7 Real-Time Operational Data Integration and Visualization

Real-time operational data integration is a fundamental prerequisite for maintaining accurate system status, supporting projections, and enabling timely actions in cyber-physical systems. Within WP4, this capability is addressed in this section through the specification of an enabler that provides a coherent foundation for collecting, synchronizing, processing, and visualizing operational data across edge and cloud environments.

This section specifies the **Capture Platform**, which acts as the primary data backbone for real-time operational awareness in WP4. The enabler addresses key challenges such as protocol heterogeneity, variable connectivity, latency constraints, and data volume by providing a unified approach to data ingestion, normalization, and contextualization. By structuring and enriching data close to its source, the Capture Platform enables downstream WP4 components—such as digital twins, dashboards, detection mechanisms, and decision-support tools—to operate on consistent and semantically meaningful system state.

A central aspect of the enabler is its support **for** flexible edge-to-cloud synchronization strategies, allowing systems to balance real-time responsiveness with robustness under intermittent connectivity. Both real-time streaming and batch-oriented synchronization mechanisms are supported, enabling deployment in diverse operational environments. In addition, the enabler specifies standardized interfaces and visualization tools through which integrated data can be accessed by human operators and external systems.

The Capture Platform is specified in the context of UC9, but is designed as a reusable and extensible enabler that can support other Sa4CPS use cases requiring reliable real-time data integration and visualization. As such, it provides a foundational capability upon which other WP4 enablers can build to support status representation, projections, and actions.

ID	Enabler Name	Related Tasks	Addressed Requirements	Related Use Case
EN_18	Capture Platform	T4.1	REQ_UC9.7	UC9

7.1 Capture platform (EN_18)

The Capture platform (being developed by Vintecc) is the core technical enabler for status, projection, and action in UC9. Capture is an Industrial Internet of Things (IIoT) and edge-to-cloud platform designed to collect, structure, contextualize, and act upon data originating from cyber-physical systems (CPS). It serves as the foundation for building and operating digital twins that represent both the real-time and historical state of industrial assets.

Capture supports data acquisition from heterogeneous industrial environments through a wide range of industrial and IT protocols, including but not limited to ADS, Siemens S7, MQTT, Redis, HTTP, MODBUS, CAN, and OPC-UA. This allows Capture to integrate with PLCs, embedded controllers, sensors, gateways, and software services without requiring architectural changes on the CPS side. All collected data is normalized into a unified internal representation and enriched with metadata such as timestamps, units, semantic tags, asset identifiers, and safety-related attributes.

Once ingested, the data is stored, processed, and organized according to logical structures such as machines, production lines, sites, or fleets. This structured and enriched data forms the operational core of the digital twin. Visualization and human interaction are supported through Grafana dashboards and the Capture Data Explorer, enabling operators and engineers to inspect real-time values, historical trends, correlations, and system states. In parallel, the same digital twin data is exposed to external systems through a REST/HTTP API and through MQTT-based data distribution, including support for a Unified Namespace (UNS) model.

7.1.1 Data Collection and Protocol Selection

The protocol used to gather data from a cyber-physical system is determined by a combination of technical and operational factors, including the capabilities of the target device, real-time requirements, network constraints, data volume, and security policies. For PLC-based systems, protocols such as ADS, S7, or OPC-UA are typically used to access internal variables and process values. For embedded or IoT-oriented devices, MQTT, HTTP, or CAN-based integrations are often preferred.

Capture uses a modular connector architecture, where protocol adapters run either on the edge device or in a nearby gateway. These connectors are responsible for polling, subscribing to, or receiving pushed data from the CPS. They also perform basic validation, timestamping, and buffering before forwarding the data into the Capture data pipeline. This architecture allows multiple protocols to coexist within the same deployment and enables gradual migration or extension of CPS connectivity without disrupting existing systems.

7.1.2 Synchronizing Data from Edge to Cloud

Capture supports multiple synchronization strategies to move data from edge environments to the cloud, depending on bandwidth, latency, and reliability requirements.

Batch Synchronization: In batch synchronization mode, data is buffered locally on the edge device and periodically uploaded to the cloud in chunks. This mode is typically used for high-volume historical data, non-time-critical measurements, or environments with intermittent connectivity. Local buffering ensures no data is lost during network outages, while controlled upload schedules prevent network congestion.

Three options are available for the local buffering:

- InfluxDB
- Timescale
- Redis Streams

Real-Time Synchronization: In real-time synchronization mode, selected signals are streamed continuously to the cloud using MQTT. This mode is used for operational monitoring, situational awareness, alerting, and decision-making loops that require up-to-date system state. Capture allows fine-grained selection of which data streams are synchronized in real time and which are handled in batch mode.

Both modes can coexist within the same deployment and are configurable per asset, per signal, or per use case.

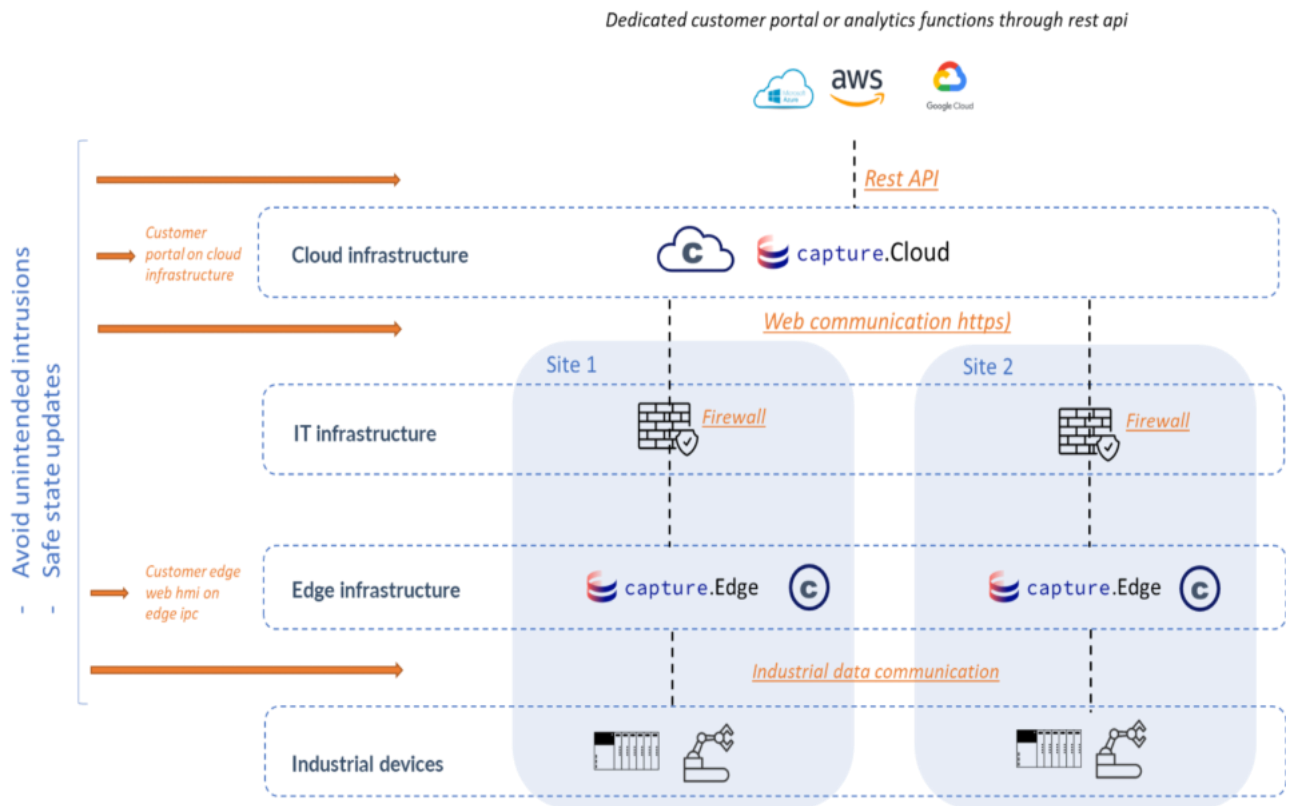


Figure 28 Edge vs. Cloud

7.1.3 Accessing Data in the Cloud

Once synchronized, digital twin data in the cloud can be accessed through multiple technical interfaces, serving different stakeholders and applications.

Grafana

Grafana is used for visualization and operational dashboards. It provides real-time and historical views on system behavior, KPIs, alarms, and correlations between signals. These dashboards support operators, engineers, and decision-makers in understanding the current and past state of the CPS.

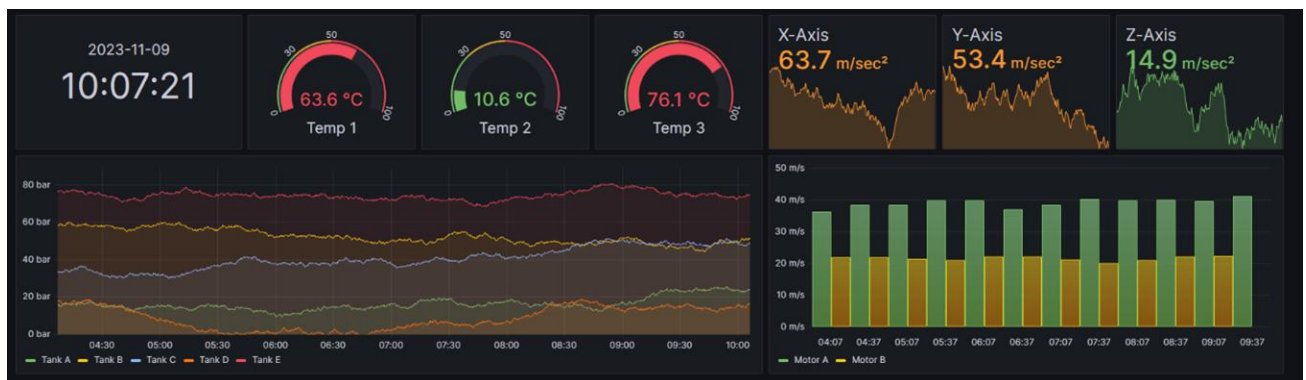


Figure 29 Grafana dashboard

Capture Data Explorer

The Capture Data Explorer provides an engineering-oriented interface for browsing assets, signals, metadata, and time series. It is used for debugging, configuration, validation, and analysis of the digital twin structure and contents.

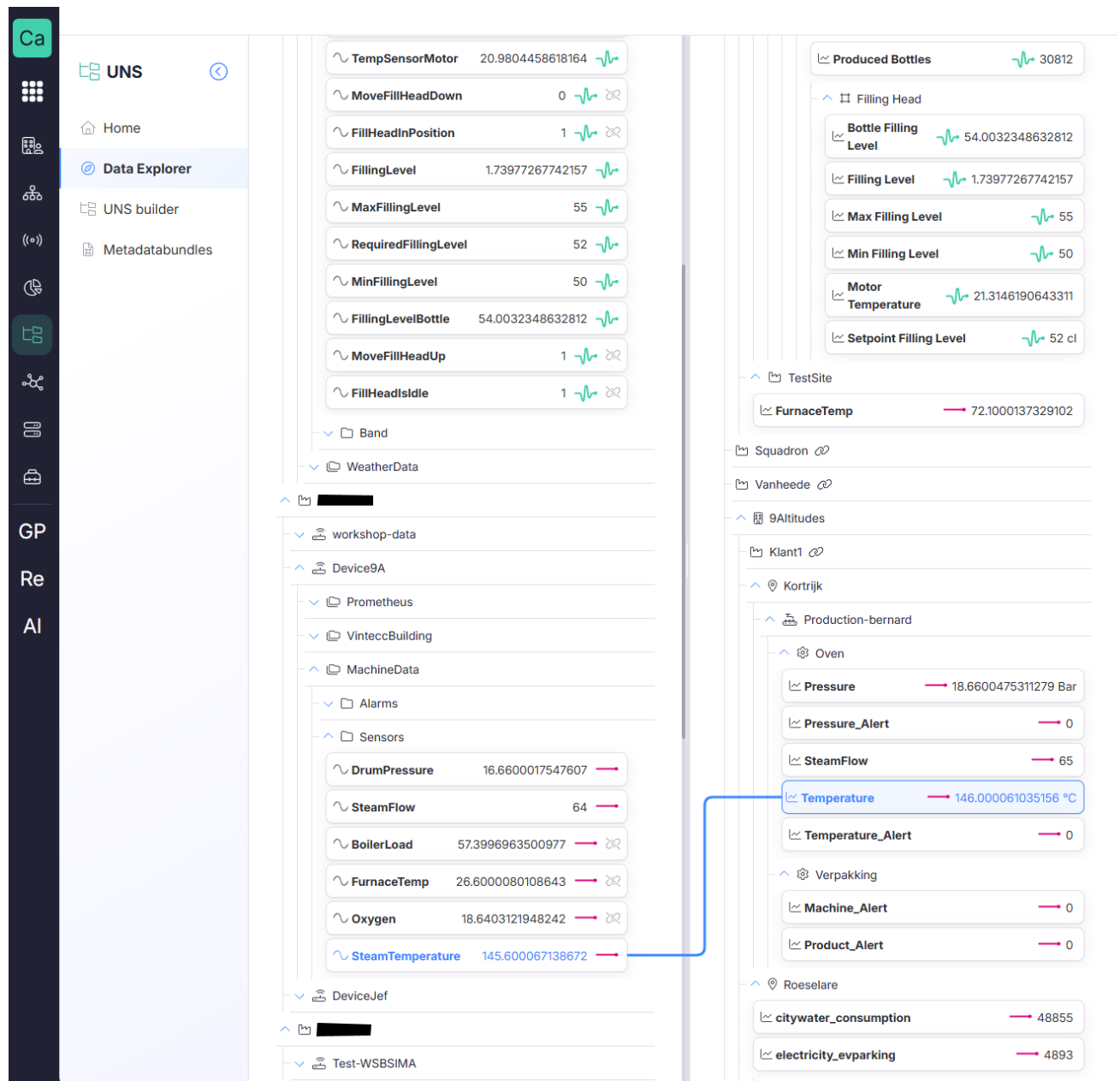


Figure 30 Capture Data Explorer

UNS over MQTT

Capture exposes selected parts of the digital twin through a Unified Namespace over MQTT. This allows other systems, services, and automation components to subscribe to structured, semantically meaningful data streams representing the live state of assets, production lines, or fleets.

REST / HTTP API

A REST API provides programmatic access to both real-time and historical data, metadata, asset models, and system state. This API is used by external applications that want to retrieve data from the CPS or trigger actions.

8 Geospatial Situational Awareness and Event Detection

Geospatial situational awareness forms a critical foundation for status representation, projections, and actions in cyber-physical systems operating in large, dynamic, and safety-critical environments. Accurate spatial context is essential for understanding the current situation, detecting relevant events, and supporting timely and informed operational responses. This is particularly important in scenarios involving distributed assets, mobile entities, heterogeneous sensor infrastructures, and multiple stakeholders, where the spatial dimension acts as the unifying reference for system state.

This section specifies the technical enablers that support **geospatial situational awareness and event detection**, with a focus on integrating heterogeneous location- and sensor-derived data into a coherent, real-time operational view. The enablers address the challenges of positioning accuracy, data fusion, visualization, and event interpretation, especially in environments where GNSS availability may be limited or unreliable and where real-time performance is essential.

The specified enablers cover the full pipeline from **data ingestion and spatial representation** to **AI-based detection and alerting**. They include map engines and visualization technologies, integration of location and detection data into a common spatial model, real-time event processing, and secure access control mechanisms. Together, these components enable the transformation of raw spatial and sensor data into actionable situational information that can be consumed by operators and downstream systems.

The section places particular emphasis on **operational usability and performance**, ensuring that spatial information can be streamed, rendered, and updated in real time, while remaining scalable and interoperable. It also addresses **security and access control requirements**, including zero-trust authentication and authorization, which are essential for deploying geospatial situational awareness capabilities in critical environments.

The technical enablers described in this section are specified in the context of **UC3**, where real-time geospatial awareness and event detection support operational decision-making and response coordination. However, the specified concepts and interfaces are designed to be reusable across other Sa4CPS use cases that rely on spatial awareness as a key element of situational understanding.

ID	Enabler Name	Related Tasks	Addressed Requirements	Related Use Case
EN_19	Map, Map Engine, and Visualization Technologies	T4.1, T4.4, T4.5	REQ_UC3.9, REQ_UC3.11, REQ_UC3.17	UC3
EN_20	AI-Enabled Integration of Location Data into the NSC3 Environment	T4.1, T4.2, T4.3, T4.4, T4.5	REQ_UC3.12, REQ_UC3.13, REQ_UC3.15, REQ_UC3.16	UC3
EN_21	AI-Based Event Detection and Real-Time Alerting	T4.2, T4.4, T4.5	REQ_UC3.18	UC3
EN_22	Integrating camera sources to NSC3 environment	T4.1, T4.2, T4.4	REQ_UC3.8, REQ_UC3.14	UC3

EN_23	Hardening NSC3 environment to Zero Trust Authentication and authorization	T4.2, T4.4, T4.5	REQ_UC3.4	UC3
-------	---	------------------	-----------	-----

8.1 Map, Map Engine, and Visualization Technologies (EN_19)

To fulfil REQ_UC3.9 (Efficient data streaming and caching 3D maps), a dedicated map service will be implemented as an open standards-based service for efficient real-time streaming and caching of 3D map data. The service will be designed around widely adopted protocols such as OGC (Open Geospatial Consortium), WMS (Web Map Service), WMTS (Web Map Tile Service), and WFS (Web Feature Service) to ensure maximum interoperability and ease of integration with both internal and external consumers. This map service will be optimized to reduce load times by employing tile-based data streaming and intelligent cache management strategies, thereby ensuring all map tiles and data features are accessible with minimal latency. Additionally, robust access control mechanisms, integrated at both the service and data layer, will be implemented to enforce authentication and authorization for all types of consumers, supporting fine-grained permissions per user, service, or external partner as required. For REQ_UC3.17 (Static camera coverage), the map visualization layer will incorporate real-time rendering of camera view cones. This will allow users to visually inspect camera fields of view and easily identify blind spots within the monitored area.

To address REQ_UC3.11 (Layers for 3D map), the NSC3 map engine will be extended to support dynamic, user-selectable map layers. Layers will include object detections, geofences, camera positions, and event overlays, all of which can be toggled on or off in the NSC3 web application interface. Published map data, including WFS (Web Feature Service) endpoints, will support query parameters for fine-grained access by authorized external services.

8.2 AI-Enabled Integration of Location Data into the NSC3 Environment (EN_20)

To comply with REQ_UC3.13 (Object detection types), a configurable object detection framework will be deployed, supporting dynamic selection of target object types such as persons and vehicles. This framework will include the creation and maintenance of tailored detection recipes for relevant scenarios.

For technical implementation, Redis Gears will be utilized to enable real-time, server-side processing of streaming detection results and metadata. Object detection outputs will be ingested into Redis streams, where Redis Gears functions will filter, enrich, and route detection events based on configurable rulesets corresponding to each detection recipe. The modular design will allow new object types or detection classes to be integrated or modified at runtime without service downtime.

To ensure minimal latency and high throughput, a high-speed queueing service (e.g., Redis Streams or a dedicated message broker such as NATS or Apache Kafka) will be deployed to handle the rapid influx of detection tasks and results between the camera ingestion services, object detection models, and downstream consumers (such as analytics and visualization services). This queueing layer will decouple detection processing from downstream event handling, supporting scalable fan-out, replay, and durable event sinks while guaranteeing delivery order and fail-safe processing, critical for high-performance AI-enabled video analytics in the NSC3 environment.

For REQ_UC3.15 (Select object detection model), a model selection and validation process will be established to ensure the optimal object detection model is integrated into the platform. Criteria will include detection accuracy, inference speed, and compatibility with other NSC3 components.

REQ_UC3.16 (Static cameras on map) will be addressed by enforcing precise geospatial calibration and input validation during camera setup, requiring static camera positions to be measured and stored in all three spatial dimensions. This enables accurate projection from camera detections to world coordinates, thereby improving reliability of detected object localization.

To address REQ_UC3.12 (Geofencing restricted and allowed areas), geofencing functionality will be embedded within the map service and detection pipeline, supporting both restricted and allowed geofence areas. The system will provide interfaces for users to define geofenced zones, and AI detection results will be automatically checked for entry/exit status with respect to those zones. Automated alerting and access control logic will react to violations as specified in operational requirements.

8.3 AI-Based Event Detection and Real-Time Alerting (EN_21)

A high-performance event-driven data pipeline will be implemented atop a pub/sub messaging bus, ensuring low-latency transfer of detection events, geolocation data, and map overlays among NSC3 services and the map backend.

To satisfy REQ_UC3.18 (Real time object detections on map), detected AI events from live camera feeds will be automatically ingested and rendered on the map interface in real time, with end-to-end latencies monitored and controlled to adhere to project requirements.

8.4 Integrating Camera Sources to NSC3 Environment (EN_22)

For REQ_UC3.14 (Object detection camera specs), a specification and onboarding workflow will be developed for integrating surveillance cameras. This workflow will enforce required minimum specifications such as resolution, lens parameters, field of view, and network stream support to ensure each integrated camera meets the standards necessary for accurate AI-assisted object detection.

Additionally, an administrative interface will be implemented to allow authorized users to view and update camera specifications after onboarding. This interface will provide an auditable and user-friendly mechanism to modify fields such as camera resolution, lens information, field of view parameters, and streaming configurations. All updates will be subject to validation to ensure ongoing compliance with AI detection requirements.

To fulfil REQ_UC3.8 (Data sources for anomaly detection), the system will catalogue and validate supported data sources for anomaly detection, supporting both live video feeds and simulated sources. To demonstrate cross-system compatibility, VTT's Unity simulation camera will be configured and integrated as a compliant IP stream via the Network Streaming Service (NSS).

8.5 Hardening NSC3 Environment to Zero Trust Authentication and Authorization (EN_23)

To implement REQ_UC3.4 (Zero trust authentication and authorization effect on performance), the NSC3 environment will be refactored to enforce comprehensive zero trust authentication and authorization patterns at every system boundary. The main challenge lies in ensuring full compliance with existing access control mechanisms while introducing enhanced zero trust capabilities. Third-party identity providers will be integrated alongside built-in user management, leveraging industry-standard protocols

(OAuth2, OIDC) to provide secure user and service authentication. This transition will be enabled by adopting a high-performance Quarkus-based application framework, which allows for extremely fast startup times and low-latency request handling. All security enforcement logic will be carefully optimized to introduce negligible additional latency, thereby preserving real-time data streaming performance and ensuring compatibility with current access control policies throughout the NSC3 platform.

8.6 NSC3 system description

The NSC3 platform is a secure video streaming solution purpose-built for deployment by national agencies, including police and firefighting organizations. It is architected to support multiple types of devices, each acting as a video source and contributing live feeds to the NSC3 system. The platform enforces a robust role-based access control model, distinguishing between administrator and operational roles to ensure that users can only perform actions appropriate to their responsibilities.

All video feeds are transmitted over secure connections using dedicated NSC3 applications, ensuring data integrity and confidentiality. NSC3 employs a task-centric workflow: both users and devices are assigned to specific tasks, which streamlines application interfaces and restricts the information and controls visible to any one user or group. This design choice promotes operational clarity and eliminates unnecessary distractions.

From a technical standpoint, the NSC3 system is composed of several modular services, all orchestrated and isolated within a Docker network. This containerized architecture provides scalability, resilience, and maintains security boundaries between critical platform components.

Access to the NSC3 system is strictly controlled; users must authenticate by logging in before they can utilize any platform features. The system enforces role-based access control (RBAC), ensuring that each user is only able to perform actions and view information aligned with their designated responsibilities. Furthermore, NSC3 supports a multi-tenant architecture, allowing multiple organizations to operate within the platform while keeping their user bases and associated data securely isolated from one another.

8.6.1 Services in the NSC3 backend

This subsection provides an overview of the core services composing the NSC3 backend and their respective responsibilities. It outlines how these services are organized to support secure video streaming, event processing, data storage, and map updates in a modular and scalable manner. The description establishes the foundation for understanding subsequent data flows and operational configurations.

Gateway service

- Reverse proxy between all the services and the public internet
- All outside connections come through here
- Some amount of connection filtering happens here to prevent malicious activities

NSC3 Core Services

- Handles communication layers between connected devices, and web apps, etc...
- Handles passing video feeds to web app users
- Maintains database
- Stores videos and images in object storage

- Handles updating the map service
- Hosts NSC3 web apps

Network Streaming Service (NSS)

- Handles network video sources like IP cameras and forwards their feeds to the NSC3 core

Messaging bus service (Redis)

- Offers an event-based publisher-subscriber bus

Object detection service (Redis gears)

- Listens to events in the 'Messaging bus service' about available material to analyze
- Supports multiple different recipes

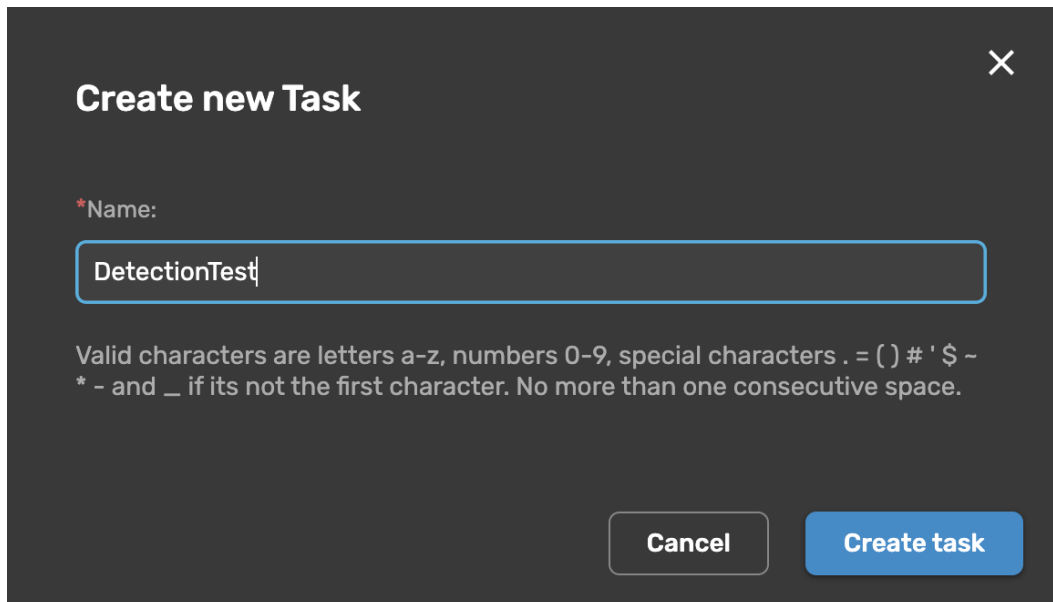
Map service

- Provides map tiles and related features

8.6.2 Setting up the system for object detections

This subsection describes the operational steps required to configure the NSC3 system for object detection. It focuses on the sequence of actions needed to associate detection models, data sources, and tasks within the platform. The goal is to clarify how detection capabilities are instantiated and prepared for real-time operation.

1. Create a task on the NSC3 system



Create new Task

*Name:

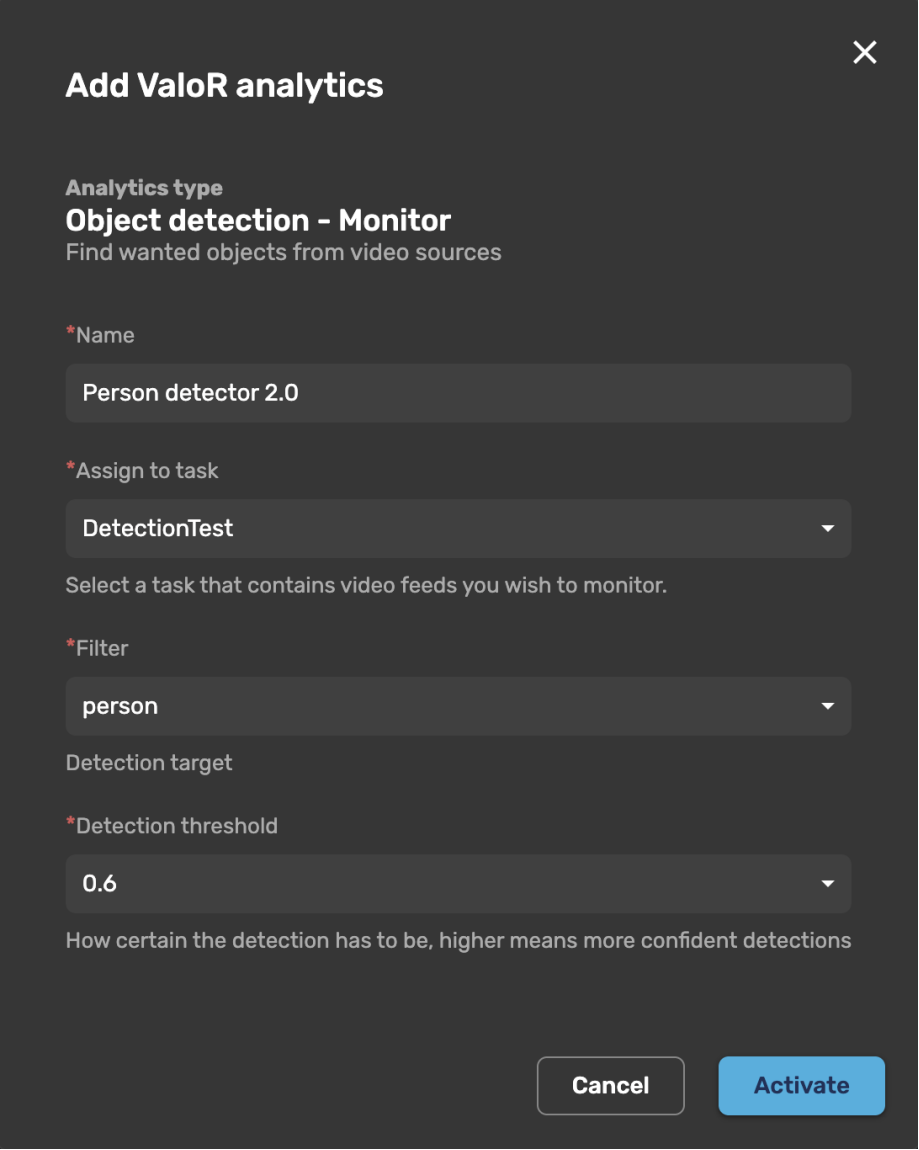
DetectionTest

Valid characters are letters a-z, numbers 0-9, special characters . = () # ' \$ ~ * - and _ if its not the first character. No more than one consecutive space.

Cancel Create task

Figure 31 Creating a task on the NSC3 system

2. Attach object detection model (car, person etc...) to the task



Add ValoR analytics ✕

Analytics type
Object detection - Monitor
Find wanted objects from video sources

*Name
Person detector 2.0

*Assign to task
DetectionTest ▼
Select a task that contains video feeds you wish to monitor.

*Filter
person ▼
Detection target

*Detection threshold
0.6 ▼
How certain the detection has to be, higher means more confident detections

Cancel Activate

Figure 32 Attaching an object detection model

3. Connect an IP camera to the NSC3 system through Network Streaming Service (NSS)

×

Add IP camera

To add an IP camera, open RTSP network stream using URL, username and password.

*Name

TestFeed

Valid characters are letters a-z, numbers 0-9, special characters . = () # ' \$ ~ * - and _ if its not the first character. No more than one consecutive space.

*RTSP URL

acme://test.cam.itea

Username

name

Password

.....|

Cancel

Create

Figure 33 Adding IP camera to the NSC3 system

4. Assign the connected camera to the created task

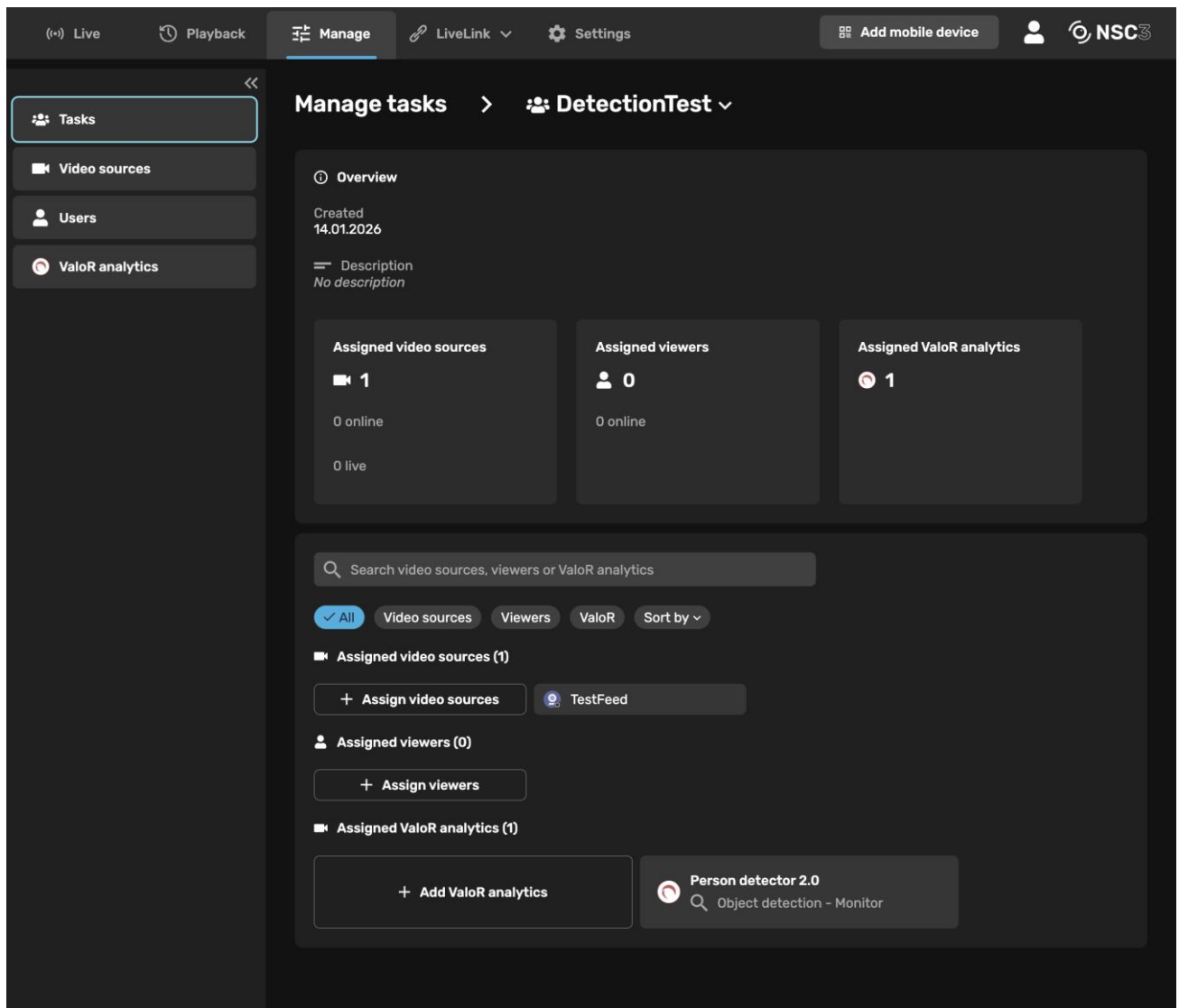


Figure 34 Assign the connected camera a task

5. Set the camera's minimum required metadata: geolocation, view angle, and field of view

8.6.3 Data flow from video feed to AI-driven detection drawn on a map

This subsection introduces the end-to-end data flow that transforms live video streams into spatially grounded detection events visualized on the map. It explains how video data, metadata, and AI detection results are processed and propagated through the NSC3 components. The description highlights the integration points that enable real-time situational awareness and event visualization. Figure 35 overviews the flow. A green line emanating from the IP camera depicts the video feed. A dotted line means reduced frame rate (1 frame per second per video feed).

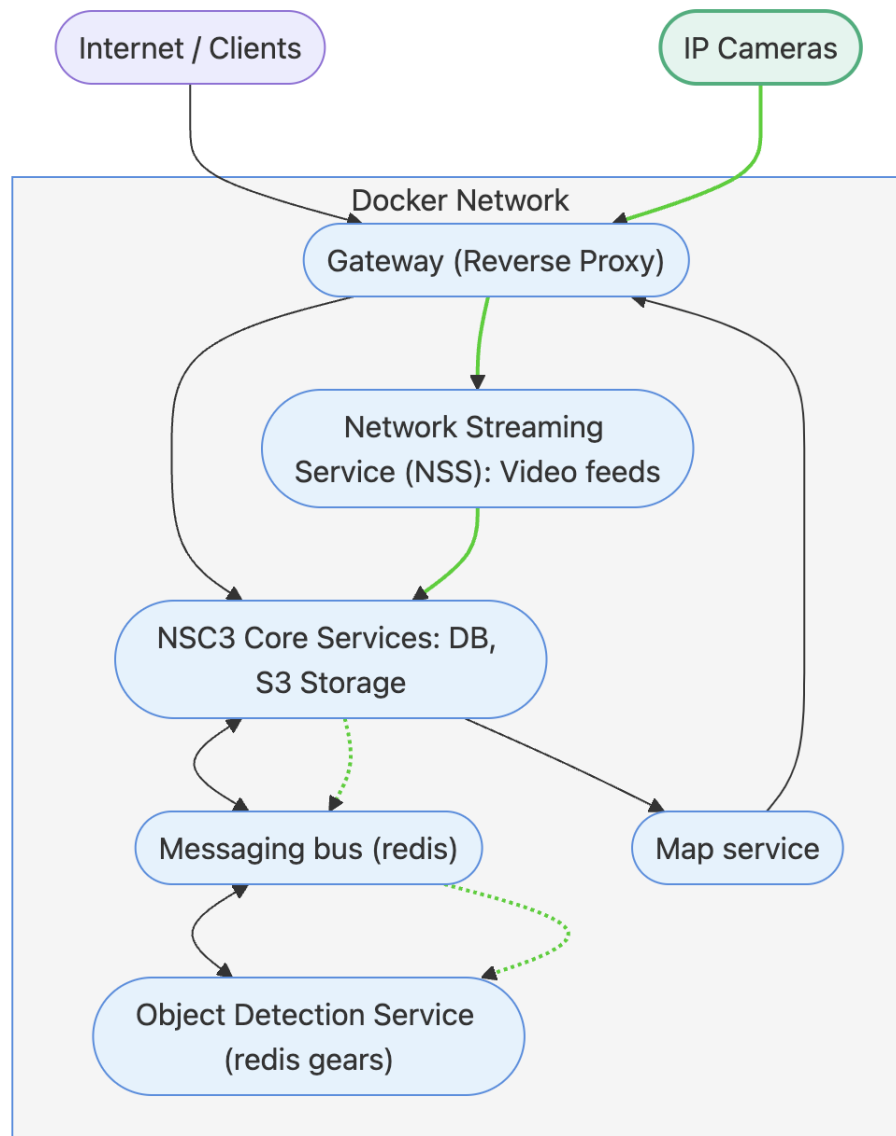


Figure 35 Data flow from video feed to AI driven detection

In detail, the object detection flow is as follows:

1. The camera sends a live video feed to the NSC3 system
 - IP camera feed is handled in the Network Streaming Service
2. NSC3 core services will store the video and its metadata
 - Once per second, an image is also directed to the messaging bus's object detection channel
3. The object detection service analyses the image
4. The analysed image and detection hit-box metadata are then passed to an algorithm that calculates coordinates for the object
 - This requires 3D coordinates, view angle, and Field of view for the camera

5. The detection details are then bundled into an event that is sent to the messaging bus
6. The event is handled by the map updater component in NSC3 core services
 - Detection location is checked for Geofence violations, and its metadata is updated accordingly
 - Detection is stored in the database and pushed to the dedicated map service

9 Integrated Status, Decision Support, and Control for Energy Communities and Distribution Networks

Energy communities and electricity distribution networks constitute a WP4 use case characterized by distributed assets, heterogeneous stakeholders, and the need to balance operational efficiency, stability, and regulatory constraints. The enablers grouped under Integrated Status, Decision Support, and Control for Energy Communities and Distribution Networks collectively address this challenge by providing coherent representations of system state, mechanisms for interpreting operational conditions, and means for executing or supervising appropriate actions.

In line with T4.1 (Digital twins & dashboards), these enablers establish authoritative digital representations of energy communities and distribution networks, supported by aggregation and visualization services that present current and historical system status to operators and stakeholders. This enables a shared understanding of operational conditions across organizational boundaries and supports WP4's objective of clearly showing the current situation in complex energy systems.

Beyond status representation, the enablers support T4.3 (Autonomous operations) and T4.4 (Detection and response) by enabling the detection of abnormal conditions, the evaluation of system state quality, and the activation of flexibility and control actions. Human-in-the-loop validation and governance mechanisms ensure that operational decisions remain transparent and aligned with regulatory and community objectives. Together, these enablers illustrate how WP4 integrates digital twins, reasoning, and action to keep energy communities and distribution networks under control in dynamic, real-world operating conditions.

ID	Enabler Name	Related Tasks	Addressed Requirements	Related Use Case
EN_24	Energy Community Operational Management and Governance Interface	T4.1, T4.3	REQ_UC4.7, REQ_UC4.13, REQ_UC4.14	UC4
EN_25	Human-in-the-Loop Community Action Validation Interface	T4.3	REQ_UC4.7	UC4
EN_26	Regulatory, Environmental and Community Performance Reporting	T4.2, T4.3	REQ_UC4.11 REQ_UC4.15 REQ_UC4.17 REQ_UC4.18	UC4
EN_27	Energy Community Digital Twin Backbone	T4.1, T4.2	REQ_UC4.1, REQ_UC4.2, REQ_UC4.10	UC4
EN_28	Energy Community Data Aggregation and Visualization Services	T4.1, T4.2	REQ_UC4.13, REQ_UC4.14	UC4
EN_29	Trust-Aware Digital Twin State Confidence Engine	T4.1, T4.2, T4.3	REQ_UC4.1 REQ_UC4.2 REQ_UC4.3 REQ_UC4.11	UC4

EN_30	Energy data sharing: Development for continuous delivery of meter data	T4.4	REQ_UC4.7	UC4
EN_31	Energy data sharing: REST APIs	T4.4	REQ_UC4.7	UC4
EN_32	Flexibility services: Load control capabilities	T4.4	REQ_UC4.7	UC4
EN_33	Digital twin for electricity distribution network with distributed generation and flexible load	T4.1	REQ_UC4.1, REQ_UC4.2	UC4
EN_34	Dashboard for energy community management	T4.1	REQ_UC4.7, REQ_UC4.13, REQ_UC4.14	UC4
EN_35	Activation of flexible resources, like storage and electric vehicles	T4.3	REQ_UC4.10	UC4
EN_36	Autonomous detection and mitigation of voltage issues	T4.4		UC4

9.1 Energy Community Operational Management and Governance Interface (EN_24)

This enabler provides the primary operational and governance interface through which CEL supervises, manages, and governs the energy community in real time. It is implemented as a web-based operational dashboard tightly coupled with the digital twin services developed in WP4, ensuring that all operator views are derived from a consistent and authoritative community state.

The interface consolidates real-time status information, historical trends, and short-term projections into a single coherent operational view. Status indicators include current consumption, production, storage levels, and energy import/export at community boundaries. These are complemented by trend visualizations and projected indicators derived from digital twin outputs, enabling operators to understand not only the current situation but also its likely evolution.

A key design objective of the interface is to reduce cognitive load while preserving traceability, accountability, and governance awareness. To this end, the dashboard emphasizes aggregated, community-level information by default, while supporting controlled drill-down to finer-grained views when required. All visualizations are aligned with the roles and responsibilities of the Energy Community Manager, ensuring that operational awareness does not expose unnecessary individual-level data.

The interface plays a central role during abnormal or stressed grid conditions, such as high peak demand, reduced local generation, or constrained import/export capacity. In such situations, the dashboard supports rapid situational comprehension by highlighting deviations from nominal behavior, boundary condition violations, and relevant KPIs, allowing CEL operators to assess the situation and coordinate appropriate responses in line with governance rules and regulatory constraints.

Key objectives are:

- Enabling day-to-day operational supervision
- Supporting governance and transparency toward community members
- Providing situational awareness during abnormal or stressed grid conditions

Functional scope

- Community member and asset registry visualization
- Aggregated consumption and production views
- Import/export visualization at community boundaries
- KPI-based summaries (self-consumption, peak contribution, flexibility usage)
- Drill-down from community to prosumer level (subject to access rights)

9.2 Human-in-the-Loop Community Action Validation Interface (EN_25)

While WP4 enables increasing levels of automation in community operation, CEL retains ultimate operational responsibility for actions affecting community members and grid interaction. This enabler provides a human-in-the-loop decision-support interface that ensures community-level actions remain transparent, governed, and accountable.

The interface acts as the final validation layer between automated system recommendations and real-world execution. Automated components may propose actions such as flexibility activation, load adjustment, or changes in operational modes based on digital twin analysis. However, these proposals are not executed automatically; instead, they are presented to CEL operators for review and validation.

Each proposed action is contextualized with its expected operational impact, including anticipated effects on community consumption, production, import/export levels, and relevant KPIs. The interface also exposes the governance context under which the action is proposed, enabling operators to assess compliance with cooperative rules, regulatory constraints, and community objectives.

Following execution, the interface supports post-action monitoring, allowing operators to verify whether the observed system behavior aligns with expectations. This closes the operational feedback loop and supports continuous improvement of automated decision logic.

By embedding human oversight directly into the action pipeline, this enabler ensures that autonomy enhances operational efficiency without undermining cooperative governance, regulatory compliance, or stakeholder trust.

The interface allows CEL operators to:

- Review proposed system actions
- Validate or reject actions based on governance rules
- Monitor post-action effects at the community level

This ensures that autonomy does not bypass cooperative governance or regulatory constraints.

9.3 Regulatory, Environmental and Community Performance Reporting (EN_26)

This enabler extends the CEL operational layer to support regulatory compliance, environmental performance assessment, and transparent communication of community outcomes, as explicitly required by UC4. It transforms operational situational awareness outputs into institutionally usable artefacts that support regulatory obligations, community governance, and stakeholder trust.

Building on digital twin state information and aggregated community-level data, the enabler enables CEL to continuously assess the performance of the energy community beyond immediate operational control. Environmental indicators, energy sharing metrics, and compliance-relevant KPIs are derived from authoritative digital twin states, ensuring consistency between operational views and reported outcomes.

The enabler supports regulatory reporting workflows by automatically generating reports aligned with energy-sector regulatory requirements. These reports provide verifiable summaries of community operation, energy exchanges, and environmental impact, reducing manual effort and the risk of inconsistencies or omissions. At the same time, the enabler supports community transparency by presenting key performance indicators and benefits in a form suitable for communication to community members.

By integrating reporting and performance assessment directly into the operational framework, this enabler ensures that situational awareness is not limited to real-time supervision but also supports accountability, long-term evaluation, and evidence-based governance of the energy community.

Functional scope

- Computation of environmental KPIs (CO₂ savings, renewable self-consumption)
- Calculation of energy sharing coefficients within the community
- Automated generation of regulatory and compliance reports
- Export of reports in regulator-compatible formats
- Visualization of community benefits for transparency

9.4 Energy Community Digital Twin Backbone (EN_27)

This enabler implements the core digital twin backbone for UC4, providing a continuously updated, authoritative virtual representation of the energy community. The digital twin models the logical and operational entities that constitute the community, including individual prosumers, shared energy assets, distributed energy resources, and grid interaction points at the community boundary.

The digital twin aggregates energy and operational data originating from heterogeneous IoT devices and metering infrastructures deployed across the community. To ensure continuity of situational awareness under real-world operating conditions, the enabler applies state estimation and baseline reconstruction techniques that compensate for missing, delayed, or degraded telemetry. This allows the system to maintain a coherent and usable community state even in the presence of partial data loss or communication disruptions.

A key design objective of the digital twin backbone is to guarantee time-bounded data availability. Energy consumption and production data are processed and made available within strict temporal limits, ensuring that 15-minute energy intervals required for community operation, analysis, and downstream services remain consistently accessible.

The digital twin persists both real-time and historical states, enabling reliable access for dashboards, trend analysis, KPI computation, and projection services implemented elsewhere in WP4. By centralizing state representation and ensuring its reliability and continuity, this enabler forms the foundational layer upon which higher-level status visualization, projections, and decision-support functions for UC4 are built.

Key capabilities:

- Secure ingestion of IoT and metering data
- Missing data replacement using historical baselines
- Time-bounded availability of 15-minute energy data
- Persistence of real-time and historical states

9.5 Energy Community Data Aggregation and Visualization Services (EN_28)

This enabler provides the data aggregation and transformation services that bridge the gap between the internal digital twin representation of the energy community and human-facing visualization and reporting components. Its primary function is to transform raw, high-frequency digital twin data into structured, interpretable representations suitable for operational dashboards, analytical views, and reporting tools.

The enabler performs aggregation across multiple dimensions, including time, assets, and community structure, enabling coherent community-level overviews while preserving the ability to analyze boundary interactions with the distribution grid. Import and export flows at the community interface are explicitly computed and exposed, supporting operational awareness and compliance with UC4 requirements.

To support governance and transparency, the aggregation services are designed to provide role-appropriate data exposure. By controlling the level of detail and abstraction applied to the data, the enabler ensures that different stakeholders (e.g., CEL operators, community members, external observers) receive views that are aligned with their roles and responsibilities, without exposing unnecessary or sensitive information.

By decoupling digital twin internals from visualization logic, this enabler contributes to a modular and scalable architecture, enabling dashboards and reporting tools to evolve independently while relying on consistent and authoritative aggregated data.

Aggregation occurs at multiple levels to support:

- Community-level overview
- Import/export boundary analysis
- Role-appropriate data exposure

9.6 Trust-Aware Digital Twin State Confidence Engine (EN_29)

This enabler augments the energy community digital twin backbone with explicit state confidence assessment, enabling trust-aware situational awareness for UC4. Rather than exposing measured, estimated, or reconstructed states as equally reliable, the digital twin associates each operational state and derived KPI with a confidence level that reflects the quality and reliability of the underlying data.

Confidence levels are computed by evaluating multiple contributing factors, including data freshness, data completeness, the reconstruction or estimation methods applied, and the reliability of data sources. By making uncertainty explicit, the enabler allows both human operators and automated components to reason about the trustworthiness of the digital twin state, which is a critical requirement in the operation of electricity distribution infrastructure.

The confidence engine is tightly integrated with the digital twin lifecycle. Confidence information is propagated alongside state updates and KPIs, ensuring that dashboards, projections, and action proposals are all qualified by an explicit indication of reliability. This prevents misleading interpretations of reconstructed data and supports informed decision-making under degraded sensing or communication conditions.

By introducing confidence-aware state representation, this enabler bridges the gap between technical state estimation and operational decision-making, ensuring that automation enhances reliability without obscuring uncertainty.

Functional scope

- Confidence scoring for digital twin states
- Visual indicators of data reliability
- Propagation of confidence into projections
- Guardrails preventing unsafe actions under low confidence
- Explainable reasoning for operators

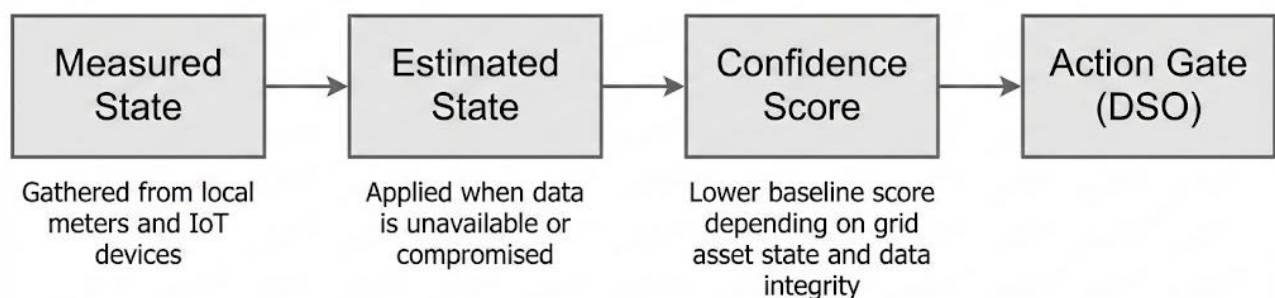


Figure 36 State processing flow

9.7 Energy data sharing: Development for continuous delivery of meter data (EN_30)

The new EU regulations will require measurement data to be transmitted at a significantly faster pace. EnerimEDM to introduce continuous measurement data collection and delivery, enabling near real-time data streams that can be leveraged across operational processes, included also for energy community members.

9.8 Energy data sharing: REST APIs (EN_31)

Extended APIs to enable flexible integration with third-party systems. These interfaces support modern microservice architecture and scale efficiently to handle large volumes of data. For example, sharing energy community data from customer information management system or receiving and retrieving measurement data via REST APIs or queue based solutions as a future-proof alternative to SOAP-based integrations.

9.9 Flexibility services: Load control capabilities (EN_32)

EnerimCIS is evolving to empower energy companies and consumers with advanced flexibility services. By introducing e.g. intelligent load control capabilities, the solution ensures compliance with EU-level regulations while supporting the dynamic transformation of electricity markets.

9.10 Digital twin for electricity distribution network with distributed generation and flexible load (EN_33)

The technical enablers within T4.1 focus on the virtualization of the distribution network and the creation of interfaces for multi-purpose visualization. The digital twin for the electricity distribution network serves as the foundational analytical engine, facilitating the missing data replacement (REQ_UC4.1) requirement. By utilizing state estimation algorithms and historical baseline datasets, the digital twin synthesizes virtual sensor data to maintain operational continuity in the event of telemetry loss. Furthermore, the twin's architecture is optimized for high-frequency synchronization to satisfy data availability (REQ_UC4.2) constraints. This ensures that 15-minute energy usage packets are processed and made available within the strict latency "hard limits" required for near real-time grid state projection.

9.11 Dashboard for energy community management (EN_34)

Adding to the virtual modeling, a dashboard for energy community management serves as the primary gateway for human oversight. This enabler fulfills the energy community management requirement (REQ_UC4.7) through a robust UI and a series of REST API calls designed to handle the lifecycle of community participants. It integrates the energy community overview (REQ_UC4.13) and energy community import/export (REQ_UC4.14) requirements by aggregating high-dimensional IoT data into simplified visual KPIs. These visualizations allow operators to monitor production and consumption trends at both the prosumer and community levels, providing the necessary transparency to validate the full potential of community-based energy services.

9.12 Activation of flexible resources, like storage and electric vehicles (EN_35)

Technical enablers, related to T4.3, address the technical requirements for localized, real-time actuation through the activation of flexible resources, specifically targeting battery energy storage systems and electric vehicle integration. This enabler is intrinsically linked to the IoT data retrieval (REQ_UC4.10) requirement, as autonomous decision-making necessitates the acquisition of status points from a heterogeneous array of participating IoT devices. The R&D in this task focuses on low-latency communication protocols that allow the system to autonomously dispatch flexible loads to balance the local grid without requiring centralized cloud intervention, thereby enhancing the resilience of the community microgrid.

9.13 Autonomous detection and mitigation of voltage issues (EN_36)

Technical enablers, related to T4.4, focus on the situational awareness frameworks capable of distinguishing between nominal grid states and emergency conditions. The enabler for the detection and responses functions as a closed-loop control system that monitors the energy community import/export data. By applying automated reasoning to detect voltage deviations, the system triggers corrective actions to keep the grid within safe operational bounds. This enabler bridges the gap between passive monitoring and active grid stabilization, ensuring that the community's net energy exchange remains harmonized with

the Distribution System Operator (DSO) specifications while moving the local cluster toward its targeted energy-efficiency state.

10 Human-Centric Digital Twins, Tracking, and Spatial Situational Awareness

Human participation is a critical factor in many WP4 use cases, requiring continuous awareness of people's location, context, and interaction with cyber-physical systems. The enablers grouped under Human-Centric Digital Twins, Tracking, and Spatial Situational Awareness address this need by combining digital twins, positioning, and spatial visualization to represent the state of individuals and shared environments in a coherent and operationally meaningful way.

In alignment with T4.1 (Digital twins & dashboards) and T4.2 (Positioning & tracking of objects), these enablers establish authoritative digital representations of people, places, and assets, supported by continuous tracking and historical state management. User interfaces and dashboards provide role-appropriate views of current and past situations, enabling operators and participants to understand system status and human presence across distributed environments.

The enablers further contribute to T4.3 (Autonomous operations) and T4.5 (Real-time deep 3D maps & tools) by enabling context-aware reactions and spatially grounded situational views. Automatic triggering based on wearable state and adaptive data dissemination ensure that relevant information remains available in real time, while 3D maps and spatial tools support intuitive understanding of complex environments. Together, these enablers illustrate how WP4 integrates human-centric digital twins, tracking, and spatial visualization to support safe and effective operation of cyber-physical systems in dynamic, real-world settings.

ID	Enabler Name	Related Tasks	Addressed Requirements	Related Use Case
EN_37	Multicommunicator to mobile, web, car and watch interface	T4.1	REQ_UC7.13, REQ_UC7.16, REQ_UC7.17, REQ_UC7.18, REQ_UC7.19	UC7
EN_38	Real-time monitoring services for Multicommunicator environment in server	T4.1	REQ_UC7.16	UC7
EN_39	Data sharing	T4.2, T4.3, T4.4	REQ_UC7.2	UC7
EN_40	Autonomous media optimized data traffic between components	T4.2, T4.4	REQ_UC7.20, REQ_UC7.21	UC7
EN_41	3D map engine, server and research	T4.1, T4.5	REQ_UC7.23, REQ_UC7.24	UC7
EN_42	Multiplace Digital Twin	T4.1	REQ_UC7.25, REQ_UC7.26, REQ_UC7.27	UC7
EN_43	Biometric Data Transfer Device	T4.1, T4.2, T4.3, T4.4, T4.5	REQ_UC7.4, REQ_UC7.25	UC7

10.1 Multicomunicator to mobile, web, car and watch interface (EN_37)

Multicomunicator to mobile, web, car and watch interface is User interface and application component within existing Natlink WeHunt X based apps utilizing the Multiplace Digital Twin system, designed to visualize and support the integration and management of real-time device data. Its primary focus is to enable the collection, storage, and presentation of operational information for devices, serving as the backbone for visualisations and dashboards created in T4.1. This enabler ensures that various data streams are efficiently combined and processed, facilitating the digital twin's ability to reflect accurate and up-to-date states, which is essential for ongoing monitoring and predictive analysis of device performance. In case there are no existing app, it is created. E.g. Car version is created within the project.

10.2 Real-time monitoring services for Multicomunicator environment in server (EN_38)

Real-time monitoring services for Multicomunicator environment in server is a web-based admin and monitoring interface designed to provide users with comprehensive control and oversight of the Multiplace Digital Twin system. It offers a user-friendly platform for administrators to configure system settings, manage devices, and monitor operational data in real time. Through intuitive dashboards and visual tools, EN_38 facilitates efficient system management, allowing for quick identification of issues and streamlined maintenance processes.

The interface supports secure access for authorised personnel, ensuring that sensitive data and system operations remain protected. Its responsive design enables access from various devices, including desktop computers and tablets, making it suitable for both onsite and remote management. Integration with the underlying digital twin infrastructure means that the EN_38 interface can display up-to-date device statuses, historical performance data, and predictive analytics, thereby enhancing decision-making and operational efficiency.

10.3 Data sharing (EN_39)

Data sharing is a data transmission enabler focused on facilitating seamless data sharing across multiple communication channels within the Multiplace Digital Twin system. It enables the transfer of data from devices to a receiver over radio frequencies, ensuring reliable connectivity even in environments with limited infrastructure. Once received, the data is relayed from the receiver to a mobile device, where it can be processed or visualised as needed. Subsequently, the mobile device transmits the information to a central server, allowing for aggregation, storage, and further analysis. The server then supports onward distribution, enabling data to be sent to other mobile devices as required, thus ensuring that up-to-date operational information is available throughout the network. This enabler is crucial for maintaining real-time synchronisation of device states, supporting collaborative operations, and enabling robust data flow across distributed teams, regardless of their physical location.

10.4 Autonomous media optimized data traffic between components (EN_40)

Autonomous media optimized data traffic between components is an autonomous media-optimised data traffic enabler designed to facilitate efficient communication between system components within the Multiplace Digital Twin environment. It addresses REQ_UC7.20 by ensuring that data connections between Multicomunicator and tracking devices are both optimised and prioritised, carefully balancing radio resource usage and battery consumption to maintain operational reliability. This enables the system to transfer media and operational data in a manner that maximises throughput while minimising unnecessary overhead, thereby supporting real-time synchronisation and responsiveness.

In line with REQ_UC7.21, EN_40 supports autonomous network operation, allowing devices developed within the project to function independently without the need for ongoing human intervention. These devices are capable of self-organising into networks, automatically establishing connections and leveraging mesh technologies to maintain robust communication even in challenging environments. This autonomy ensures that network operations remain resilient and adaptive, with devices intelligently managing their connectivity and resource allocation to support mission-critical tasks and collaborative operations.

10.5 3D Map Engine, Server, and Research (EN_41)

3D Map Engine, Server, and Research covers both backend and frontend development for 3D map engine integration in the Multiplace Digital Twin system, fulfilling REQ_UC7.23 and REQ_UC7.24. It involves researching, adopting, or creating tools to display situational information—such as device position, status, maps, tracked units, images, and media—on computers and mobile devices. Where existing engines fall short, custom backend and user-facing frontend solutions will be developed.

This enabler supports spatial data visualisation in real time and historically, driving user awareness and collaboration. The process evaluates available market options, with adoption or bespoke development depending on project needs.

EN_41 also enables media sharing and analysis, enhancing situational awareness for users and authorities. The combined backend and frontend approach ensures advanced 3D visualisation, robust server performance, and intuitive interfaces for actionable insights.

10.6 Multiplace Digital Twin (EN_42)

Multiplace Digital Twin is a comprehensive solution that brings together the capabilities of mobile devices, servers, and, if technically feasible, portable wearables such as smartwatches that can operate independently of a phone. The aim is to ensure that users can access, monitor, and manage their digital twin assets seamlessly across a variety of platforms, supporting real-time device tracking and operational visibility wherever they are.

Within the Sa4CPS project, EN_42 forms the backbone of digital twin research and development in WP4. Natlink's work focuses on dashboards and visualisations that show the system's operational status, historical data, and even predictive insights, using both server and mobile applications, as well as integrations with vehicles and wearables. The system is built to collect real-time device data from many possible sources and messages, bringing them together into a unified digital twin that always reflects the latest information and settings. From the manufacturing phase onward, every detail—from hardware specifications and software versions to component records and device identifiers—is stored and tracked, with each change logged by source and timestamp to ensure transparency and effective control throughout the device's lifecycle.

A key feature of EN_42 is its historical data management. Users can query the system for the status of any device, dog, or person as it was at a specific moment in the past, retrieving all settings, locations, and sensor data that were valid at that exact timestamp. The system relies on efficient storage, registering only changes and their update times, making it possible to reconstruct device states accurately without unnecessary repetition. It manages data that may arrive out of order, always honouring the true timestamp of the information, and provides controlled access to history depending on user roles and privacy needs—so, for example, maintenance staff can access previous location data, but new owners cannot.

EN_42 also facilitates operational flexibility and collaboration. It handles the complexities of simultaneous settings changes from different users or interfaces, making sure that changes are tracked and conflicts resolved before actions are finalised. Alerts and insights are easily generated when anomalies occur, such as gaps in tracking, battery drops, or hardware changes.

Overall, EN_42 Multiplace digital twin stands as a central hub for lifecycle management, monitoring, and collaborative operations, always maintaining a clear, up-to-date, and historically rich model of every asset in the field.

10.7 Biometric Data Transfer Device (EN_43)

The purpose of this section is to define the role of a secure data pipeline that addresses situational awareness in building a digital twin of a person using wearable devices. A digital twin includes both location and motion data as well as physiological functional capacity data. Based on this information, it is possible to create a digital twin in both real-time and temporal longitudinal continuum.

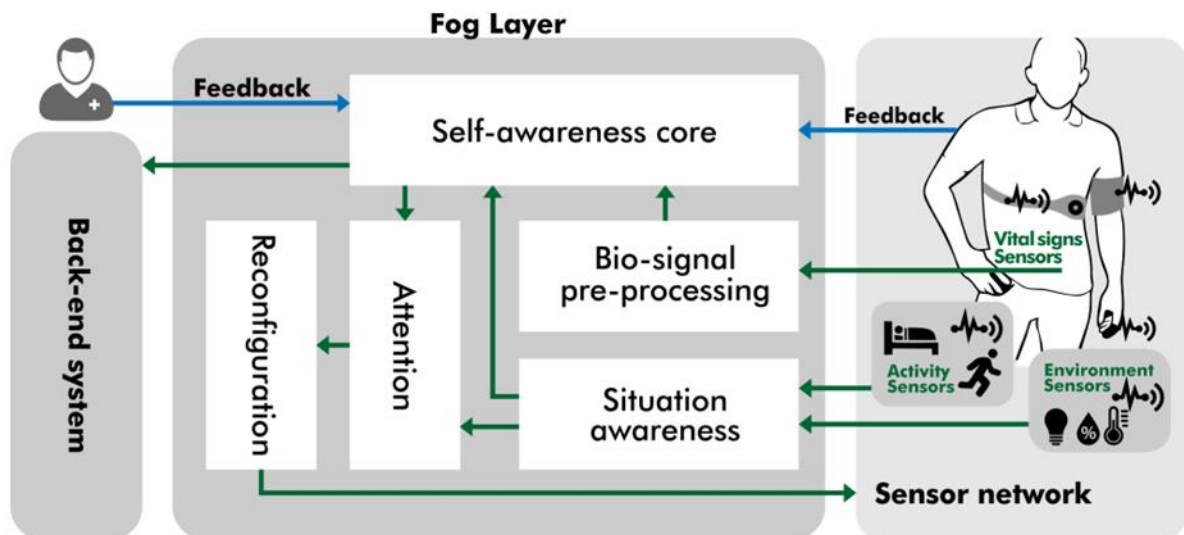


Figure 37 Architecture of the Data Transfer Device service

10.7.1 Wearable devices and a data pipeline to create situational awareness (Task 4.2)

Earlier in task 4.1 (REQ_UC7.25 "Mobile, Server, and possibly also portable devices as watch implementation in case watch is working without phone"), Polar introduced a **Data Transfer Device** (DTD) as a "FOG layer" that is intended to transfer biometric data in real time to a monitoring system over the Internet (UC7 requirements).

The following requirements in task 4.1 has not been met yet:

- Historical stored to view the state of the Digital Twin at any given time. As a result of the query to twin history, you get situation of the device / person at time stamp, you get all settings, location, all sensor data which was valid in this specific timestamp, taking care also privacy and data level considerations. Digital twin history data viewing rights is dependent on the viewer, time of the data and level of the user

- Create Digital Twin, of the device, person and handheld, which is stored in Mobile app and in Server, and is Read / Write and handles user rights and flow control of the updates and visibility.

The significance of a digital twin of a person becomes evident in a situation where previously collected biometric data is needed for real-time operational control and replanning. The digital twin consists of both biometric data and a set of goals used in the management of operations, which are compared from the perspective of the planned content and implementation of the activities. This review includes parameters that can be derived from biometric data and the basics of the physiology of strain and rest. Such control logic requires the use of AI technology to coordinate the planning, implementation, and biometric data of operations.

The Task 4.1 support is created when the data is transferred to the Guidance system for processing groups and content. The architectural specification is underway. The requirements specification of the system mainly has been done, and the preliminary user interface logic has been developed for the browser, but it does not store data yet. Digital Twin of the device and person stored in mobile app and server is also underway. Security and user rights definitions are included into the data transfer device and the Guidance server.

The enabler has the following requirements to be met:

1. Data sharing over wearable to DTD, from DTD to mobile and from mobile to server
 - The BLE-SDK interface implementation from the wearable device to the DTD has been implemented in the manner described, but the control of a person's actions and digital twin requires the creation of a digital twin of the wearable device as well, in which case the DTD is required to have two-way synchronization capability: Receives the location, motion and biometric data of the wearable device, transmits the data and transfers the setpoints required for the operation to the wearable device, e.g. mobile application or guidance system
 - Digital Twin of the DTD will be created and monitored from the mobile and web application
2. Mechanism to be WiFi router that binds IoT radio and WiFi (or bluetooth) together by giving restricted access and distance extension for mobile phone for specific data and apps.
 - Instead of WiFi, DTD uses the BLE SDK
3. Real-time and history administration and monitoring / dashboard system for Multi communicator environment, in server / web environment
 - The definition of requirements for the pedagogical guidance process, including the earning models, has mainly been done. The architectural design of the system is underway and is going to be implemented by both Polar and the subcontractor. The requirement specification and design of the mobile application for real-time operation control have not been done. Its purpose is to carry out the necessary control operations based on location, motion and biometric data collected in real time. Automatic performance testing protocols and implementation are also the task of the application but will be implemented later.
4. Optimized and prioritized data connections, which are using reasonable amount of radio resources and battery (T4.2-T4.4)
 - To determine the security levels of DTD data connections, requests for quotations have been requested from various suppliers who have the AQAP certificate for the delivery

capability of devices and secure systems suitable for data transfer in accordance with the MIL standard. The definition of information security levels starts from fulfilling the minimum EU data security requirements for civil data traffic towards MIL requirements

5. The devices created in the project must be autonomous and be able to operate without continuous human controlling and monitoring, they could also network themselves automatically and utilize e.g. mesh technologies (T4.3)
 - The DTD is not fully autonomous. It works autonomously only when collecting data from the wearable device and transferring it to monitoring and control systems. For other operations, the user is required to take steps to set and start various protocols for wearable devices. The need for DTD devices or the ability to create Mesh networks has not yet been investigated, but such a capability could be created if it is considered tactically important to implement
6. Research available 3D map engines and possibly implement one if not available and tools to present the situational information, including devices operational status, requirements for services, maps, tracked units, images and other media, in computers and mobile devices. Polar provides biometrical data for 2nd tier parties and maps for smart watches (T4.5)

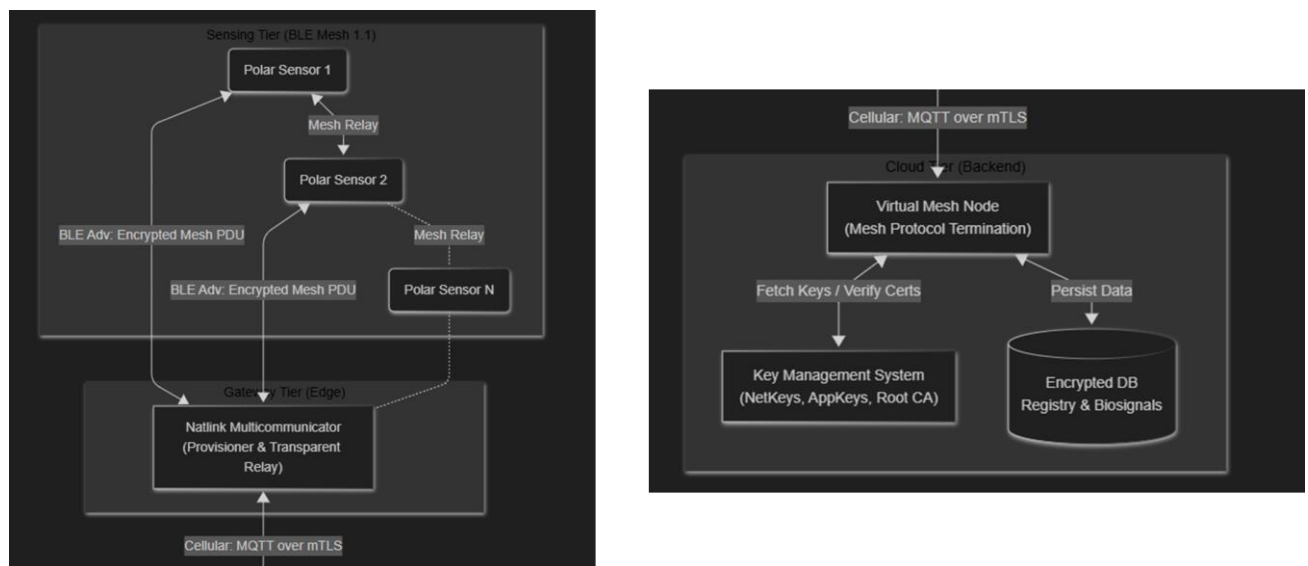


Figure 38 DTD Processes

Polar wearable wrist devices already have map and orienteering functionality. The Flow service also includes the service in question. It may be necessary to specify separately if 3D map services are needed for the management of group activities with the same kind of technology that Natlink has planned to implement. In principle, MIL situational awareness systems include map information, but the matter remains unresolved in the case of civilian sports and exercise.

11 Conclusions

This deliverable establishes the specification baseline for the technical enablers addressing status, projections, and actions in WP4. The specifications defined in D4.2 provide the necessary input for the subsequent WP4 deliverables, which will progressively move the enablers from specification to realization and validation.

Building directly on D4.2, Deliverable D4.3 will focus on the design and architectural realization of the specified enablers, detailing their internal structure, interfaces, and integration within the overall Sa4CPS architecture. Particular emphasis will be placed on cross-enabler interoperability and on alignment with the trust, security, and privacy mechanisms defined in WP5. Following this, Deliverable D4.4 will address the final implementation, integration, and demonstration readiness of the WP4 enablers, consolidating them into operational solutions that are evaluated across the project use cases.

12 References

All the references have been provided as footnotes in this document.